SOCIETY FOR ECOSYSTEM RESTORATION
IN NORTHERN BRITISH COLUMBIA

Restoring Fish Passage in the Peace Region - 2025

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Prepared for
Fish and Wildlife Compensation Program
and
Fish Passage Technical Working Group

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Society for Ecosystem Restoration in Northern BC

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new graph environment

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Acknowledgement

We understand our well-being as inseparable from the health of the land and waters we work within. When we care for ecosystems, we care for ourselves and for the communities connected to them.

Executive Summary

Executive Summary (PDF)

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_peace_2025_reporting. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from fish_passage_peace_2025_reporting.pdf.

The Society for Ecosystem Restoration in Northern BC (SERNbc) is working together with the McLeod Lake Indian Band, the Peace Region Fish and Wildlife Compensation Program (FWCP), the Provincial Fish Passage Technical Working Group (FPTWG), road/rail tenure holders and other stakeholders/partners to prioritize, plan and fund the restoration of fish passage at road and rail crossing structure barriers within the FWCP Peace Region.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

The project engages FWCP partners and stakeholders to clearly communicate fish passage issues in FWCP Peace Region watersheds while collaboratively planning and executing the steps necessary to achieve fish passage restorations. The work completed and ongoing aligns with the Fish and Wildlife Compensation Program Rivers, Lakes and Reservoirs Action Plan sub-objective 6 of addressing fish passage issues in streams to enhance the productivity of priority species. Project activities undertaken address the following actions:

- PEA.RLR.S06.RI.20 — Conducting engagement to prioritize options for fish passage improvement (P1)
- PEA.RLR.S06.RI.19 — Conducting research to prioritize fish passage actions (P1)
- PEA.RLR.S06.HB.21 — Restoring fish access to streams (P1)

Connectivity modelling now runs weekly with automated updates across all 16 watershed groups in the FWCP Peace area, driven by `bcfishpass` outputs using channel width, gradient, and provincial / federal barrier inventories. Field assessments were completed this year within five of those watershed groups — Parsnip River, Crooked River, Carp River, Nation River, and Parsnip Arm — to maintain momentum and identify candidate sites and partners. In parallel, we contribute desktop and field corrections back to the provincial `bcfishpass` modelled-crossing inventory — 282 records since April 1, 2025 across the FWCP Peace Region — refining the inputs the weekly model consumes and the same inventory other analyses across BC rely on (see the Methods chapter for details). The work and its forward proposal have the support of the Provincial Fish Passage Technical Working Group (FPTWG) — a partnership between the Province of British Columbia (WLRS, MOF, MOE, MOTI) and Fisheries and Oceans Canada.

Alongside the automated weekly updates, we use our open-source R packages `fresh` and `link` to expand, experiment with, and develop new project-specific habitat-modelling outputs — most recently an initial Arctic grayling intrinsic-habitat layer for the FWCP Peace area, parameterized on stream size and gradient. The framework is designed to take additional covariates as data and methods mature: stream-temperature and Growing-Season Degree Days predictions from collaborative modelling with Poisson Consulting, water-temperature observations via `water-temp-bc`, and calibration of predictions against fish-presence observations in `bcfishobs` together with research activities (some commissioned for FWCP Peace) that identify areas of known high-value spawning and rearing habitat. Each release of this report is tagged so reviewers can navigate between the submitted snapshot and current state at the report URL referenced above.

Although the main purpose of this report is to document 2025 field work data and results, it also builds on multiple prior years of SERNbc fish passage reporting; per-site links to the relevant prior reports are in the Assessment Data Summary appendix.

Fish passage assessment procedures conducted through SERNbc in the FWCP Peace Region since 2020 are amalgamated online within the Results and Discussion section of the report found here, which includes links to project reporting for each site.

In 2025, fieldwork activities took place within the Parsnip River, Parsnip Arm, Carp River, and Nation River watershed groups. Fish passage assessments and habitat confirmations were completed at new and previously identified crossings, with site-level results documented per-crossing in the appendices. Two crossings advanced to FWCP-funded 2026 implementation following this year's work: PSCIS 199663 (Kennedy Siding, tributary to the Parsnip River — replacement with a streambed simulation in coordination with BC Timber Sales) and PSCIS 203597 (THUT15000 Deactivation, tributary to the Nation River — riparian enhancements including soil decompaction, native planting, and coarse woody debris placement).

Environmental DNA (eDNA) sampling was incorporated at program scale this year. A total of 35 samples were collected across 10 streams in partnership with the University of Northern British Columbia (UNBC) for ddPCR analysis. Per-species detection results informed crossing prioritization and partner conversations, and are summarized in Appendix - Environmental DNA Results and visualized in the interactive Peace eDNA map.

Fish sampling and PIT-tagging was completed at three monitoring sites using a custom fish-passage effectiveness monitoring form standardized across the program. Ongoing effectiveness monitoring continued at PSCIS 125179 (Tributary to Missinka River) and PSCIS 125131 (Tributary to Table River), extending multi-year baselines from prior reports; baseline monitoring was initiated this year at PSCIS 198692 (Kerry Lake) — originally framed as pre-replacement, now repurposed as pre-removal: Sinclar Forest Products, as the Kerry Lake FSR tenure holder, has pivoted from structure replacement (24/25) to road decommissioning and removal of PSCIS 198692 (km 9) and the adjacent PSCIS 198693 (km 8) in 25/26. eDNA sampling was integrated at all three sites as part of both the baseline and the ongoing effectiveness components of the monitoring protocol.

Two analytical dimensions are new to the program this year. A floodplain delineation pilot was completed for the Parsnip River Watershed Group using the `flooded` R package, mapping the functional floodplain footprint and the off-channel habitat units (lakes, wetlands, side channels) that lateral connectivity supports — see Appendix - Floodplain Delineation. A climate departure analysis was completed for the FWCP Peace Region using the `cd` R package, quantifying temperature, precipitation, snowpack, and atmospheric drying trends since 1951 across the region's ecoregions — see Appendix - Climate Departure. Both layers complement the structural barrier inventory by providing the watershed context that prioritization decisions depend on.

Fish passage work in this region operates inside the constraints described in the Approach section of the background: thousands of structures across multiple jurisdictions and tenure holders, six-to-eight-figure remediation costs, and a finite funding envelope. Decisions about which crossings to advance are made opportunistically — pursuing simpler, lower-cost options to maintain momentum, while building the multi-year relationships and shared data that larger commitments require. The breadth of 2025 partner engagement is summarised in Engage Partners.

Looking ahead, the Recommendations chapter outlines a coordinated set of next steps. Headline themes:

- Advance site-specific work in 2026/27 — replacement of crossing 199663 (Tributary to the Parsnip River) on the Chuchinka-Colbourne FSR with BC Timber Sales (BCTS); above-minimum-standards riparian restoration on the Tributary to Nation River deactivations with BCTS; and support for Sinclar Forest Products' planned road decommissioning and structure removals at PSCIS 198692 and 198693 on the Kerry Lake FSR — with both the Kerry Lake and Tributary to Nation River riparian / deactivation sites acting as field sites for a native-seed and plant-propagation collaboration with the University of Northern British Columbia (UNBC) and the College of New Caledonia (CNC).
- Continue effectiveness monitoring (fish sampling, electrofishing with PIT tagging, eDNA, water-temperature data, and aerial imagery) extending multi-year baselines at the Missinka, Table, and Kerry Lake sites.
- Push the climate-departure and floodplain-delineation analyses forward — extending the watershed-group ecoregion mapping that anchors barrier prioritisation in regional climate

signal, expanding the functional-floodplain footprint to additional FWCP Peace watersheds, and building partner capacity to interpret and act on what these layers reveal.
- Develop a public webmap interface for fish-passage priorities and outcomes.
- Position `water-temp-bc` as an example of a data-portal model that can be extended or replicated to incorporate other past and future datasets collected by UNBC, cumulative-effects monitoring initiatives, FWCP-funded research, and others.
- Model and extend the same open-access pattern for eDNA data across the broader research community.

The intent across all of these is to anchor per-crossing fish-passage decisions inside the watershed-scale context they belong in, and to build a community of practice that can address basin-scale questions together.

1 Introduction

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The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

This project builds on prior years of Society for Ecosystem Restoration Northern BC (SERNbc) fish passage reporting (Irvine 2020; Irvine 2022; Irvine and Winterscheidt 2023, 2024; Irvine and Schick 2025); per-site links to the relevant prior reports are in the Assessment Data Summary appendix.

The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al. 2002; Swales and Levings 1989; Diebel et al. 2015). Culverts can present barriers to fish migration due to low water depth, increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney and Zaldokas 1997; Cote et al. 2005). As road crossing structures are commonly upgraded or removed there are numerous opportunities to restore connectivity by ensuring that fish passage considerations are incorporated into repair, replacement, relocation and deactivation designs.

Although remediation and replacement of stream crossing structures can have benefits to local fish populations, the costs of remedial works can be significant and the impacts of the work often complex to evaluate and quantify. Additionally, allocation of ecosystem restoration funding towards infrastructure upgrades on transportation right of ways are not always considered ethical under all circumstances from all perspectives. When funds are finite and invested groups are engaged in fund raising, cost benefits and the ethics of crossing replacements should be explored collaboratively alongside the cost benefits and ethics of alternative investment activities including transportation corridor relocation/deactivation, land procurement/covenant, cattle exclusion, riparian/floodplain restoration, habitat complexing, water conservation, commercial/recreational fishing management, salt water interventions and research.

Please note that at the time of reporting this document can be considered a living document. See the Open Source Reporting Framework section for details on version tracking and project documentation.

2 Background

2.1 Approach

Fish passage assessments in British Columbia follow the provincial protocol (MoE 2011), which scores barrier risk at road-stream crossings from structural parameters — culvert embedment, slope, outlet drop, and culvert diameter relative to channel width. The “Barrier” and “Potential Barrier” classifications represent passability for juvenile salmon or small resident rainbow trout under any flow conditions that may occur throughout the year (Clarkin et al. 2005; Bell 1991; Thompson 2013). The thresholds are deliberately conservative, so the classifications are risk scores rather than determinations of whether fish actually move through. Many structures classified as “Barrier” still pass some species and life stages at some flows, and natural features in adjacent reaches can be as limiting as the structure itself. Passability can be quantified in other ways — fish length, swim speed, and species physiology all affect connectivity outcomes — and finer-grained scoring methods exist for structures already classified as barriers (Bourne et al. 2011; Kemp and O’Hanley 2010; Washington Department of Fish & Wildlife 2009). Whether a remediation changes outcomes for fish therefore depends on the habitat upstream, the species and life stages in question, and the broader condition of the reach the structure connects to — not the classification alone.

The provincial inventory contains thousands of structures on fish-bearing streams. Replacement at a typical forest road culvert is in the six-figure range; structural replacements on highway and rail corridors run into seven and eight figures. Funding for restoration is finite and split across multiple programs and partners. Prioritization across the inventory therefore carries more weight than any individual assessment — the assessment establishes what could be fixed, while prioritization decides what should be.

Authority and funding for these structures are distributed across rail authorities, highway ministries, forestry licensees, and private landowners. Each operates under its own mandate, performance metrics, and budget cycle. Advancing restoration involves working inside organizations whose primary mandate is moving goods, maintaining roads, or harvesting timber — not aquatic ecological outcomes. The work proceeds through shared data, sustained relationships, and incremental progress across multiple budget years and field seasons.

Many road and rail corridors were built without consideration of fish passage or floodplain function. Damage frequently extends beyond the structure at the crossing — channelized streams, dykes that sever lateral connection, drained wetlands, cleared riparian buffers. Replacing a culvert in-kind can restore longitudinal passage while leaving the broader corridor-scale disconnection unaddressed. Understanding where infrastructure constrains watershed function helps distinguish cases where structural replacement is sufficient from cases where remediation needs to address more than the structure itself.

This report compiles the layers used for prioritization in the watershed groups where the work is active: structural assessment data, habitat modelling, climate trajectory, and floodplain extent. Each layer contributes a piece of the picture — what is present, what it could support, how it is changing, and the surrounding watershed context. The work happens over many years across many partners; this report documents one year's contribution to that effort.

2.2 Project Location

The study area includes the FWCP Peace Region with a focus to date on traditional territories of the Tse'khene First Nations. In 2025, field assessments were completed with the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups (Figure 2.1).

In 2019/2020, following a literature review, analysis of fish habitat modelling data, the Provincial Stream Crossing Inventory System (PSCIS) and a community scoping exercise within the McLeod Lake Indian Band habitat confirmation assessments were conducted at 17 sites throughout the Parsnip River watershed with 10 crossings rated as high priorities for rehabilitation and three crossings rated as moderate priorities for restoration. An engineering design for site 125179 on a tributary to the Missinka River was also completed through the 2019/2020 project. In 2021/2022, project activities reconvened through FWCP directed project PEA-F22-F-3577-DCA. Partners were engaged, funding was raised, planning was conducted and reporting was completed to initiate restoration activities of high priority crossings. Materials were purchased and permitting was put in place to prepare for replacement of the twin culverts on the Missinka River tributary with a clear-span bridge. In 2025/2026, this collaborative project leveraged ongoing connectivity restoration initiatives in the province and engaged multiple partners to catalyze fish passage restoration activities at high-priority sites identified in from 2019-2024.

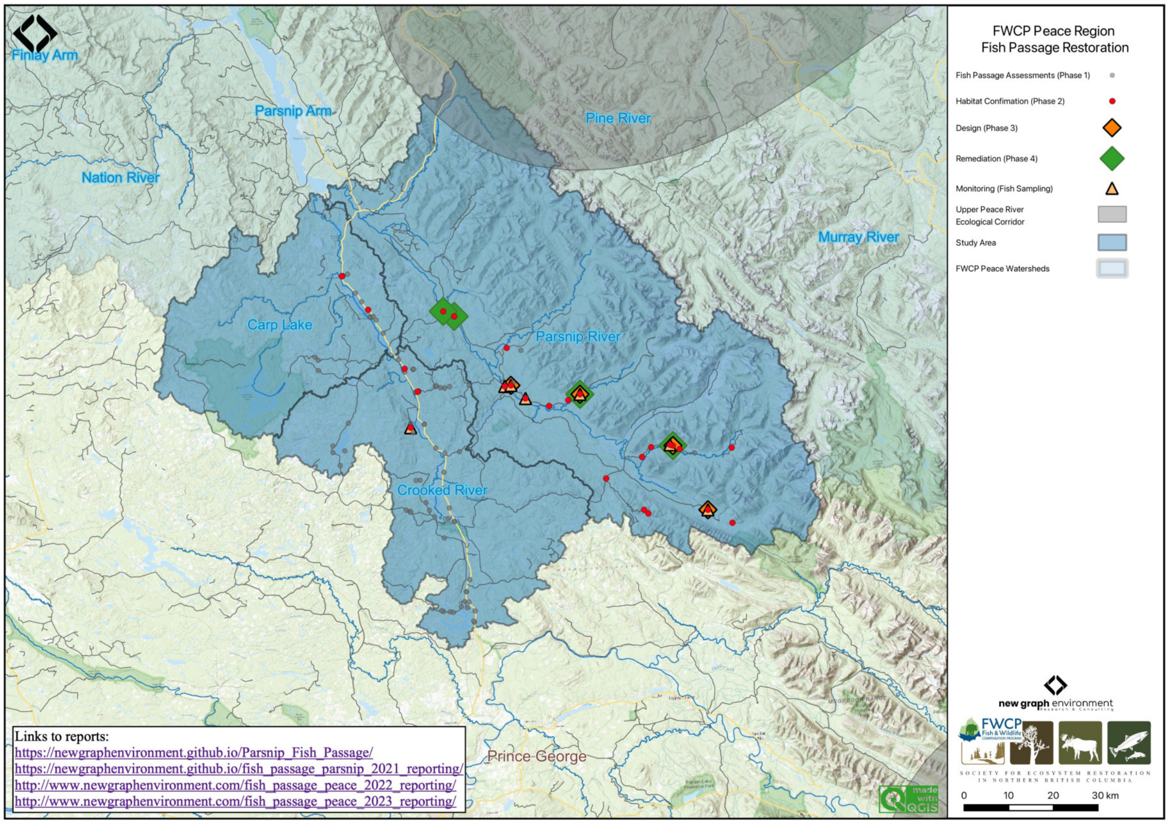


Figure 2.1: Overview map of Study Area

2.3 Tse'khene

The Parsnip River watershed is located within the south-eastern portion of the 108,000 km^2 traditional territory of the Tse'khene from the McLeod Lake Indian Band. The Tse'khene *"People of the Rocks"* are a south westerly extension of the Athabascan speaking people of northern Canada. They were nomadic hunters whose language belongs to the Beaver-Sarcee-Tse'khene branch of Athapaskan ("History Who We Are" 2023). Extensive work is underway to preserve the language with resources such as First Voices available online and in app form for iphone and ipad devices.

The continental divide separates watersheds flowing north into the Arctic Ocean via the Mackenzie River and south and west into the Pacific Ocean via the Fraser River (Figure 2.1). The Parsnip River is a 6th order stream with a watershed that drains an area of 5597km^2. The mainstem of the river flows within the Rocky Mountain Trench in a north direction into Williston Reservoir starting from the continental divide adjacent to Arctic Lakes. Major tributaries include the Misinchinka, Colbourne, Reynolds, Anzac, Table, Hominka and Missinka sub-basins which drain the western slopes of the Hart Ranges of the Rocky Mountains. The Parsnip River has a mean annual discharge of 150.5 m^3/s with flow patterns typical of high elevation watersheds on the west side of the northern Rocky Mountains which receive large amounts of precipitation as snow leading to peak levels of discharge during snowmelt, typically from May to July (Figure 2.2).

Construction of the 183m high and 2134m long W.A.C. Bennett Dam was completed in 1967 at Hudson's Hope, BC, creating the Williston Reservoir (Hirst 1991). Filling of the 375km^2 reservoir was complete in 1972 and flooded a substantial portion of the Parsnip River and major tributary valleys forming what is now known as the Peace and Parsnip reaches. The replacement of riverine habitat with an inundated reservoir environment resulted in profound changes to the ecology, resource use and human settlement patterns in these systems (Hagen et al. 2015; Pearce et al. 2019; Stamford et al. 2017). Prior to the filling of the reservoir, the Pack River, into which McLeod Lake flows, was a major tributary to the Parsnip River. The Pack River currently enters the Williston Reservoir directly as the historic location of the confluence of the two rivers lies within the reservoir's footprint.

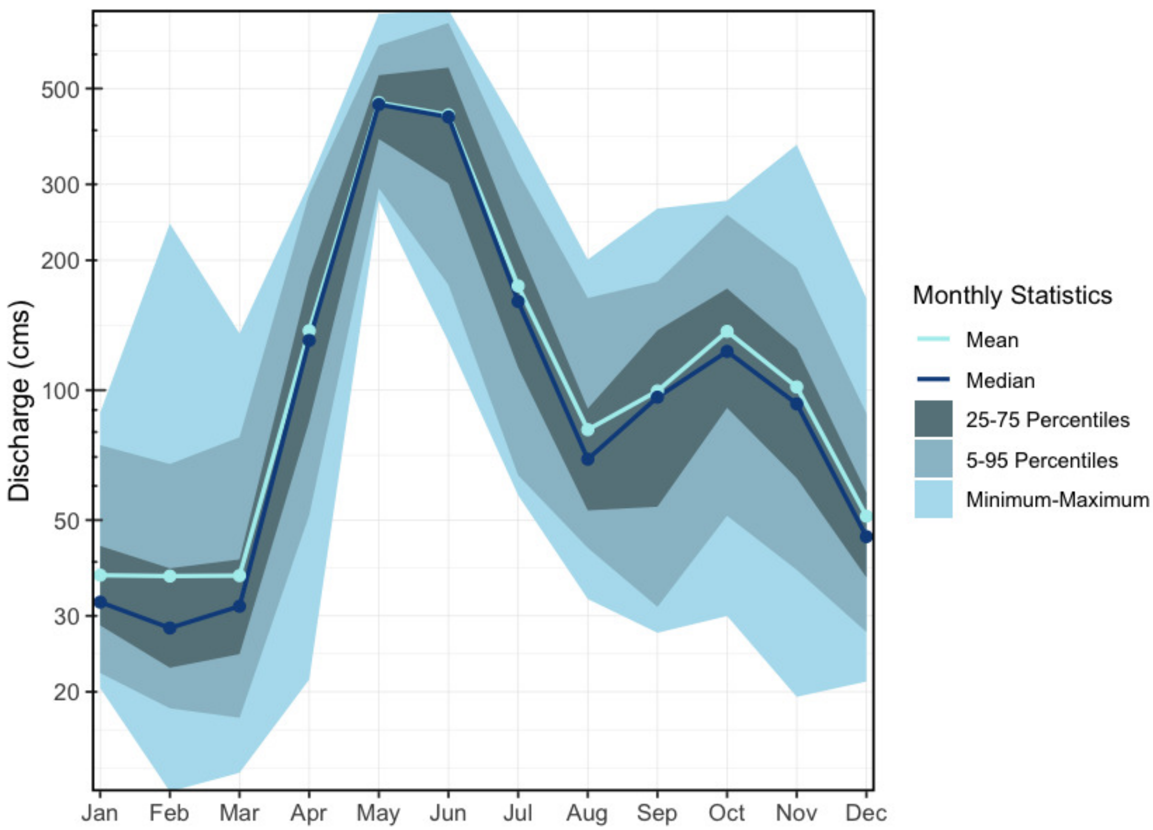


Figure 2.2: Hydrograph for Parsnip River above Misinchinka River (Station #07EE007).

2.4 Floodplains and Watershed Function

Floodplains are the low-lying ground adjacent to stream channels that water reclaims during high-flow events. They are not idle margins — they are where rivers do geomorphic work: storing and sorting sediment, exchanging nutrients between terrestrial and aquatic systems, recruiting riparian vegetation, and delivering large woody debris to the channel. For fish, floodplain habitats — side channels, sloughs, beaver complexes, oxbow ponds, and connected wetlands — provide the off-channel conditions that drive freshwater productivity. Juvenile salmonids depend on these areas for low-velocity rearing, overwinter refuge when mainstem conditions become harsh, and high-flow shelter during freshet. In many systems, floodplain habitats are where freshwater survival is won or lost.

Floodplains are often the part of the watershed most compromised by the infrastructure legacy described above. Dykes sever lateral connection; channelization confines the active footprint; drainage eliminates wetland complexes. Mapping floodplain extent at watershed scale therefore complements the linear barrier inventory — prioritization can consider not just whether a structure passes fish but the condition and extent of the off-channel habitat in the reach the structure connects to.

2.5 Fisheries

Fish species recorded in the BC Fisheries Information Summary System (FISS) for the watershed groups in this study area are summarised in Appendix - Fish Species by Watershed Group (MoE 2024a). In addition to flooding related to the formation of the Williston Reservoir, transmission lines, gas pipelines, rail, road networks, forestry, elevated water temperatures, interrupted connectivity, invasion from non-native species and insect infestations affecting forested areas pose threats to fisheries values in the Parsnip River watershed (Hagen et al. 2015; Stamford et al. 2017; Hagen and Weber 2019; COSEWIC 2012). A brief summary of trends and knowledge status related to Arctic grayling, bull trout, kokanee, mountain whitefish and rainbow trout in Williston Watershed streams is provided in Fish and Wildlife Compensation Program (2020) with a more detailed review of the state of knowledge for Parsnip River watershed populations of Arctic grayling and bull trout provided below.

Bull Trout - sa’ba

Tse’khene Elders from the McLeod Lake Indian Band report that sa’ba (bull trout) size and abundance has decreased in all rivers and tributaries from the reservoir with more injured and diseased fish captured in recent history than was common in the past (Pearce et al. 2019) .

Bull trout populations of the Williston Reservoir watershed are included within the Western Arctic population ‘Designatable Unit 10’, which, in 2012, received a ranking of ‘Special Concern’ by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC 2012). They were added to Schedule 1 under the Species of Risk Act in 2019 (Species Registry Canada 2020) and are also considered of special concern (blue-listed) provincially (BC Species & Ecosystem Explorer 2020). Some or all of the long-term foot survey index sections of four Williston Reservoir spawning tributaries (Davis Creek, Misinchinka River, Point Creek, and Scott Creek), have been surveyed within 16 of the 19 years between 2001 and 2019 (16 of 19 in Davis River, 10 years over a 13-year period in the Misinchinka River, 11 years over a 14-year period for Point Creek, and 9 years over an 11-year period for Scott Creek (Hagen et al. 2020).

A study of bull trout critical habitats in the Parsnip River was conducted in 2014 with the Misinchinka and Anzac systems identified as the most important systems for adfluvial (large bodied) bull trout spawners. The Table River was also highlighted as an important spawning destination. Other watersheds identified as containing runs of large bodied bull trout spawners included the Colbourne, Reynolds, Hominka and Missinka River with potentially less than 50 spawners utilizing each sub-basin (Hagen et al. 2015). Hagen and Weber (2019) have synthesized a large body of information regarding limiting factors, enhancement potential, critical habitats and conservation status for bull trout of the Williston Reservoir and the reader is encouraged to review this work for context. They have recommended experimental enhancements within a monitoring framework for Williston Reservoir bull trout (some spawning and rearing in Parsnip River mainstem and tributaries) which include stream fertilization, side channel development, riparian restoration and fish access improvement.

In 2018, sub-basins of the Anzac River watershed, Homininka River, Missinka River and Table River watersheds were designated as fisheries sensitive watersheds under the authority of the *Forest and Range Practices Act* due to significant downstream fisheries values and significant watershed sensitivity (Beaudry 2013a, 2014, 2013b; P. Beaudry 2014). Special management is required in these watersheds to protect habitat for fish species including bull trout and Arctic grayling including measures (among others) to limit equivalent clearcut area, reduce impacts to natural stream channel morphology, retain old growth attributes and maintain fish habitat/movement (Forest and Range Practices Act 2018).

Arctic Grayling - dusk'ihje

Tse'khene Elders from the McLeod Lake Indian Band report that dusk'ihje (Arctic grayling) numbers have declined dramatically since the flooding of the reservoir and that few dusk'ihje have been caught in the territory in the past 30 years (Pearce et al. 2019).

Since impoundment of the Williston Reservoir, it appears that physical habitat and ecological changes have been the most significant factors limiting Arctic grayling productivity. Although these changes are not well understood they have likely resulted in the inundation of key low gradient juvenile rearing and overwintering habitats, isolation of previously connected populations and increases in abundance of predators such as bull trout (Shrimpton et al. 2012; Hagen et al. 2018). Rapid increases in industrial activity and angler access in the Parsnip River watershed pose significant risks to Arctic Grayling productivity with these threats primarily linked to forestry and pipeline initiatives (Hagen and Stamford 2021).

A detailed review of Arctic grayling life history is available in Stamford et al. (2017). Migration of mature adult Arctic grayling occurs in the spring, typically coinciding with water temperatures of 4° C. In the Parsnip watershed, spawning appears to occur between late May and late June, primarily within the lower reaches of the Anzac and Table rivers and the Parsnip River mainstem. Spawning habitat is generally associated with side-channel and multi-channel areas containing small gravels. The current distribution of Williston Arctic grayling is concentrated in fourth-order and larger streams (Williamson and Zimmerman 2005; Stamford et al. 2017). According to Stewart et al. (2007), Arctic grayling spawn in both large and small tributaries to rivers and lakes, intermittent streams, mainstem rivers, and lakes—most commonly at tributary mouths. Although earlier studies suggest that 0+ grayling overwinter in the lower reaches of larger tributaries (e.g., Table and Anzac rivers) and the Parsnip River, and that few age-1+ grayling have been sampled in tributaries, habitat use in small tributaries and the extent to which they are connected with mainstem habitats across all core areas remains poorly understood. Between 1995 and 2019, population monitoring of Arctic grayling was conducted in the Table River in 9 out of 25 years, and in the Anzac River for 8 years, using snorkel surveys. Results from 2018 and 2019 support efforts to assess the conservation status of the species in the Parsnip Core Area (Hagen et al. 2018). Preliminary telemetry data from 2019 indicate that both Arctic grayling and bull trout rely on the Parsnip River mainstem for overwinter residency. Arctic grayling typically begin moving into tributaries in April and are widespread across the watershed by June, with larger adults observed in headwater habitats by August.

A five-year study on Arctic grayling abundance and trends in the Parsnip River watershed is presented in Hagen and Stamford (2023), which reports that the most productive summer rearing habitats for Arctic grayling are within the Anzac and Table rivers. Although estimated abundance is lower than in the Anzac and Table, productive summer rearing habitats for adult Arctic grayling in the upper Parsnip River watershed are distributed between 36–25km of the Missinka River and 48–32km of the Hominka River. In the Anzac River, Hagen and Stamford (2021) report that a 30km stretch from 47km to 16km supports high Arctic grayling abundance and is assumed to provide productive rearing habitat for adults. A chute obstruction located at 47km likely limits further upstream movement. Although the spatial distribution of high Arctic grayling abundance in the Table River has not been delineated through reconnaissance surveys, it has been observed to span at least a 20km zone from the waterfall migration barrier at 37km to 18km.

Spatial ecology studies in the Parsnip River between 2018 and 2021 has been reported on by Martins et al. (2022) with results related to:

- temperature modeling and spatio-temporal patterns in thermal habitat,
- telemetry data modeling and arctic grayling spatial ecology, and
- trophic relationships between Arctic grayling and bull trout

Fish presence by Habitat Type

A review of available fisheries data for the Parsnip River watershed stratified by different habitat characteristics (stream gradient, channel width, and watershed size) can provide insight into which habitats may provide the highest intrinsic value for fish species based on the number of fish captured in those habitats in past assessment work (Tables 2.1 - 2.3 and Figures 2.3 - 2.5). It should be noted however that it should not be assumed that all habitat types have been sampled in a non-biased fashion or that particular sites selected do not have a disproportionate influence on the overall dataset composition (ie. fish salvage sites are often located adjacent to construction sites which are more commonly located near lower gradient stream reaches).

Table 2.1: Summary of historic fish observations vs. stream gradient category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

species_code	Gradient	Count	total_spp	Percent
BT	0 - 3 %	411	614	67
BT	03 - 5 %	86	614	14
BT	05 - 8 %	58	614	9
BT	08 - 15 %	50	614	8
BT	15 - 22 %	9	614	1
GR	0 - 3 %	249	251	99
GR	03 - 5 %	2	251	1
KO	0 - 3 %	145	150	97
KO	03 - 5 %	2	150	1
KO	05 - 8 %	2	150	1
KO	22+ %	1	150	1
RB	0 - 3 %	2035	2619	78
RB	03 - 5 %	275	2619	11
RB	05 - 8 %	186	2619	7
RB	08 - 15 %	103	2619	4
RB	15 - 22 %	14	2619	1
RB	22+ %	6	2619	0

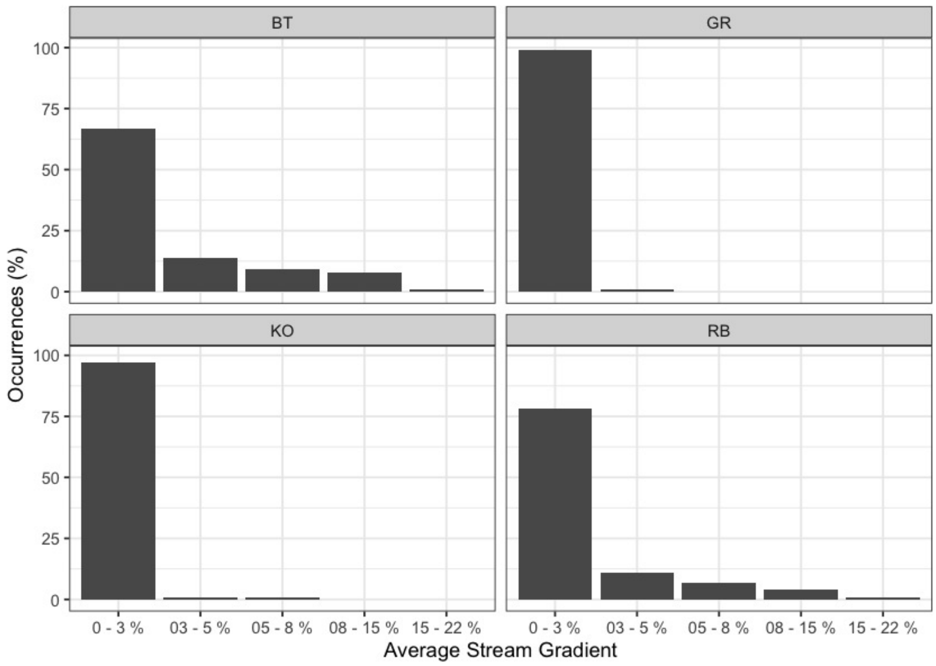


Figure 2.3: Summary of historic fish observations vs. stream gradient category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

Table 2.2: Summary of historic fish observations vs. channel width category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

species_code	Width	Count	total_spp	Percent
BT	0 - 2m	30	614	5
BT	02 - 04m	96	614	16
BT	04 - 06m	81	614	13
BT	06 - 10m	82	614	13
BT	10 - 15m	59	614	10
BT	15m+	248	614	40
BT	–	18	614	3
GR	02 - 04m	6	251	2
GR	06 - 10m	6	251	2

species_code	Width	Count	total_spp	Percent
GR	10 - 15m	6	251	2
GR	15m+	231	251	92
GR	–	2	251	1
KO	0 - 2m	4	150	3
KO	02 - 04m	22	150	15
KO	04 - 06m	3	150	2
KO	06 - 10m	11	150	7
KO	10 - 15m	5	150	3
KO	15m+	31	150	21
KO	–	74	150	49
RB	0 - 2m	656	2619	25
RB	02 - 04m	520	2619	20
RB	04 - 06m	257	2619	10
RB	06 - 10m	222	2619	8
RB	10 - 15m	122	2619	5
RB	15m+	454	2619	17
RB	–	388	2619	15

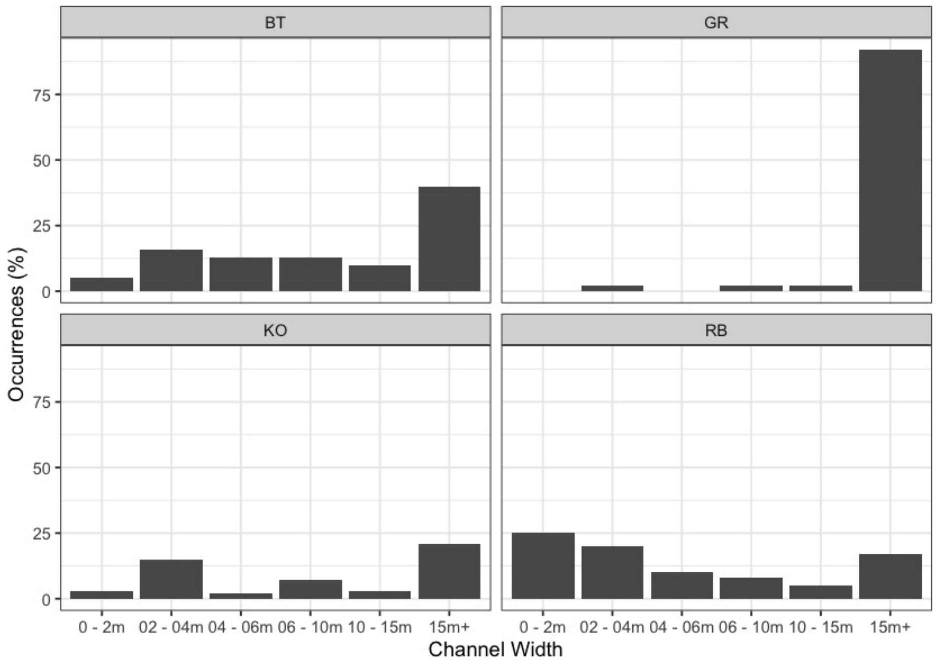


Figure 2.4: Summary of historic fish observations vs. channel width category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

Table 2.3: Summary of historic fish observations vs. watershed size category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

species_code	Watershed	count_wshd	total_spp	Percent
BT	0 - 25km2	252	614	41
BT	25 - 50km2	93	614	15
BT	50 - 75km2	40	614	7
BT	75 - 100km2	11	614	2
BT	100km2+	218	614	36
GR	0 - 25km2	8	251	3
GR	50 - 75km2	2	251	1
GR	75 - 100km2	4	251	2
GR	100km2+	237	251	94

species_code	Watershed	count_wshd	total_spp	Percent
KO	0 - 25km2	105	150	70
KO	25 - 50km2	6	150	4
KO	50 - 75km2	5	150	3
KO	75 - 100km2	3	150	2
KO	100km2+	31	150	21
RB	0 - 25km2	1802	2619	69
RB	25 - 50km2	171	2619	7
RB	50 - 75km2	97	2619	4
RB	75 - 100km2	48	2619	2
RB	100km2+	501	2619	19

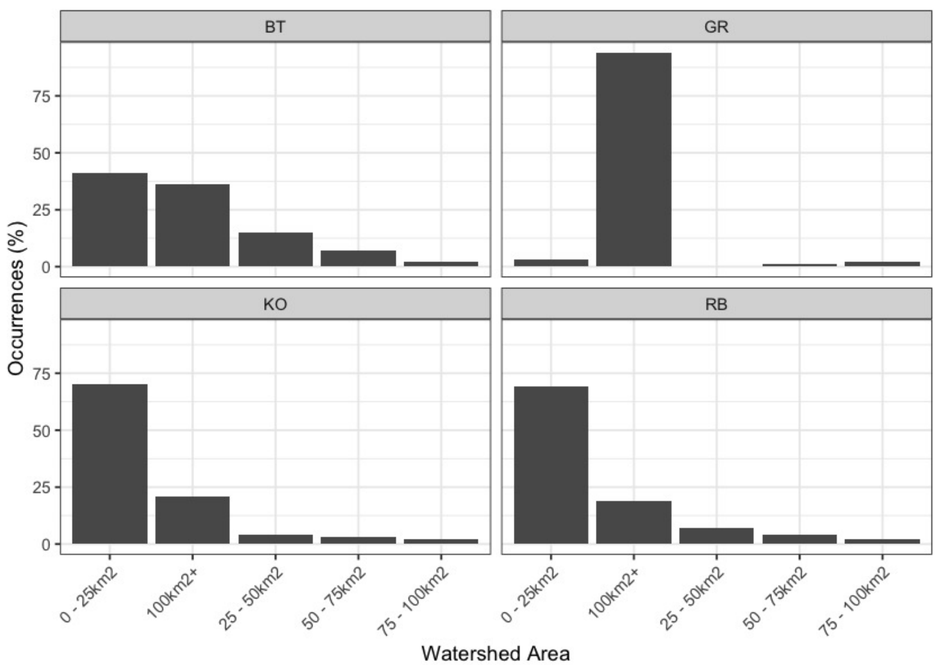


Figure 2.5: Summary of historic fish observations vs. watershed size category for the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

3 Methods

3.1 Climate Departure

Climate departure measures how far recent climate has shifted from a historical baseline — in this case, how much warmer, drier, or snowpack-altered the FWCP Peace Region has become relative to the pre-warming period (1951–1980). Where traditional fish-passage assessments focus on the physical structures in and around streams, climate departure adds the atmospheric and hydrologic context those structures sit inside: warming air temperatures that drive warmer water, shifting snowpack that reshapes the freshet, and atmospheric drying that reduces late-summer flows. Understanding these departures at the watershed scale lets us ask whether the habitat above a barrier is gaining or losing value as climate changes — and weight barrier prioritisation accordingly.

We examined 15 climate variables spanning temperature (annual mean, daytime maximum, overnight minimum), moisture (precipitation, vapour pressure deficit, soil moisture, relative humidity), and snowpack (snow water equivalent, snowfall, snowmelt, snow cover, snowfall fraction, peak snow water equivalent, snowmelt timing, and peak melt rate). Each variable was assessed at annual and seasonal scales, across the full region and within each of five ecoregions.

Climate departures were computed with the cd R package, which reads ERA5-Land hourly reanalysis (1950–present, ~9 km native grid) aggregated to monthly, seasonal, and annual periods on a single grid covering British Columbia, served as Cloud-Optimised GeoTIFFs from a public S3 catalog. For each variable the package crops the published rasters to the area of interest, takes a yearly spatial mean, and computes anomalies against the 1951–1980 pre-warming reference period — the same base period Hansen et al. (2012) used for global temperature emergence. Long-term trends are tested with Mann-Kendall significance and quantified with Theil-Sen slopes; the recent-decade (2015–2025) versus pre-warming window comparison uses a Welch two-sample t-test on the annual values. The same pipeline runs at any polygon scale — region, ecoregion, or watershed group. The regional and ecoregion-level analyses are summarised in this section; the watershed-group breakdown feeds into the per-watershed barrier prioritisation that follows. Appendix - Climate Departure carries the full set of plots and tables.

3.2 Planning — Habitat, Connectivity and Floodplain Modelling

3.2.1 Habitat Modelling

Habitat modelling used to help guide planning for field assessments is generated by `bcfishpass` (Norris [2020] 2024) which has been designed to prioritize potential fish passage barriers for assessment or remediation by generating a simple model of aquatic habitat connectivity. We utilize the `bcfishpass` access model, linear spawning/rearing habitat model and lateral habitat connectivity for planning purposes. These models provide a valuable starting point, but their results are not definitive and should always be considered with professional judgment. Detailed information regarding model methodology, select parameters and known model limitations are detailed in Norris ([2020] 2024) with key documentation linked below:

- Access model
- Linear spawning/rearing habitat models
- Lateral habitat model

3.2.1.1 Contributions to Provincial Connectivity Models

`bcfishpass` operates on a province-wide modelled-crossing inventory — stream-line / road-network intersections that may or may not correspond to a real structure on the ground. To reduce field follow-up at sites that are non-issues, we contribute desktop and field corrections to two open tables in the canonical `bcfishpass` repository. The same tables also feed the `fresh` and `link` framework — one source of truth across provincial and project-specific connectivity models.

- `user_modelled_crossing_fixes.csv` — most commonly used to flag modelled crossings that are not field issues. Using Google satellite, Bing aerial, and Esri satellite imagery we identify records where no structure exists (the road or crossing is absent) or where the structure is an open-bottom crossing — bridge, arch — that does not warrant a fish-passage assessment. These corrections remove false-positive sites from field-followup lists before assessors are deployed.
- `pscis_modelledcrossings_streams_xref.csv` — ties field-assessed PSCIS crossings to the correct modelled-crossing record when the spatial alignment is off. The modelled inventory catches most assessed crossings directly; this xref captures the remainder.

Each entry is signed by the reviewer and dated; corrections flow back into the next `bcfishpass` build and into the `fresh` and `link` workflows. Because the inventory is provincial, the work is available for any subsequent analysis across BC. Since April 1, 2025, the project has contributed 282 records to `user_modelled_crossing_fixes` and 11 to `pscis_modelledcrossings_streams_xref` across the FWCP Peace Region — concentrated in the Nation, Peace, Parsnip Arm, and Parsnip watershed groups.

Table 3.1 documents the custom species-specific thresholds for stream gradient and channel width applied to the linear spawning and rearing habitat model for this year's project planning.

Table 3.1: Stream gradient and channel width thresholds used to model potentially highest value fish habitat.

Variable	Bull Trout	Arctic Grayling
Spawning Gradient Max (%)	5.5	2.5
Spawning Width Min (m)	2	4
Rearing Width Min (m)	1.5	1.5
Rearing Gradient Max (%)	10.5	3.5

*

3.2.2 Bull Trout and Arctic Grayling Critical Habitat

We collaborated with Eduardo Martins (Associate Professor at the University of Northern British Columbia, UNBC) to incorporate Arctic grayling into our collaborative GIS project so it could be used to inform field work activities and potential remediation site prioritization. Similarly, critical bull trout habitat data was added to the GIS project after being obtained from Luc Turcotte (WLRS - Regional Aquatic Specialist) and stored securely. Bull trout data acquisition required prior completion of the Access to Secure Species and Ecosystem Data and Information (SEDIS) training conducted through WLRS.

3.2.3 Floodplain Delineation

To complement the linear barrier inventory, we mapped functional floodplain extent at watershed scale for the Parsnip River Watershed Group. Overlaying floodplain footprint on the crossing inventory lets us ask not just whether fish can pass a structure, but how much off-channel habitat sits within the affected corridor and whether infrastructure has compromised the lateral connectivity that sustains it.

We used the `flooded` R package (Nagel et al. 2014), which models the area a river actively inundates during regular high-flow events — the **functional floodplain**. This is distinct from the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain is the scale where off-channel habitat is created and maintained: where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Bankfull depth is estimated from the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams. Inputs were the MRDEM-30 30 m DEM, the bull trout accessible stream network (stream order ≥ 3) from bcfishpass, and BC Freshwater Atlas lakes and wetlands on that network. Full methodology and watershed-scale mapping are in Appendix - Floodplain Delineation.

3.3 Statistical Support for Habitat Modelling

This project — together with prior SERNbc fish-passage work across the Peace, Skeena, and Fraser regions — has provided the field data, multi-year partnerships, and statistical-modelling support relationships that drive ongoing refinement of `bcfishpass` habitat parameters and the Bayesian Spatial Stream Network analyses described below. The work underpins our intrinsic-habitat-potential and connectivity-prioritization outputs. It **extends** — rather than supersedes — the canonical BC freshwater modelling stack (`fwapg` for the Freshwater Atlas in PostgreSQL, `bcfishpass` for habitat and barrier modelling, and `bcfishobs` for fish-observation records). Our role is two-sided: contributor — providing the funding context, field data, and project-specific motivation that shapes which questions get asked — and downstream user — taking the model outputs into project-specific parameterization of habitat thresholds, integration of water-temperature and water-quality covariates, and calibration against observed fish-presence evidence.

3.3.1 Channel Width

Channel width is a primary determinant of habitat suitability in `bcfishpass`. In early 2021, Bayesian statistical methods were developed to predict channel width across BC Freshwater Atlas stream segments where field measurements were unavailable. The original model related channel width to upstream watershed area and watershed-area-weighted mean annual precipitation (Thorley and Irvine 2021). In December 2021 the methodology was updated using the power-law form derived by Finnegan et al. (2005), relating stream discharge to watershed area and precipitation; the updated model (Thorley et al. 2021) is the channel-width estimator used in `bcfishpass` at the time of reporting. Detailed documentation of the data-collection and statistical-analysis methodology is given in Irvine ([2021] 2022) and Thorley et al. (2021).

3.3.2 Stream Temperature and Growing-Season Degree Days

Water temperature drives productivity for cold-water salmonids — too cold and growth stalls; too warm and thermal stress reduces survival. We work with Poisson Consulting on Bayesian Spatial Stream Network (SSN) models that predict stream temperature at fine spatial resolution across the FWA network.

The 2024 Nechako modelling effort (Hill et al. 2024) mapped stream discharge and temperature causal-effects pathways across the Nechako Watershed with the intent of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values — including elimination of stream reaches from intrinsic-habitat models where water temperatures are likely too cold to support fish production. The project began with a custom mechanistic model of discharge and temperature interactions (visually represented here), but the model struggled to converge. The project then pivoted to the `air2stream` framework (Toffolon and Piccolroaz 2015), a hybrid formulation that bridges fully mechanistic models (data-intensive, hard to parameterize) and purely statistical approaches (no physical justification, extrapolate poorly). After several adaptations, expected stream temperatures were best modelled using the four-parameter version of `air2stream` with site-level random effects on each of the four parameters. Inputs spanned years 2019–2021 and combined Nechako water-temperature observations, ERA5-Land hourly air temperature (Muñoz Sabater 2019), and PCIC-gridded discharge from the ACCESS1-0 RCP8.5 scenario.

The 2025 Skeena modelling effort (Hill et al. 2025) extended this approach to a larger spatial network and added Growing-Season Degree Days (GSDD) — accumulated thermal units during the season when stream temperature remains at or above 5 °C — as the productivity-relevant metric for age-0 salmonid growth. The Skeena work also moved to `water-temp-bc` as the canonical water-temperature data source (described below), replacing per-project Zenodo and ECCC downloads.

Stream-temperature and discharge observations are now organized centrally in `water-temp-bc` — a monthly-automated ingestion pipeline that publishes Environment Canada hydrometric and water-quality records as a partitioned-parquet layer on Amazon S3. Stations metadata, observations by parameter (water temperature, discharge, water level), and per-snapshot dedupe (canonical value retained at the latest `harvested_at` timestamp) are queryable via the package's `query_canonical()` helper without database setup. This replaces the per-project Zenodo and ECCC downloads used in earlier modelling work — one accessible source for any subsequent SSN or temperature-suitability analysis in BC.

3.3.3 Habitat-Modelling Framework — Trajectory

We are building a parameterizable habitat-suitability and connectivity framework using `fresh` and `link` — open-source R packages that operationalize FWA network operations for project-specific habitat queries. The framework is currently wired up to the same input classes that `bcfishpass` uses today: channel width, gradient thresholds, and provincial / federal obstacle and barrier inventories. An initial Arctic grayling intrinsic-habitat model for the FWCP Peace region — based on stream size and gradient only — has been produced on that basis and is in development at the time of this report. Because the framework is designed to be agnostic of input variables, additional covariates can be wired in as data, methods, and validation mature: statistical stream-temperature and GSDD predictions from the Poisson Consulting work described above, water-temperature and water-quality observations from `water-temp-bc`, jurisdictional overlays, and calibration of model predictions against fish-presence observations from `bcfishobs` together with research activities (some commissioned for FWCP Peace) that identify areas of known high-value spawning and rearing habitat. Water-temperature integration and the calibration loop are further down the pipeline; the framework is built to accept them when ready. Outputs will be published in subsequent versions of this report and as standalone GIS layers shared via the project's collaborative GIS environment. This work complements and extends — rather than supersedes — the canonical `bcfishpass` outputs from Hillcrest Geographics. The latest results are served at the report URL referenced in the executive summary; each report version is tagged so reviewers can navigate between the submitted snapshot and current state. We demonstrate this for the Parsnip River Watershed Group — reproducing `bcfishpass`'s per-segment habitat and connectivity classification, then re-parameterising the rules to extend it to Arctic grayling, a species `bcfishpass` does not yet model — in Appendix - Parsnip River Habitat and Connectivity Modelling.

3.4 Fish Passage Assessments

3.4.1 Natural Barriers to Fish Passage

Our assessments may include natural features such as waterfalls that could limit fish passage. This informs whether upstream culvert upgrades would restore access for anadromous species (e.g., salmon) or primarily benefit resident fish already upstream. We document these features by measuring height, gradient, and pool depth, recording field observations, capturing site photographs, and reviewing background sources for context.

3.4.2 Road Stream Crossings

In the field, crossings prioritized for follow-up were first assessed for fish passage following the procedures outlined in "Field Assessment for Determining Fish Passage Status of Closed Bottomed Structures" (MoE 2011). The reader is referred to (MoE 2011) for detailed methodology. Crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered "other" (i.e. fords). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any other relevant features. The following information was recorded for all surveyed crossings: date of inspection, crossing reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for OBS. A more detailed "full assessment" was completed for all closed bottom structures and included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, road fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.2).

Table 3.2: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	**Fish Habitat Criteria**
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

Fish passage potential was determined for each stream crossing identified as a closed bottom structure as per MoE (2011). The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier to some fish species and life stages (Table 3.3, Table 3.4). These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al. 2005; Bell 1991; Thompson 2013). For crossings determined to be potential barriers or barriers based on the data, a culvert fix and recommended diameter/span was proposed.

Table 3.3: Fish Barrier Risk Assessment (MoE 2011).

Risk	LOW	MOD	HIGH
Embedded	>30cm or >20% of diameter and continuous	<30cm or 20% of diameter but continuous	No embedment or discontinuous
Value	0	5	10
Outlet Drop (cm)	<15	15-30	>30
Value	0	5	10
SWR	<1.0	1.0-1.3	>1.3
Value	0	3	6
Slope (%)	<1	1-3	>3
Value	0	5	10
Length (m)	<15	15-30	>30
Value	0	3	6

Table 3.4: Fish Barrier Scoring Results (MoE 2011).

Cumlative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project, a gradient threshold between accessible and non-accessible habitat was set at 25% (for a minimimum length of 100m) intended to represent the maximum gradient of which the strongest swimmers of anadromous species (bull trout) are likely to be able to migrate upstream.

For reporting of Phase 1 - fish passage assessments within the body of this report (Table 3.3), a "total" value of habitat <20% output from `bcfishpass` was used to estimate the amount of habitat upstream of each crossing less than 25% gradient before a falls of height >5m - as recorded in MoE (2024b) or documented in other `bcfishpass` online documentation. For Phase 2 - habitat confirmation sites, conservative estimates of the linear quantity of habitat to be potentially gained by fish passage restoration, bull trout rearing maximum gradient threshold (10.5%) was used. To generate estimates for area of habitat upstream (m^2), the estimated linear length was multiplied by half the downstream channel width measured (overall triangular channel shape) as part of the fish passage assessment protocol. Although these estimates are not generally conservative, have low accuracy and do not account for upstream stream crossing structures they allow a rough idea of the best candidates for follow up.

Potential options to remediate fish passage were selected from MoE (2011) and included:

- Removal (RM) - Complete removal of the structure and deactivation of the road.
- Open Bottom Structure (OBS) - Replacement of the culvert with a bridge or other open bottom structure. Based on consultation with FLNR road crossing engineering experts, for this project we considered bridges as the only viable option for OBS type .
- Streambed Simulation (SS) - Replacement of the structure with a streambed simulation design culvert. Often achieved by embedding the culvert by 40% or more. Based on consultation with FLNR engineering experts, we considered crossings on streams with a channel width of <2m and a stream gradient of <8% as candidates for replacement with streambed simulations.
- Additional Substrate Material (EM) - Add additional substrate to the culvert and/or downstream weir to embed culvert and reduce overall velocity/turbulence. This option was considered only when outlet drop = 0, culvert slope <1.0% and stream width ratio < 1.0.
- Backwater (BW) - Backwatering of the structure to reduce velocity and turbulence. This option was considered only when outlet drop < 0.3m, culvert slope <2.0%, stream width ratio < 1.2 and stream profiling indicates it would be effective..

3.4.3 Cost Estimates

Cost estimates for structure replacement with bridges and embedded culverts were generated based on the channel width, slope of the culvert, depth of fill, road class and road surface type. Road details were sourced from FLNRORD (2020b) and FLNRORD (2020a) through `bcfishpass`. Interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay, Steve Page, Area Engineer - FLNR - Northern Engineering Group and Matt Hawkins - MoTi - Design Supervisor for Highway Design and Survey - Nelson were utilized to help refine estimates which have since been adjusted for inflation in 2020 and based on past experience.

Base costs for installation of bridges on forest service roads and permit roads with surfaces specified in provincial GIS road layers as rough and loose was estimated at $30000/linear m and assumed that the road could be closed during construction and a minimum bridge span of 15m. For streams with channel widths <2m, embedded culverts were reported as an effective solution with total installation costs estimated at $100k/crossing (pers. comm. Phil MacDonald, Steve Page then adjusted for inflation in 2020). For larger streams (>6m), estimated span width increased proportionally to the size of the stream. For crossings with large amounts of fill (>3m), the replacement bridge span was increased by an additional 3m for each 1m of fill >3m to account for cutslopes to the stream at a 1.5:1 ratio. To account for road type, a multiplier table was generated to estimate incremental cost increases with costs estimated for structure replacement on paved surfaces, railways and arterial/highways costing up to 15 times more than forest service roads due to expenses associate with design/engineering requirements, traffic control and paving. The cost multiplier table (Table 3.5) should be considered very approximate with refinement recommended for future projects.

Table 3.5: Cost multiplier table based on road class and surface type.

Class	Surface	Class Multiplier	Surface Multiplier	Bridge $/15m	Streambed Simulation $
FSR	Rough	1	1	450,000	100,000
FSR	Loose	1	1	450,000	100,000
Resource	Loose	1	1	450,000	100,000
Collector	Loose	1	1	450,000	100,000
Recreation	Loose	1	1	450,000	100,000
Resource	Rough	1	1	450,000	100,000
Permit	Unknown	1	1	450,000	100,000
Permit	Loose	1	1	450,000	100,000
Permit	Rough	1	1	450,000	100,000
Unclassified	Loose	1	1	450,000	100,000
Unclassified	Rough	1	1	450,000	100,000
Unclassified	Paved	1	2	750,000	150,000
Unclassified	Unknown	1	2	750,000	150,000
Local	Loose	4	1	1,500,000	200,000
Local	Paved	4	2	3,000,000	400,000
Collector	Paved	4	2	3,000,000	400,000
Arterial	Paved	15	2	11,250,000	1,500,000
Highway	Paved	15	2	11,250,000	1,500,000
Rail	Rail	15	2	11,250,000	1,500,000

3.4.4 Climate Change Risk Assessment

In collaboration with the Ministry of Transportation and Infrastructure (MoTi), a new climate change replacement program aims to prioritize vulnerable culverts for replacement (pers. comm Sean Wong, 2022) based on data collected and ranked related to three categories - culvert condition, vulnerability and priority. Within the “condition” risk category - data was collected and crossings were ranked based on erosion, embankment and blockage issues. The “climate” risk category included ranked assessments of the likelihood of both a flood event affecting the culvert as well as the consequence of a flood event affecting the culvert. Within the “priority” category the following factors were ranked - traffic volume, community access, cost, constructability, fish bearing status and environmental impacts (Table 3.6). This project is still in its early stages with methodology changes going forward.

Table 3.6: Climate change data collected at MoTi culvert sites

Parameter	Description
erosion_issues	Erosion (scale 1 low - 5 high)
embankment_fill_issues	Embankment fill issues 1 (low) 2 (medium) 3 (high)
blockage_issues	Blockage Issues 1 (0-30%) 2 (>30-75%) 3 (>75%)
condition_rank	Condition Rank = embankment + blockage + erosion
condition_notes	Describe details and rational for condition rankings
likelihood_flood_event_affecting_culvert	Likelihood Flood Event Affecting Culvert (scale 1 low - 5 high)
consequence_flood_event_affecting_culvert	Consequence Flood Event Affecting Culvert (scale 1 low - 5 high)
climate_change_flood_risk	Climate Change Flood Risk (likelihood x consequence) 1-6 (low) 6-12 (medium) 10-25 (high)
vulnerability_rank	Vulnerability Rank = Condition Rank + Climate Rank
climate_notes	Describe details and rational for climate risk rankings
traffic_volume	Traffic Volume 1 (low) 5 (medium) 10 (high)
community_access	Community Access - Scale - 1 (high - multiple road access) 5 (medium - some road access) 10 (low - one road access)
cost	Cost (scale: 1 high - 10 low)
constructability	Constructibility (scale: 1 difficult -10 easy)
fish_bearing	Fish Bearing 10 (Yes) 0 (No) - see maps for fish points
environmental_impacts	Environmental Impacts (scale: 1 high -10 low)
priority_rank	Priority Rank = traffic volume + community access + cost + constructability + fish bearing + environmental impacts
overall_rank	Overall Rank = Vulnerability Rank + Priority Rank
priority_notes	Describe details and rational for priority rankings

3.5 Habitat Confirmation Assessments

Following fish passage assessments, habitat confirmations were completed in accordance with procedures outlined in the document “A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation of a Stream Crossing” (Fish Passage Technical Working Group 2011). The main objective of the field surveys was to document upstream habitat quantity and quality and to determine if any other obstructions exist above or below the crossing. Habitat value was assessed based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation. Criteria used to rank habitat value was based on guidelines in Fish Passage Technical Working Group (2011) (Table 3.2).

During habitat confirmations, to standardize data collected and facilitate submission of the data to provincial databases, information was collected on digital field forms adapted from provincial “Site Cards”. Habitat characteristics recorded included channel widths, wetted widths, residual pool depths, gradients, bankfull depths, stage, temperature, conductivity, pH, cover by type, substrate and channel morphology (among others). When possible, the crew surveyed downstream of the crossing to a minimum distance 300m and upstream to a minimum distance of 500 - 600m. Any potential obstacles to fish passage were inventoried with photos, physical descriptions and locations recorded on site cards. Surveyed routes were recorded with time-signatures on handheld GPS units.

3.5.1 Fish Sampling

3.5.1.1 Electrofishing

Fish sampling was conducted on a subset of sites when biological data was considered to add significant value to the physical habitat assessment information. Electrofishing was utilized for fish sampling according to stream inventory standards and procedures found in the Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Manual (Resource Inventory Standards Committee 2001). A Haltech 2000 backpack electrofisher was used within discrete site units both upstream and downstream of the subject crossing with electrofisher settings and seconds, water quality parameters (i.e. conductivity, temperature and ph), start and end locations, length of site and wetted widths (average of a minimum of three) recorded.

3.5.1.2 Fish Handling and Processing

Captured fish were held in buckets with sufficient water to minimize stress until processing, and multiple buckets were used when catch numbers were high. For each fish captured, fork length, weight and species was recorded with results documented in the fish data submission spreadsheet.

3.5.1.3 Pit Tagging

Fish with a fork length greater than 60 mm and belonging to species approved under the scientific fish collection permit PG25-983916 were tagged with Passive Integrated Transponders (PIT tags) using the [Abdominal Cavity] method outlined by Biomark. To anesthetize fish prior to pit tagging, we used a solution of approximately 0.1 mL of clove oil per 1 L of water (1:10,000). This concentration was selected for its efficiency in providing effective sedation with minimal residual effects, making it ideal for studies in which fish are released back into their natural habitats (Fernandes et al. 2017). The clove oil solution was prepared in advance by dissolving pure clove oil in ethyl alcohol in a 1:9 ratio (clove oil: ethyl alcohol) to enhance solubility, then mixed into the water bucket (Fernandes et al. 2017). Fish were immersed in this solution until they reached an appropriate level of anesthesia for handling and then were tagged. To maintain needle sharpness and minimize injury risk, needles were replaced approximately every 10 fish. Each tagged fish was scanned with the PIT reader, and both the PIT tag ID and row ID were recorded. Once tagged, fish were placed into a bucket of fresh water and allowed to recover before being released back into the stream. Fish information and habitat data will be submitted to the province under scientific fish collection permit PG25-983916.

3.5.2 Environmental DNA (eDNA) Sampling

3.5.2.1 Sample Collection and Filtration

Samples were collected using two approaches informed by Hobbs et al. (2025): in-stream filtration with a Haltech EMOS sampler and grab-and-go bottle collection. For both methods, all water collected at a site (typically 1–3L) was filtered onto a single filter, yielding one replicate per site. Mixed cellulose ester (MCE) filters with 1μm pore size were used most commonly, with 0.45μm and 1.5μm filters of other materials used occasionally. Samples were typically collected upstream and downstream of each road–stream crossing, though in some cases only downstream samples were obtained.

For in-stream filtration, Haltech reusable aluminum housings were pre-loaded with filters. In high-turbidity conditions, a Haltech pre-filter with a 100μm nitex screen was installed upstream to remove coarse material. For grab-and-go sampling, stream water was collected in sanitized 1L HDPE Nalgene bottles and filtered at the end of each day.

After filtration, filters were removed from housings in a clean environment, placed in coin envelopes with silica desiccant inside labelled ziplock bags, and frozen until shipment. Aluminum housings and Nalgene bottles were sanitized with bleach solution between uses.

3.5.2.2 Quality Assurance

Approximately 10% of samples were collected for quality assurance. Field blanks using distilled water were collected daily. Office blanks were processed at field accommodations by filtering clean water when processing grab-and-go samples. Positive controls were obtained at select sites where electrofishing confirmed species presence prior to eDNA sampling.

3.5.2.3 Laboratory Analysis

Samples were shipped frozen to UNBC for extraction and analysis. DNA was extracted and analyzed using quantitative polymerase chain reaction (qPCR) with species-specific assays following protocols developed by Murray and Booth (2023). Assays targeted mitochondrial genes (COI, CytB) using fluorescent dyes (FAM, VIC, HEX), allowing multiple species to be tested per sample through multiplexing. Target species included Chinook, Coho, Sockeye, Rainbow Trout, Bull Trout, Arctic Grayling, and Kokanee, with additional assays for Coastal Cutthroat and White Sturgeon at select sites.

Due to close evolutionary relationships and hybridization, some species pairs cannot be distinguished using mtDNA-based assays. The Bull Trout assay also detects Dolly Varden, requiring geographic context to interpret results in areas where ranges overlap. Similarly, Rainbow Trout and Coastal Cutthroat assays may cross-react where these species co-occur or hybridize.

To optimize laboratory efficiency, we collaborated with UNBC to develop a sample run plan that grouped samples by compatible dye types, minimizing the number of qPCR runs required. An assay mapping table cross-referencing species codes to assay identifiers and dye types was developed, along with processing scripts to generate sample groupings and a run order configuration specifying species order within each group.

3.5.2.4 eDNA Extraction

eDNA was extracted in the dedicated eDNA lab at UNBC and analysed with the protocols developed by Murray and Booth (2023).

3.5.3 Open Source Reporting Framework

This report is produced using open-source tools (R, bookdown) with full version control via git, enabling iterative updates. All code, data, and revision history are publicly accessible in the project repository, with ongoing tasks tracked via GitHub issues and version history maintained in the project changelog.

3.5.4 Aerial Imagery

Scripted processing and serving of UAV imagery collected during the project is available at https://github.com/NewGraphEnvironment/stac_uav_bc/ (Irvine [2025] 2025). `OpenDroneMap` was utilized to produce orthomosaics, digital surface models (DSMs), and digital terrain models (DTMs) (OpenDroneMap Authors [2014] 2025). To support efficient web-based access - imagery products were converted to cloud-optimized GeoTIFFs (COGs) using `rio-cogeo`, then collated accordiong to the SpatioTemporal Asset Catalog (STAC) specification with `pystac` and uploaded to S3 storage Amazon Web Services (2025). A `titiler` tile server was set up to facilitate interactive viewing of the orthoimagery and an Application Program Interface (API) leveraging `stac-fastapi-pgstac` is served at https://images.a11s.one to enable linking of collection images through QGIS as well as remote spatial and temporal querying using open source software such as `rstac` (Development

Seed [2019] 2025; stac-utils 2025; Simoes et al. 2021). Orthoimagery for sites where drone surveys were flown is presented in Appendix - UAV Imagery.

3.6 Engineering Design

Engineering designs were signed and sealed by professional engineers. When possible - completed designs are loaded to the PSCIS data portal.

3.7 Remediations

Structure replacement was conducted by project specific contractors. If not already completed, as-built drawings will be loaded to the PSCIS data portal.

3.8 Monitoring

Monitoring of fish passage restoration sites — both proposed and completed - is essential to ensure restoration investments lead to meaningful ecological outcomes. Monitoring enables evaluation of whether remediation actions improve connectivity for fish and provides critical feedback to refine future prioritization and restoration strategies.

Baseline data collection, including fish sampling and aerial surveys via drone, are core components of this monitoring. While detailed methods for these activities are included in the previous habitat confirmation section, they are also fundamental to effectiveness monitoring, as they provide context to not only facilitate prioritization and communications but also for detecting change following restoration.

3.9 Collaborative GIS Environment

A collaborative GIS environment (sern_peace_fwcp_2023) was established using QGIS and Mergin Maps to enable project team members to view, edit, and analyze shared spatial data on desktop computers and on mobile devices in the field. The environment integrates background layers from the BC Data Catalogue and other sources with project-specific field data, providing a central platform for planning, standardized data collection, and progress tracking. A summary of layers loaded to the project is provided in Appendix - Collaborative GIS Layers.

To support consistent and targeted assessment, a custom field form was developed for routine effectiveness monitoring based loosely on Forest Investment Account (2003) but tailored specifically for fish passage projects. Table 3.7) outlines the monitoring metrics used.

Table 3.7: Description of monitoring metrics used for effectiveness monitoring.

Parameter	Description
Dewatering	Have the remediation works led to dewatering of the channel due to substrate aggradation or other factors?
Velocity	Are flow velocities similar to those within the natural channel? Are they expected to exceed swim speeds of particular fish species/life stages of interest?

Parameter	Description
Constriction	Have the remediation works led to constriction of the channel. Compare channel width underneath structure and within construction footprint to average channel widths upstream and downstream?
Substrate	Is the substrate within/under and adjacent to the remediated structure generally equivalent to that found upstream and downstream where natural channel conditions exist?
Riparian	What is the condition of the riparian area within the construction footprint?
UAV Flight	Was a flight conducted with unmmanned aerial vehicle to document conditions at time of monitoring?
Flow_depth	What are the flow depths at the time of assessment within project footprint. Are depths expected to be sufficient to facilitat upstream passage for specific species/life stages of interest?
Stability	Does the structure appear to be stable or is there evidence of erosion/shifting?
Revegetation	How were riparian areas rehabillitated and are they improving fish habitat value?
Cover	Is cover available for fish within the construction footprint in the form of overhanging vegetation, large/small woody debris, boulders, undercut banks, etc?
Maintenance	If required, provide maintenance recommendations.
Recommendations	General recommendations for follow up. Could include revegetation, addition of substrate, fish sampling, etc.

4 Results and Discussion

4.1 Engage Partners

Project advancement in 2025 depended on engagement with partners across multiple agencies, Nations, industrial tenure holders, funders, and collaborators. Touchpoints during the year (email, video calls, meetings, field-based coordination, and contract administration) are summarised in Table 4.1.

Table 4.1: Project partners engaged in 2025.

Organization / Role	Topic / Focus
BC Timber Sales — Crossing CRP / bridges site plans	Kennedy Siding engineering design standards and site plan
BC Timber Sales — Engineering	Review of road-deactivation plans to support site prioritization based on modelled fish habitat and connectivity metrics
BC Timber Sales — Engineering Officer	Kennedy Siding (PSCIS 199663) FWCP-funded replacement; THUT15000 (PSCIS 203597) deactivation and riparian rehab
Canfor (prior-year coordination thread)	Fern Creek / Table FSR coordination
Chu Cho Environmental	Mapping support for Mesilinka River and Finlay River watershed groups FWCP restoration planning, and Upper Sustut Salmon Resiliency Fund proposal; set up with collaborative QGIS/Mergin project access
Eclipse Geomatics / Skeena Knowledge Trust	Temperature data sharing
Independent fisheries biologists	eDNA methodology for bull trout and Arctic grayling; Arctic Grayling synthesis report collaboration
McLeod Lake Indian Band — TLU Coordinator, Land Stewardship, LRO, Referrals, Director	NE-004 contract; land-guardian field participation; electrofishing course; collaborative QGIS/Mergin project access; council meeting planning; Azouzetta bull trout discussion
Ministry of Transportation and Infrastructure	Review of floodplain modelling and fish habitat/connectivity modelling tools; eDNA field planning and results
Poisson Consulting	Stream Network Temperature Modelling 2025
Sekani Forest Products (formerly Canfor)	Mackenzie and Peace NRD coordination; SUP spatial data; in-ground bridges KML
Sinclar Forest Group	Kerry Lake FSR; Arctic FSR; PSCIS 125000 (Trib to Parsnip) coordination
Stantec / Vitreo Minerals	Offsetting work consultation related to fish passage impacts in McLeod Lake Territory
UNBC	eDNA ddPCR analysis (program-scale 2025 submission)
Upper Peace Bull Trout CEMPRA Workshop (multi-agency: BC Hydro, WLRS, MoE BC, gov.ab.ca, Chu Cho, ESSA, independent consultants)	Bull trout stressor modeling and expert elicitation (March 2025)
WLRS / Provincial Fish Passage Technical Working Group	Letters of support; FWCP F27 proposal; FPTWG and MoTi funding 2025/26 coordination
WLRS — Bull Trout WHA and Fisheries Sensitive Watershed program	Bull trout critical habitat; meeting follow-up and 2025 re-engagement

4.2 Site Assessment Data

A summary of fish passage assessment procedures conducted through SERNbc since 2020 — including assessments, habitat confirmations, design, remediation, monitoring, fish sampling, eDNA sampling, and drone imagery — is provided in Appendix - Assessment Data Summary.

4.3 Climate Departure

Four findings from the regional climate-departure analysis carry to fish-passage prioritisation.

Warming is broad, fast, and significant region-wide. All five ecoregions of the FWCP Peace Region warmed about 1.8 °C since 1951 — annual mean (+1.8 °C), daytime maximum (+1.7 °C), and overnight minimum (+1.9 °C) all tell the same story. The trend is highly significant (Mann-Kendall $p < 0.001$ in every ecoregion). A subtle west-of-Rockies gradient layers on top, with the western and central ecoregions running about 0.2 °C warmer than the southeast — see Appendix - Climate Departure Figure 5.11.

The atmosphere is drying despite locally rising precipitation. Vapour pressure deficit rose 0.28 hPa region-wide, significant in every ecoregion. Annual precipitation rose +4 % regionally but the long-term trend is significant only in the two northernmost ecoregions; soil moisture is essentially flat everywhere because the warmer atmosphere is drinking the extra water back.

The snowpack signal is about timing, not quantity. Total annual snowfall is essentially unchanged (-6 %); annual snow water equivalent is down 10 %. But the seasonal redistribution is sharp — summer snow water equivalent collapsed 75 %, spring snowmelt rose +37 %, and the snowmelt midpoint shifted 11 days earlier. The freshet is arriving weeks earlier and the high-elevation snowpack that historically lingered into summer no longer does.

Day-night asymmetry is present. Overnight minimums warmed 1.9 °C while daytime maximums warmed 1.7 °C — overnight outpacing daytime is the textbook fingerprint of greenhouse warming (Karl et al. 1993). Mean diurnal range has narrowed by 0.3 °C across the record.

For barrier prioritisation these signals cut both ways. Where upstream reaches are cold-limited, added growing-degree-days can raise the intrinsic potential of habitat above the barrier and *increase* the value of restoring passage; where upstream reaches sit closer to the upper edge of the cold-water thermal niche, the same warming reduces usable habitat and lowers the payoff. The watershed-group breakdown in Appendix - Climate Departure maps the five-ecoregion signal onto the 16 FWCP Peace watershed groups.

4.4 Planning — Habitat, Connectivity and Floodplain Modelling

4.4.1 Habitat Modelling

Habitat modelling from `bcfishpass` including access model, linear spawning/rearing habitat model and lateral habitat connectivity models for watershed groups within our study area were updated for the spring of 2025 and are included spatially in the collaborative GIS project. A snapshot of these outputs related to each modeled and PSCIS stream crossing structure are also included within an `sqlite` database within this year's project reporting/code repository here.

4.4.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

The statistical-modelling methodology — channel-width prediction, stream-temperature SSN modelling with Poisson Consulting (Hill et al. 2024, 2025), the broader `fresh` + `link` habitat-modelling framework, and the calibration trajectory against `bcfishobs` and research activities targeting known high-value spawning and rearing — is described in the Methods chapter under Statistical Support for Habitat Modelling. The outputs feeding intrinsic-habitat and connectivity prioritization are summarised below under Statistical Habitat Modelling Outputs.

4.4.2 Floodplain Delineation

Modelled functional floodplain in the Parsnip River Watershed Group covers 48,116 ha — 8.6 % of the 5,597 km² watershed group — attached to 1,580 km of bull trout accessible order-3+ streams. Within that footprint sit 198 lakes and 464 mapped wetland polygons — the off-channel inventory that crossing remediations in these reaches stand to reconnect or leave severed, depending on whether design accounts for the floodplain context around the structure. The full metric table, watershed-scale map, and south-east detail map are in Appendix - Floodplain Delineation.

4.5 Statistical Habitat Modelling Outputs

Connectivity modelling now runs weekly with automated updates across all 16 watershed groups in the FWCP Peace area, driven by `bcfishpass` outputs using channel width, gradient, and provincial / federal barrier inventories — channel-width estimates are themselves a product of the Bayesian statistical work described in the methods (Thorley and Irvine 2021; Thorley et al. 2021). Field assessments were completed this year in five of those watershed groups (Parsnip River, Crooked River, Carp River, Nation River, and Parsnip Arm) to maintain momentum and surface candidate sites and partners. The work has the support of the Provincial Fish Passage Technical Working Group (FPTWG) — a partnership between the Province of British Columbia (WLRS, MOF, MOE, MOTI) and Fisheries and Oceans Canada.

Alongside the weekly automated updates, we use our open-source R packages `fresh` and `link` to expand, experiment with, and develop new project-specific habitat-modelling outputs. The first concrete application is an Arctic grayling intrinsic-habitat and connectivity model for the FWCP Peace area, parameterized on stream size and gradient, with further covariates such as stream temperature (`water-temp-bc`) available to incorporate as the framework develops. We present our first outputs — reproducing `bcfishpass`'s bull-trout classification and extending it to Arctic grayling for the Parsnip River Watershed Group — in Appendix - Parsnip River Habitat and Connectivity Modelling (Figures 5.19–5.21).

4.6 Fish Passage Assessments

Field assessments were conducted between September 21, 2025 and September 23, 2025 by Allan Irvine, R.P.Bio. and Lucy Schick, B.Sc. A total of 16 Fish Passage Assessments were completed, including 15 Phase 1 assessments and 1 reassessments.

Of the 16 sites where fish passage assessments were completed, 15 were not yet inventoried in the PSCIS system. This included 3 crossings considered “passable”, 1 crossings considered a “potential” barrier, and 11 crossings were considered “barriers” according to threshold values based on culvert embedment, outlet drop, slope, diameter (relative to channel size) and length (MoE 2011).

A summary of crossings assessed, a rough cost estimate for remediation, and a priority ranking for follow-up for Phase 1 sites is presented in Table 4.2. Detailed data with photos are presented in Attachment - Phase 1 Data and Photos.

Table 4.2: Upstream habitat estimates and cost benefit analysis for Phase 1 assessments ranked as a ‘barrier’ or ‘potential’ barrier. Bull trout network model used for habitat estimates (total length of stream network <25% gradient).

PSCIS ID	External ID	Priority	Stream	Road	Barrier Result	Habitat value	Habitat Upstream (km)	Stream Width (m)	Fix	Cost Est ($K)
6559	–	low	Tributary To Williston Reservoir	Finlay Community Connector	Barrier	Low	17.4	1.0	SS-CBS	100
203597	15201563	high	Tributary To Nation River	Thut 15000	Barrier	High	16.8	2.5	OBS	450
203598	15201146	moderate	Tributary To Nation River	Thut 15000	Barrier	Medium	9.2	1.6	SS-CBS	100
203599	15202249	moderate	Tributary To Nation River	Thut 14000	Barrier	Medium	14.2	3.0	OBS	450
203600	15200985	moderate	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	30.6	4.1	OBS	450
203601	15201007	moderate	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	15.9	3.5	OBS	450
203602	15201006	high	Tributary To Nation River	Finlay-Nation FSR	Barrier	High	17.5	3.4	OBS	495
203603	16400714	moderate	Tributary To Williston Reservoir	Finlay-Nation FSR	Barrier	High	106.7	7.4	OBS	450
203604	16401990	moderate	Tributary To Williston Reservoir	Finlay FSR	Barrier	Medium	26.6	10.0	OBS	810

4.7 Habitat Confirmation Assessments

During the 2025 field assessments, habitat confirmation assessments were conducted at 3 sites within the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups. A total of approximately 2km of stream was assessed.

As collaborative decision-making was ongoing at the time of reporting, site prioritization can be considered preliminary. Results are summarized in Table 4.3 with raw habitat and fish sampling data included in Attachment - Data. A summary of preliminary modelling results illustrating quantities of bull trout spawning and rearing habitat potentially available upstream of each crossing as estimated by measured/modelled channel width and upstream accessible stream length are presented in Figure 4.1. Detailed information for each site assessed with Phase 2 assessments (including maps) is presented in the per-site habitat-confirmation appendices: Tributary to Parsnip

River (PSCIS 199663), Tributary to Nation River (PSCIS 203597), and Tributary to Williston Reservoir (PSCIS 203605).

Table 4.3: Overview of habitat confirmation sites. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Tenure	UTM	UTM zone	Fish Species	Habitat Gain (km)	Habitat Value	Priority	Comments
199663	Tributary To Parsnip River	Chco 12000	BCTS	515416 6094963	10	RB	0.0	Medium	Moderate	Low gradient stream with significant riparian tree blow down throughout. Very difficult walking so most was done outside of the active channel with spot visits to the stream for measurements/photos completed intermittently for the 600m that was assessed. Wide floodplain within well defined canyon (approximately 15m deep with slopes of ~45°). Healthy riparian, consisting of herb and shrub cover within sparse spruce forest that appears to have been impacted by some sort of beetle or other affliction. Most of the spruce trees were dead with many already fallen down.
203597	Tributary To Nation River	Thut 15000	Sekani R072148	425959 6122703	10	RB	5.6	High	High	Surveyed 350m upstream to the beginning of the wetlands. Very nice stream with good flow, although recent rainfall may have had an influence. Pockets of gravels throughout the area surveyed. Although the area viewed was limited, the wetlands appeared shallow so freezing to the bottom may occur during the winter.
203605	Tributary To Williston Reservoir	Finlay FSR	MoF	486504 6124990	10	RB	35.9	High	High	Large stream with complex habitat and diverse cover including deep pools, boulders, large and small woody debris, and undercut banks. A canyon section approximately 100m long was located 100m upstream of the Modeste FSR bridge, containing multiple cascade steps up to 1m high with deep downstream pools. At the upper end of the site, the channel deflected against steep gully walls. Abundant gravel was present, suitable for adfluvial and resident rainbow trout as well as bull trout. eDNA sample collected.

Table 4.4: Summary of Phase 2 fish passage reassessments.

PSCIS ID	Embedded	Outlet Drop (m)	Diameter (m)	SWR	Slope (%)	Length (m)	Final score	Barrier Result
199663	No	0.60	1.2	0.00	3	13	30	Barrier
203597	No	0.25	0.8	3.12	1	19	29	Barrier
203605	No	0.95	3.9	1.54	2	46	37	Barrier

Table 4.5: Cost benefit analysis for Phase 2 assessments. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Barrier Result	Habitat value	Stream Width (m)	Fix	Cost Est ($K)	Habitat Upstream (m)	Cost Benefit (m / $K)	Cost Benefit (m2 / $K)
199663	Tributary To Parsnip River	Chco 12000	Barrier	Medium	3.0	SS-CBS	–	0	–	–
203597	Tributary To Nation River	Thut 15000	Barrier	High	2.5	OBS	450	5570	12377.8	15472.2
203605	Tributary To Williston Reservoir	Finlay FSR	Barrier	High	6.7	OBS	450	35932	79848.9	239546.7

Table 4.6: Summary of Phase 2 habitat confirmation details.

PSCIS ID	Length surveyed upstream (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
–	–	–	–	–	–	–	–
——:	——————————:	—————————:	—————————:	————————:	———————:	:————	:————-

Table 4.7: Summary of watershed area statistics upstream of Phase 2 crossings.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
199663	1.0	713	707	849	784	774	SW
203597	8.2	900	907	1065	980	971	S
203605	55.6	681	671	1644	980	937	SE
203611	55.6	682	671	1644	980	937	SE

* Elev P60 = Elevation at which 60% of the watershed area is above

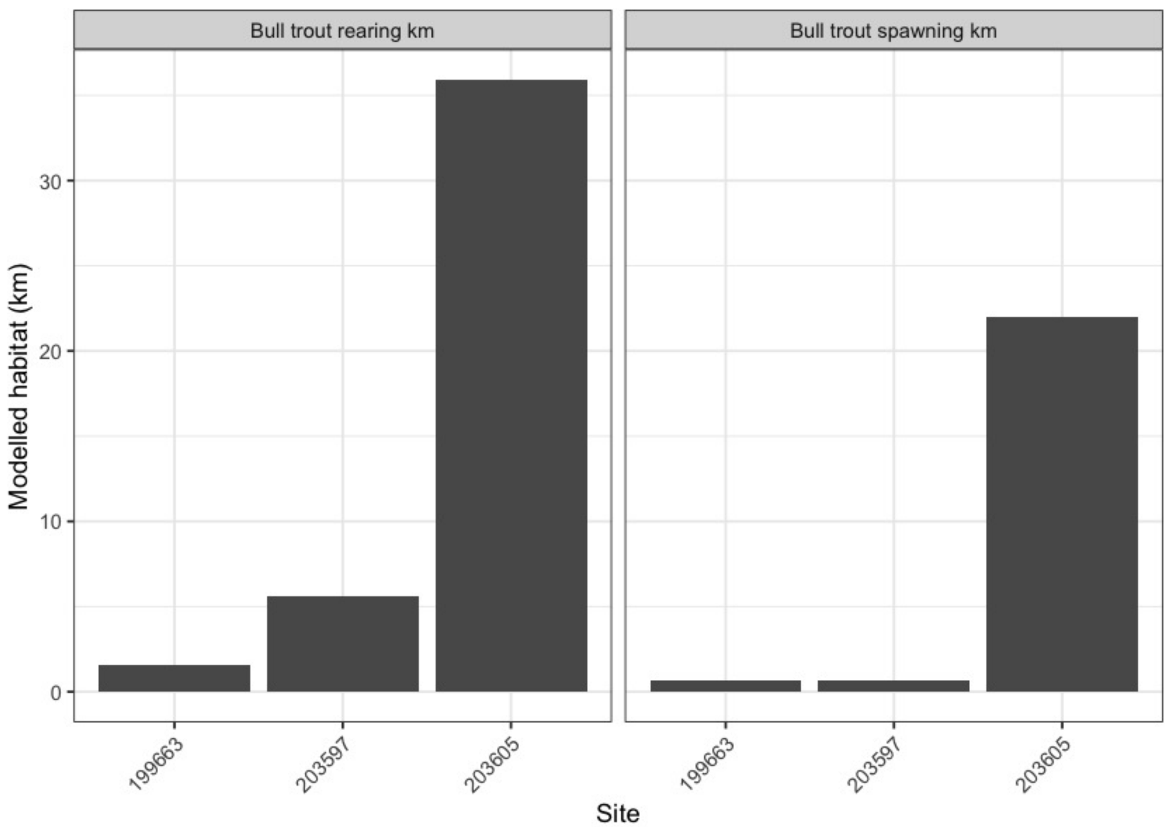


Figure 4.1: Summary of potential rearing and spawning habitat upstream of habitat confirmation assessment sites. Bull trout rearing and spawning models used for habitat estimates (total length of stream network <10.5% and <5.5% gradient, respectively).

4.7.1 Fish Sampling

Fish sampling was conducted at 8 sites within 6 streams, with a total of 85 fish captured. Fork length, weight, and species were documented for each fish. Salmonids with fork lengths >60mm were PIT-tagged to facilitate long-term tracking of health and movement. Fork length data was used to delineate salmonids based on life stages: fry (0 to 65mm), parr (>65 to 110mm), juvenile (>110mm to 140mm), and adult (>140mm) by visually assessing the histograms presented in Figure 4.3. A summary of sites assessed is included in Table 4.8, and raw data is provided in Attachment - Data. A summary of density results is also presented in Figure 4.2.

Table 4.8: Summary of electrofishing sites.

site	stream	passes	ef_length_m	ef_width_m	area_m2	enclosure
125179_us_ef1	Tributary To Missinka River	1	45	2.2	99.0	partial enclosure
125179_ds_ef1	Tributary To Missinka River	1	35	2.7	94.5	partial enclosure
125231_ds_ef1	Tributary To Table River	1	50	2.8	140.0	partial enclosure
125231_us_ef1	Tributary To Table River	1	50	1.6	80.0	partial enclosure
198692_us_ef1	Tributary To Kerry Lake	1	16	–	–	partial enclosure
198692_us_ef2	Tributary To Kerry Lake	1	12	–	–	open
198692_ds_ef1	Tributary To Kerry Lake	1	34	–	–	partial enclosure
198692_ds_ef2	Tributary To Kerry Lake	1	4	–	–	open

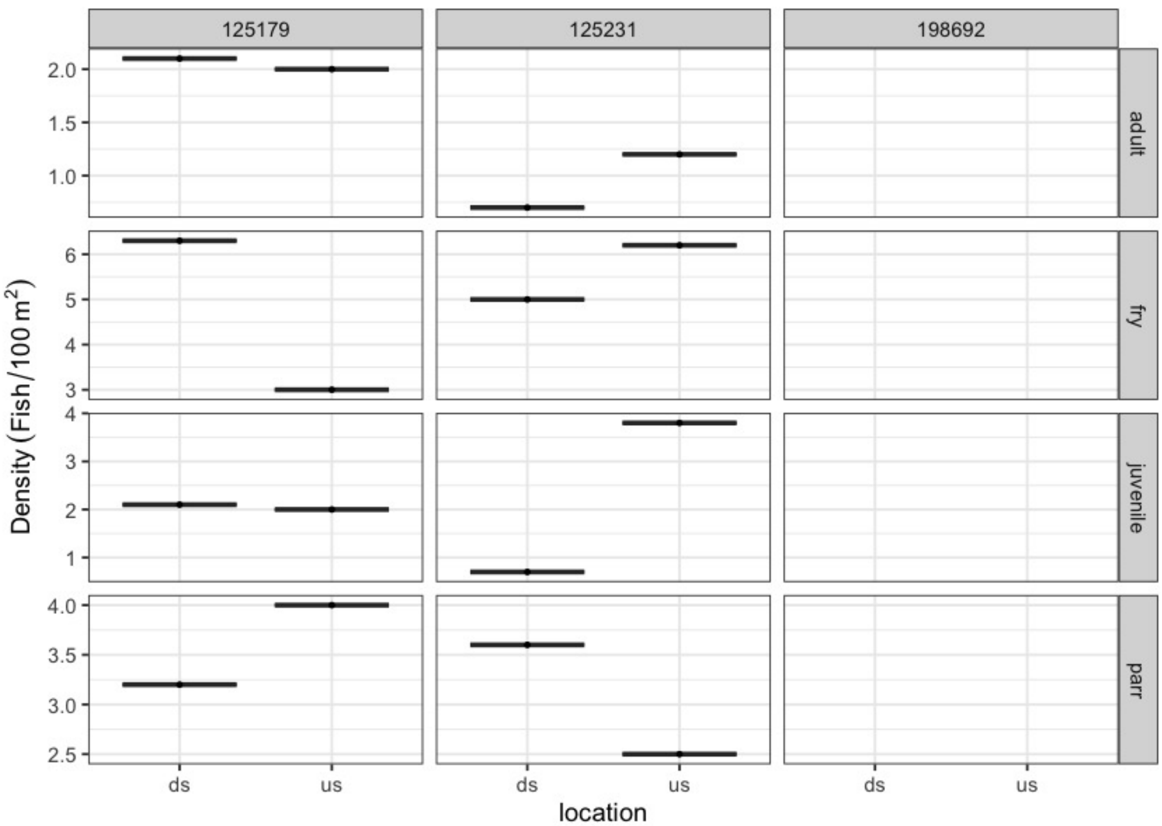


Figure 4.2: Boxplots of densities (fish/100m2) of fish captured downstream (ds) and upstream (us) by electrofishing.

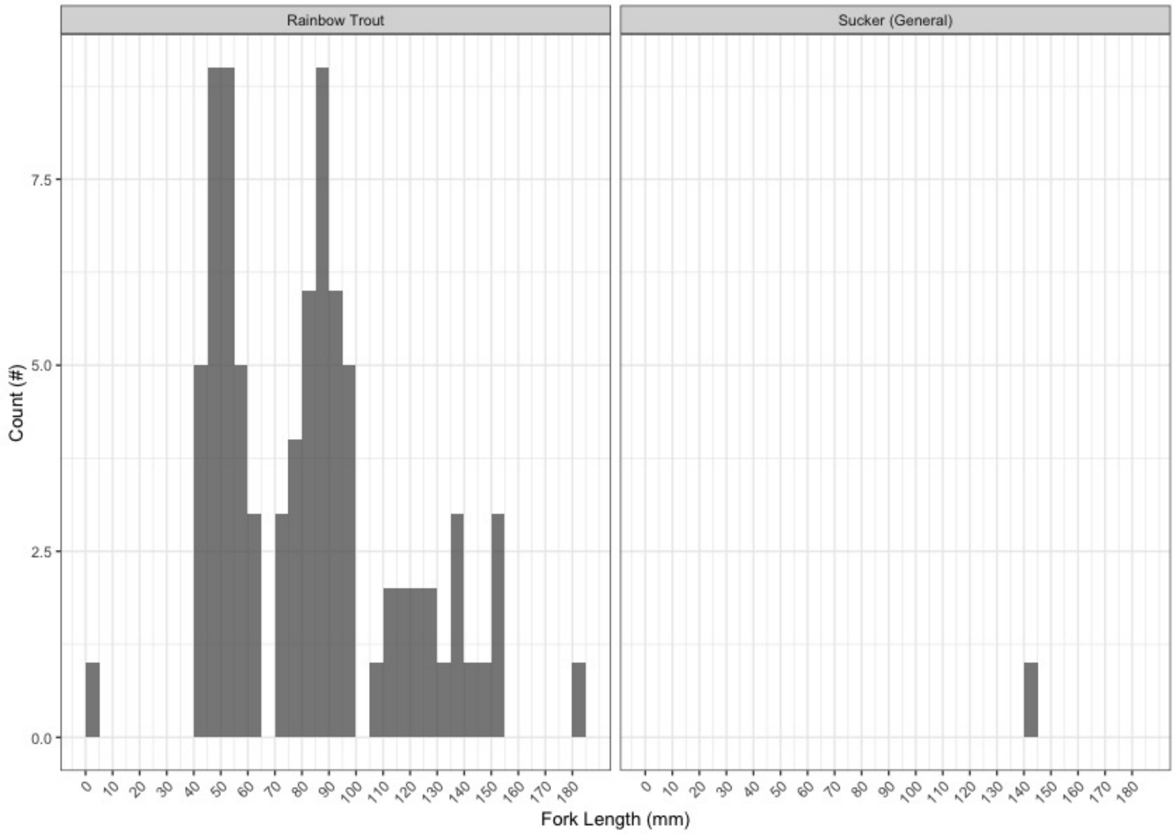


Figure 4.3: Histograms of fish lengths by species. Fish captured by electrofishing during habitat confirmation assessments.

4.7.2 Environmental DNA (eDNA) sampling

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 35 collection events were undertaken across 10 streams, including 28 real environmental samples, 3 field blanks, and 4 office blanks for quality assurance.

Per-species detection results across the 28 real environmental samples are summarized in Table 4.9. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; sub-threshold signals (1-3 droplets) are tracked separately as they are not interpretable as species presence per ddPCR convention. Site-level detail (per-site detection table, field blanks, and retests) is provided in Appendix - Environmental DNA Results, and the interactive Peace eDNA map shows toggleable per-species layers.

Table 4.9: Per-species summary of eDNA results across real environmental samples in the Peace project area (excludes field and office blanks).

Species	Code	Sites tested	Detected (≥4 droplets)	Sub-threshold only (1-3 droplets)	No detection (0 droplets)
Species	Code	Sites tested	Detected (≥4 droplets)	Sub-threshold only (1-3 droplets)	No detection (0 droplets)
Rainbow Trout	RAIN	28	22	2	4
Bull Trout	BULT	28	1	8	19
Burbot	BURB	2	1	0	1
Arctic Grayling	GRAY	24	0	1	23
Kokanee	SOCK	22	0	0	22

4.8 Engineering Design

In 2025, the engineering-design pipeline shifted in two ways. The proposed Kerry Lake FSR structure replacement at PSCIS 198692, originally scoped for 2024/25, was shelved after Sinclar Forest Products and project partners agreed that road decommissioning and structure removal at both PSCIS 198692 (km 9) and PSCIS 198693 (km 8) is a better outcome on both economic and ecological grounds; Sinclar will cover removal and deactivation to minimum provincial standards in 2025/26 (baseline-monitoring context in Appendix - Tributary to Kerry Lake).

BC Timber Sales (BCTS) shared road-deactivation plans for the Tributary to Nation River crossings deactivated in fall 2025 — used to scope above-minimum-standards riparian restoration (soil decompaction, riparian planting, coarse woody debris placement) for 2026/27. BCTS is also lining up Kennedy Siding (PSCIS 199663) on the Chuchinka-Colbourne FSR for a site plan and engineering design in spring 2026, with the project providing modelled habitat and connectivity context to inform the design scope.

All sites with a design for remediation of fish passage can be found online within the Results and Discussion section of the report found here.

4.9 Monitoring

In 2025, effectiveness monitoring was conducted at two previously remediated crossings, and baseline data collection (fish sampling and eDNA only — no effectiveness form) was completed at one site ahead of planned remediation:

- **Tributary to Missinka River (PSCIS 125179)** on Chuchinka-Missinka FSR — effectiveness monitoring (form + fish sampling + eDNA), extending the multi-year baseline established in Irvine and Winterscheidt (2023) and Irvine and Schick (2025). Detail in Appendix - Tributary to Missinka River.

- **Tributary to the Table River (PSCIS 125231)** on Chuchinka-Table FSR — effectiveness monitoring (form + fish sampling + eDNA) following the 2023 replacement of the crossing with a clear-span bridge by Canfor with FWCP funding via SERNbc coordination (Irvine and Winterscheidt 2024). Detail in Appendix - Tributary to the Table River.

- **Tributary to Kerry Lake (PSCIS 198692)** — pre-remediation baseline (fish sampling and eDNA) to inform future effectiveness monitoring. The remediation plan has pivoted: where the 2024/25 work envisaged a structure replacement, Sinclar Forest Products — as the tenure holder for the Kerry Lake FSR — has now committed to road decommissioning and removal of both PSCIS 198692 (km 9) and PSCIS 198693 (km 8) in 2025/26 to minimum provincial deactivation standards. The 2024 habitat confirmation is documented in Irvine and Schick (2025); the 2024–2025 baseline at km 8 (this site) becomes the pre-removal reference for post-removal effectiveness monitoring. Restoration above the minimum deactivation standards (soil decompaction, riparian planting, coarse woody debris placement) is being scoped through the 2026/27 proposal process. Detail in Appendix - Tributary to Kerry Lake.

A summary of effectiveness monitoring form metrics across the two effectiveness sites is presented in Table 4.10.

Table 4.10: Summary of monitoring sites.

PSCIS ID	Stream	Road	Crossing type	Crossing subtype	Diameter or span (m)	Length or width (m)	Assessment comments
125231	Tributary to Table River	Table FSR	OBS	BRIDGE	–	–	Overall, the site looks very similar to last year, in good condition with no evidence of any structural issues and the instream portion doing quite well. As there is a large amount of rock rip wrap adjacent to the stream that was not planted the riparian vegetation within the footprint is basically nonexistent. Electrofishing was conducted at sites downstream and upstream of the bridge with fish over 65mm PIT tagged. eDNA samples were collected as well.
125179	Tributary to Missinka River	Chuchinka-Missinka FSR	OBS	BRIDGE	16	4.8	Routine effectiveness evaluation. Structure was replaced several years ago and monitoring is ongoing. Electrofishing with fish over 65mm in length PIT tagged, as well as EDNA sampling was conducted in 2025.

4.10 Collaborative GIS Environment

In addition to numerous layers documenting fieldwork activities since , a summary of background information spatial layers and tables loaded to the collaborative GIS project (sern_peace_fwcp_2023) at the time of writing (2026-06-26) is provided in Appendix - Collaborative GIS Layers.

5 Recommendations

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices—picking battles carefully, building trust, and staying committed to a longer-term transformation.

While preliminary top remediation priorities are provided by watershed group, these rankings are inherently subjective and can depend on the capacity and willingness of infrastructure owners and tenure holders to support implementation—both financially and over the often multi-year project timelines. In practice, we must often act opportunistically, pursuing simpler, lower-cost options to maintain momentum and achieve near-term progress.

Government, community groups, landowners, non-profits, industry and other stakeholders should work collaboratively to address high and moderate priority barriers identified online within the Results and Discussion section of the report found here. Although the table presents many options - currently - linked reports specify whether each site is a low, moderate, or high priority. Progress on any front is meaningful, and aiming to remediate at least one high-priority site per year per watershed group—regardless of its overall rank—is a practical and effective approach.

To enhance fish passage restoration in the FWCP Peace Region:

Site-specific restoration and remediation actions

- Advance the replacement of crossing 199663 (Tributary to the Parsnip River) on the Chuchinka-Colbourne FSR in partnership with BC Timber Sales (BCTS), building on the 2025 assessment work and ongoing prioritization conversations with the licensee.
- Deliver above-minimum-standards habitat restoration on the two Tributary to Nation River crossings (PSCIS 15201563 and 15201146) deactivated by BCTS in fall 2025 — soil

decompaction, riparian planting, and coarse woody debris placement to convert minimum deactivation outcomes into measurable aquatic-habitat improvement.
- Support Sinclar Forest Products' planned road decommissioning and removal of crossings 198692 and 198693 on the Kerry Lake FSR (km 9 and km 8) in 2025/26, layering riparian restoration and habitat enhancement on top of the minimum provincial deactivation standards Sinclar will cover. Use both the Kerry Lake and Tributary to Nation River riparian / deactivation sites as field sites for the University of Northern British Columbia (UNBC) and College of New Caledonia (CNC) seed-bank and native-plant-propagation collaboration described below.
- Maintain progress on Fern Creek (PSCIS 125261) — replacement is a high priority given the size of the system and the absence of other downstream anthropogenic barriers. Canfor has procured preliminary engineering designs; finalize partnership financing despite harvest-planning uncertainty.

Capacity building and partner integration

- Build a local native-seed and plant-propagation supply chain for FWCP Peace fish-passage restoration sites in collaboration with Shelby Roberts (CNC / UNBC) — connected to native seed collection, nursery stock, and plant production programs in the Peace region — and with Lisa Wood (UNBC) who runs the UNBC seed bank and Food HUB initiative. Locally-sourced conifer seedlings, understory species, and cottonwood / willow material for riparian restoration at Kerry Lake and Nation River tributary sites strengthens the restoration outcomes while giving these UNBC / CNC programs a concrete field application.
- Continue to collaborate with WLRS, UNBC, local fisheries experts, FWCP, the Provincial Fish Passage Technical Working Group (FPTWG), and the CEMPRA Project working group to align prioritization, monitoring, and restoration practice across the region.
- Maintain strong partnerships with rail, highway, forestry, mining, and Nations tenure holders to support funding, site selection, remediation, and monitoring through adaptive management informed by traditional knowledge and real-time data.

Prioritization and modelling integration

- Continue using climate modelling (climate-departure analyses, this report's Appendix - Climate Departure; thermal-regime modelling) to prioritize crossings that enable access to cold, drought-resistant habitats and to anchor adaptation strategies to the watershed-scale climate signal.
- Integrate stream-temperature predictions and Growing-Season Degree Days from the collaborative Bayesian Spatial Stream Network modelling (Hill et al. 2024, 2025) into intrinsic-habitat modelling for prioritization, surfacing the thermal regime alongside connectivity, gradient, and channel-width signals.
- Calibrate bull trout and Arctic grayling intrinsic-habitat models against areas of known high-value spawning and rearing identified through fish-presence observations in `bcfishobs` together with research activities (some commissioned for FWCP Peace) — quantifying model uncertainty and refining prioritization.
- Dovetail cumulative-effects modelling (CEMPRA framework and related provincial efforts) into the prioritization process so cross-watershed pressures — forestry, mining, climate, hydro — are weighted alongside per-crossing barrier evaluation.
- Prioritize detailed assessments in areas with blockages and high habitat potential.

Watershed-scale context — climate departure and floodplain delineation

- Continue the climate-departure analysis (Appendix - Climate Departure) — extend the watershed-group ecoregion mapping that bridges regional climate signal to per-watershed barrier prioritisation, refine the per-WSG weighting, and add the variables and seasonal-period breakdowns that surface the climate-driven differences in habitat trajectory across the 16 watershed groups.
- Extend functional-floodplain mapping (Appendix - Floodplain Delineation) from the Parsnip River Watershed Group to additional FWCP Peace watersheds so the off-channel habitat inventory — side channels, wetlands, riparian recruitment areas — is available everywhere barrier prioritisation gets done in the region.
- Build capacity in partner organizations (Nations, industry, regulators, researchers, community groups) to interpret and use the climate-departure and floodplain layers — through workshops, partner-facing outreach materials, and the public webmap interface — so the opportunities and challenges these analyses surface are visible to the people whose decisions shape watershed-scale outcomes.
- Use these watershed-scale context layers to position fish passage where it belongs — inside the broader atmospheric, hydrologic, and geomorphic system that determines whether the habitat above a barrier is gaining or losing value. The per-crossing inventory is the action surface; climate departure and floodplain context provide the weighting and the narrative arc that lets reviewers, funders, and partners understand why a specific barrier matters in its watershed.

Monitoring continuation

- Continue effectiveness monitoring at key sites using fish sampling, eDNA, PIT tagging, water-temperature data, and aerial imagery — extending the multi-year baselines at Tributary to Missinka River (PSCIS 125179), Tributary to the Table River (PSCIS 125231), and Tributary to Kerry Lake (PSCIS 198692, now repurposed as pre-removal baseline per Sinclar's pivot).
- Expand water-temperature monitoring as the foundation for understanding climate-change impacts and shaping adaptation strategies — both within our own program and in support of the largest regional initiatives (FWCP-funded thermal-ecology research on grayling and bull trout, UNBC-led work on critical habitats, and partner monitoring programs).
- Continue to develop a cost-effective monitoring framework that ties baseline + post-remediation data to measurable productivity gains from improved passage.

Data infrastructure for research and collaboration

- Develop a public webmap interface to communicate fish-passage priorities, baseline conditions, restoration outcomes, and partner activities to FWCP collaborators, First Nations, industry, regulators, and the public.
- Position `water-temp-bc` as an example of a data-portal model that can be extended or replicated to incorporate other past and future datasets collected by UNBC, cumulative-effects monitoring initiatives, FWCP-funded research, regional Nations, industry tenure holders, and others. Many programs have been generating water-temperature and water-quality records in the region for years; the goal is not to ask partners to hand over their data

but to make the data discoverable and queryable through a shared interface so researchers can ask basin-scale questions without per-source data hunts or permission gates.

- Build a community of practice around regional water-temperature data with UNBC, cumulative-effects monitoring leads, FWCP partners, and other researchers — focused on collective capacity to ask and answer holistic regional questions about thermal regime, climate-change response, and habitat suitability, rather than every program reconstructing the same regional picture from fragmented sources.
- Model and extend the same open-access pattern for eDNA data across UNBC, FWCP partners, and the broader research community — making detections, samples, methods, and metadata queryable in a shared interface rather than scattered across project repositories.

Appendix - Assessment Data Summary

Fish passage assessment procedures conducted through SERNbc since 2020 are amalgamated within Table 5.1 — with detailed per-site outputs located online within the Results and Discussion section of the report found here. Since 2020, orthoimagery and elevation model rasters have been generated and stored as Cloud Optimized Geotiffs on a cloud service provider (AWS) with select imagery linked to in the collaborative GIS project. Additionally, a tile service has been set up to facilitate viewing and downloading of individual images, provided in Appendix - UAV Imagery.

Table 5.1: Summary of fish passage assessment procedures conducted by SERNbc in the FWCP Peace Region.

Watershed Group	Assessment	Reassessment	Habitat Confirmation	Design	Remediation	Fish Sampling	Drone Imagery
Carp Lake	15	3	3	0	0	0	0
Crooked River	57	6	4	0	0	1	0
Parsnip River	30	10	20	5	3	6	4
Total	102	19	27	5	3	7	4

Appendix - Fish Species by Watershed Group

Historical fish observations by watershed group are recorded in the BC Fisheries Information Summary System (FISS) (MoE 2024a). Per-species presence across the watershed groups in this study area is summarised in Table 5.2. In the interactive (gitbook) version of this report the table shows one column per watershed group; in the PDF version the watershed-group columns are collapsed to a single comma-separated list of watershed-group codes per species to fit page width.

Table 5.2: Fish species recorded in the Carp Lake, Crooked River, Finlay Arm, Finlay River, Firesteel River, Fox River, Ingenika River, Lower Omineca River, Mesilinka River, Nation River, Ospika River, Parsnip Arm, Parsnip River, Peace Arm, Toodoggone River, and Upper Omineca River watershed groups.

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
Catostomus catostomus	Longnose Sucker	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Catostomus columbianus	Bridgelip Sucker	Yellow	–	Crooked
Catostomus commersonii	White Sucker	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Fox, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Upper Omineca
Catostomus macrocheilus	Largescale Sucker	Yellow	–	Carp Lake, Crooked, Finlay Arm, Ingenika, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone
Coregonus clupeaformis	Lake Whitefish	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Cottus aleuticus	Coastrange Sculpin (formerly Aleutian Sculpin)	Yellow	–	Parsnip
Cottus asper	Prickly Sculpin	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Cottus cognatus	Slimy Sculpin	Yellow	–	Carp Lake, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Cottus ricei	Spoonhead Sculpin	Yellow	NAR (May 1989)	Finlay Arm, Lower Omineca, Parsnip Arm, Upper Omineca
Couesius plumbeus	Lake Chub	Yellow	DD	Carp Lake, Crooked, Finlay Arm, Finlay, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Esox lucius	Northern Pike	Yellow	–	Crooked
Hybognathus hankinsoni	Brassy Minnow	No Status	–	Carp Lake, Crooked, Finlay Arm, Nation, Parsnip Arm

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
Lota lota	Burbot	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Mylocheilus caurinus	Peamouth Chub	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Ingenika, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Upper Omineca
Oncorhynchus clarkii	Cutthroat Trout	No Status	–	Ospika
Oncorhynchus clarkii clarkii	Coastal Cutthroat Trout	Blue	–	Finlay, Lower Omineca
Oncorhynchus kisutch	Coho Salmon	Not Reviewed	–	Mesilinka
Oncorhynchus mykiss	Rainbow Trout	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Oncorhynchus mykiss	Steelhead	Yellow	–	Mesilinka
Oncorhynchus mykiss	Steelhead (Winter-run)	Yellow	–	Finlay Arm
Oncorhynchus nerka	Kokanee	Not Reviewed	–	Finlay Arm, Finlay, Firesteel, Ingenika, Lower Omineca, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone
Oncorhynchus nerka	Sockeye Salmon	Not Reviewed	–	Mesilinka
Osmerus dentex	Rainbow Smelt	Unknown	–	Firesteel, Nation, Parsnip
Prosopium coulterii	Pygmy Whitefish	Yellow	NAR (Nov 2016)	Finlay Arm, Fox, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Upper Omineca
Prosopium cylindraceum	Round Whitefish	Yellow	–	Parsnip
Prosopium williamsoni	Mountain Whitefish	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Ptychocheilus oregonensis	Northern Pikeminnow	Yellow	–	Carp Lake, Crooked, Finlay Arm, Lower Omineca, Nation, Parsnip Arm, Parsnip, Peace Arm
Rhinichthys cataractae	Longnose Dace	Yellow	–	Crooked, Finlay Arm, Finlay, Firesteel, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Toodoggone, Upper Omineca
Richardsonius balteatus	Redside Shiner	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Fox, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Upper Omineca
Salvelinus confluentus	Bull Trout	Blue	SC (Nov 2012)	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Salvelinus fontinalis	Brook Trout	Exotic	–	Crooked, Parsnip Arm, Parsnip
Salvelinus malma	Dolly Varden	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
Salvelinus namaycush	Lake Trout	Yellow	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
Thymallus arcticus	Arctic Grayling	Yellow	–	Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone, Upper Omineca
–	Arctic Char	–	–	Finlay
–	Chub (General)	–	–	Carp Lake, Crooked, Parsnip Arm, Parsnip
–	Dace (General)	–	–	Finlay Arm, Parsnip Arm, Parsnip
–	Lamprey (General)	–	–	Nation
–	Minnow (General)	–	–	Carp Lake, Crooked, Finlay Arm, Fox, Parsnip Arm, Parsnip, Upper Omineca
–	Mottled Sculpin	–	–	Firesteel, Nation, Parsnip
–	Salmon (General)	–	–	Nation
–	Sculpin (General)	–	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Fox, Ingenika, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm, Toodoggone
–	Smelt (General)	–	–	Finlay Arm, Parsnip Arm
–	Squanga	–	–	Crooked
–	Sucker (General)	–	–	Carp Lake, Crooked, Finlay Arm, Finlay, Firesteel, Lower Omineca, Mesilinka, Nation, Parsnip Arm, Parsnip, Peace Arm, Toodoggone
–	Whitefish (General)	–	–	Carp Lake, Crooked, Lower Omineca, Mesilinka, Nation, Ospika, Parsnip Arm, Parsnip, Peace Arm

* COSEWIC abbreviations :
SC - Special concern
DD - Data deficient
NAR - Not at risk
E - Endangered
T - Threatened

BC List definitions :
Yellow - Species that is apparently secure
Blue - Species that is of special concern
Exotic - Species that have been moved beyond their natural range as a result of human activity

Dolly Varden are documented in all three study watersheds — Carp, Crooked, and Parsnip — but these records are likely misidentifications of bull trout (COSEWIC 2012).

Appendix - Climate Departure

Climate departure and fish passage

Stream crossings are the visible structures in a fish-passage inventory: culverts, bridges, weirs, fords. The streams they cross sit inside a climate that is changing — warmer air, shifting snowpack, longer shoulder seasons — and those climate changes set the hydrologic and thermal context the structures pass fish *through*. A culvert that worked when it was installed in 1985 is now passing a different hydrograph: earlier freshet, lower late-summer flows, warmer water. The barrier inventory does not see that on its own.

Warmer air drives warmer streams. For the cold-water salmonids the Peace supports — bull trout, Arctic grayling, mountain whitefish — that can move habitat in either direction: cold-limited headwaters gain growing-degree-days (larger fish, better overwinter survival, stronger out-migration), while reaches near the upper edge of the thermal niche lose usable habitat. The recovery value of removing a downstream barrier scales with which side of that line each upstream reach sits on. Climate departure tells you which side that is, watershed by watershed.

Beyond temperature, hydrologic timing matters. Earlier and faster snowmelt shifts the freshet weeks earlier in the calendar and compresses the high-flow pulse. That changes design assumptions for crossing structures, and it changes the timing of migration, gravel mobilisation, and off-channel rearing — the in-stream processes barrier removal is meant to restore.

Quantifying departure at the watershed scale lets us carry climate context into barrier prioritisation alongside the structural and biological inputs.

Area of interest

The FWCP Peace Region is a 72,811 km² administrative boundary in northeastern British Columbia covering Williston Reservoir and surrounding watersheds. All watersheds drain to the Mackenzie River — the Peace flows east into the Slave, which flows into the Mackenzie, which empties into the Beaufort Sea on the Arctic Ocean. The region spans 5 ecoregions (Figure 5.1) — broad climate-physiography zones based on latitude, elevation, and dominant vegetation — and we carry that ecoregion partition through the analysis because climate departure can vary across them in ways the regional average hides.

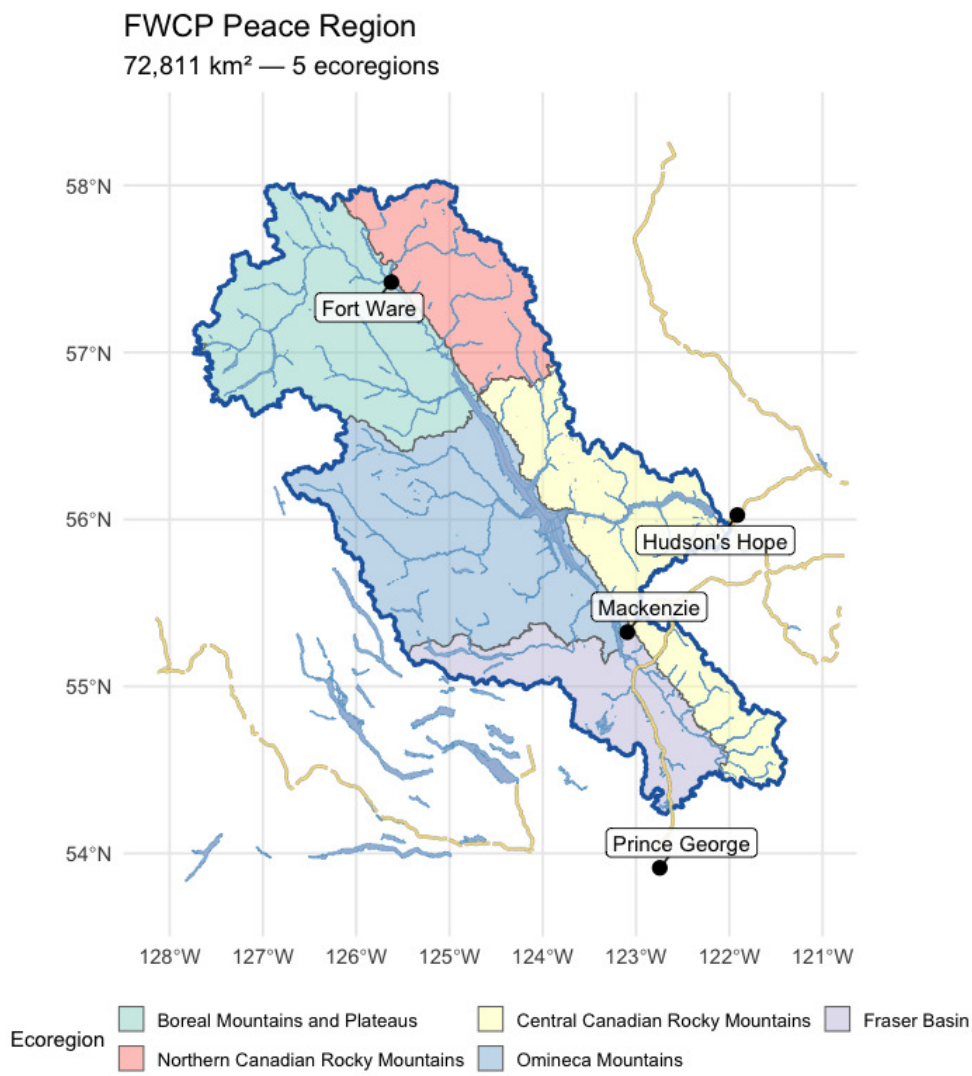


Figure 5.1: FWCP Peace Region area of interest in northeastern British Columbia, coloured by ecoregion. Williston Reservoir dominates the basin; the climate-departure pipeline is run on the regional outline (heavy blue) and again on each ecoregion polygon.

Recent decade versus pre-warming reference

The table below compares two windows directly — the recent decade (2015–2025) against the pre-warming reference (1951–1980). The difference is reported in the variable's native units, alongside percent change where it is meaningful (precipitation, soil moisture, vapour pressure deficit, relative humidity). Two p-values appear, answering different questions. **Δ p (windows)** is the Welch t-test on the annual values within each window — this tells us if the recent decade differs from the pre-warming reference. **Trend p (75-yr)** is the Mann-Kendall test on the full 1951–present series — this tells us if there is a steady year-on-year ramp. Step changes show up as significant Δ p with non-significant trend p; gradual ramps show up as both significant.

Table 5.3: Recent decade (2015-2025) versus pre-warming reference (1951-1980) for all 15 climate variables and all available periods, FWCP Peace Region. Two p-values answer different questions — see text.

Variable	Period	Recent (2015–2025)	Pre-warming (1951–1980)	Δ absolute	Δ %	Δ p (windows)	Trend p (75-yr)
prcp	annual	835.20	803.11	32.08	4.0	0.308	0.197
prcp	fall	239.18	230.76	8.42	3.7	0.622	0.493
prcp	spring	155.72	150.62	5.10	3.4	0.643	0.164
prcp	summer	266.04	242.15	23.89	9.9	0.283	0.185
prcp	winter	174.26	179.59	-5.33	-3.0	0.674	0.661
rh	annual	74.08	74.08	0.00	0.0	0.992	0.862
rh	fall	79.38	80.16	-0.78	-1.0	0.279	0.091
rh	spring	68.43	69.49	-1.06	-1.5	0.159	0.385
rh	summer	67.25	67.63	-0.38	-0.6	0.760	0.756

The recent decade ran 1.8 °C warmer than the pre-warming reference for annual mean temperature, with both window and trend p-values below 0.001. Vapour pressure deficit rose significantly on both tests. Annual precipitation rose about 4 % but neither test confirms the shift. Soil moisture and relative humidity show no meaningful change. The snow rows (swe summer, snowmelt spring, snowmelt_doy_50) carry the headline snowpack-timing signals discussed below.

Trends over the long record

Anomalies are computed against the 1951–1980 pre-warming reference. Saying a year is "+1.5 °C" means it was 1.5 °C warmer than the average year between 1951 and 1980. The trend table below has two rows per variable — trends computed from two different start years:

- **1951–present (75 years)** — the long view. Captures the full magnitude of warming since the pre-warming reference.
- **1981–present (45 years)** — starts at the beginning of the current 30-year climate normal (1981–2010), the reference used in most published climate products.

Comparing the two slopes is informative. If the 45-year slope is steeper than the 75-year slope, warming has accelerated; if shallower, warming has slowed (though "slower" almost never means "stopped"). Similar slopes mean a roughly steady rate of change across the full record. Total change is slope multiplied by the number of years — the cumulative shift over the trend window.

Table 5.4: Trend statistics for all variables and periods, FWCP Peace Region — Theil-Sen slope, Mann-Kendall p-value, and cumulative change over each trend window.

Parameter	Period	Slope	Years	Total Change	Unit	p-value
Precipitation	Annual	0.085	75	6.4	%	0.1971
Relative humidity	Annual	0.001	75	0.1	%	0.8620
Snow cover	Annual	-0.053	75	-4.0	%	0.0026
Snowfall	Annual	-0.088	75	-6.6	%	0.2528
Snowfall fraction	Annual	-0.079	75	-5.9	%	0.0078
Snowmelt	Annual	-0.039	75	-2.9	%	0.5279
Day of 50% melt	Annual	-0.150	75	-11.3	day	0.0018
Peak weekly melt rate	Annual	0.101	75	7.6	mm/wk	0.3848

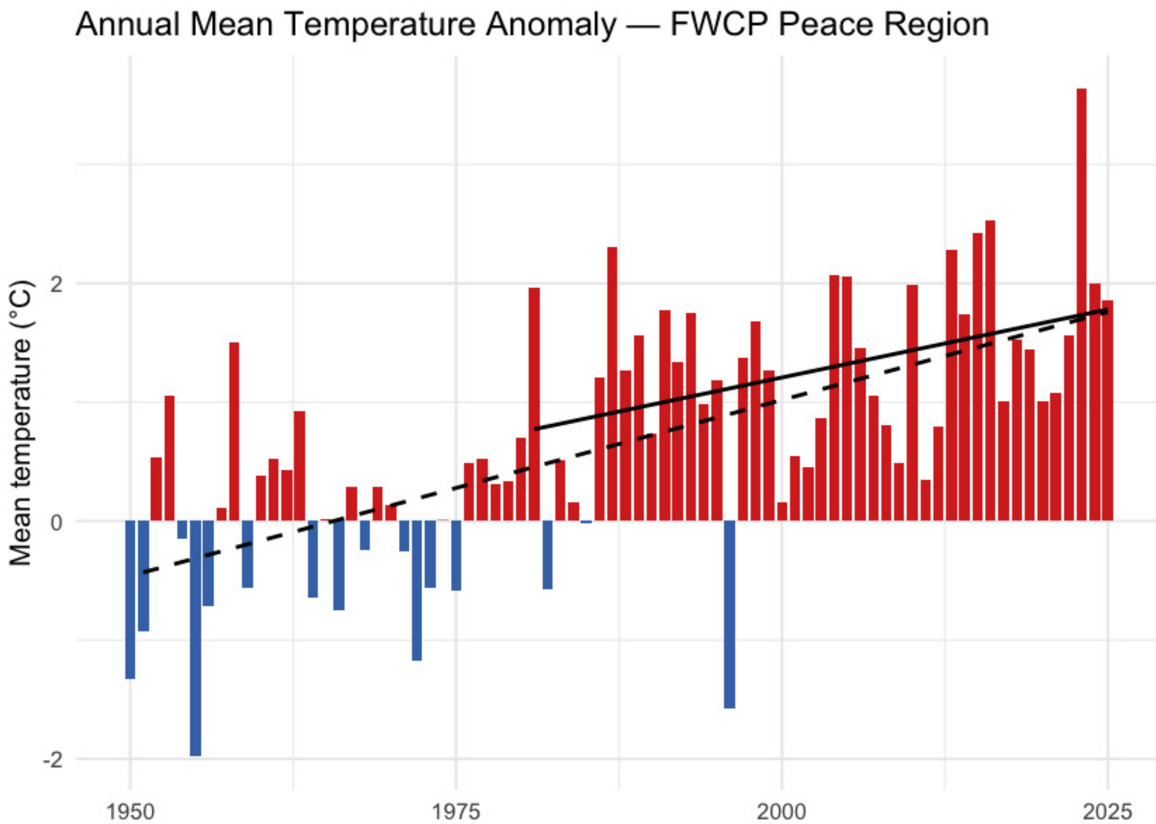


Figure 5.2: Annual mean temperature anomaly for the FWCP Peace Region relative to the 1951-1980 baseline. Bars are degrees above (red) or below (blue) the pre-warming average; the two trend lines are the 75-year (1951-present, dashed) and 45-year (1981-present, solid) Theil-Sen slopes.

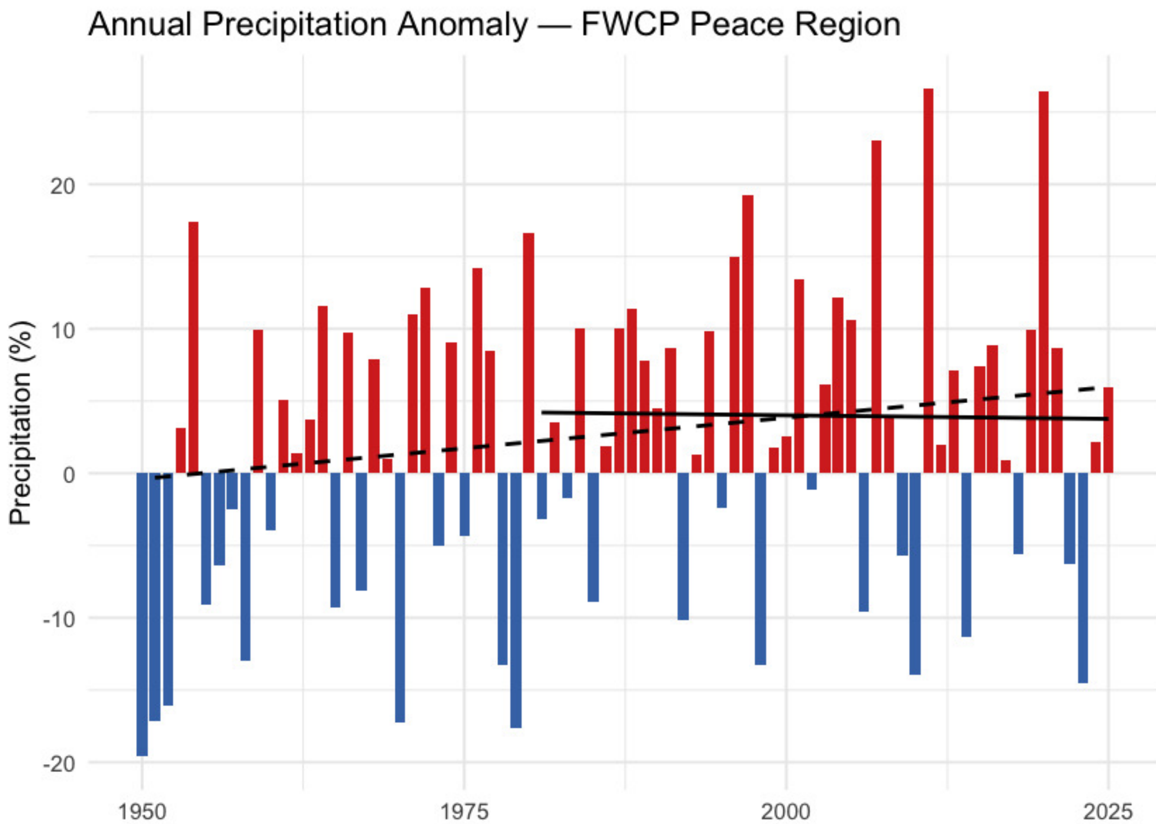


Figure 5.3: Annual precipitation anomaly (% of the 1951-1980 baseline) for the FWCP Peace Region. The long-term trend is positive but not statistically significant at the regional scale; significance does appear in the two northernmost ecoregions (see Per-Ecoregion section).

Daytime highs and overnight lows

The `cd` package ships daytime maximum (tmax) and overnight minimum (tmin) temperatures alongside the daily mean. They carry distinct information. Overnight minimums warming faster than daytime maximums — the "day-night asymmetry" — is one of the textbook fingerprints of greenhouse warming. Karl et al. (1993) documented this first at the global scale, finding overnight minimums rose at roughly three times the rate of daytime maximums between 1951 and 1990.

For the FWCP Peace Region, overnight minimums are warming faster than daytime maximums. Daytime maximums warmed about +0.027 °C per year since 1951 (+2.0 °C cumulative); overnight minimums warmed about +0.032 °C per year (+2.4 °C cumulative). The overnight side warmed roughly 0.4 °C more than the daytime side over the full record, narrowing the diurnal temperature range by the same amount (Figure 5.6).

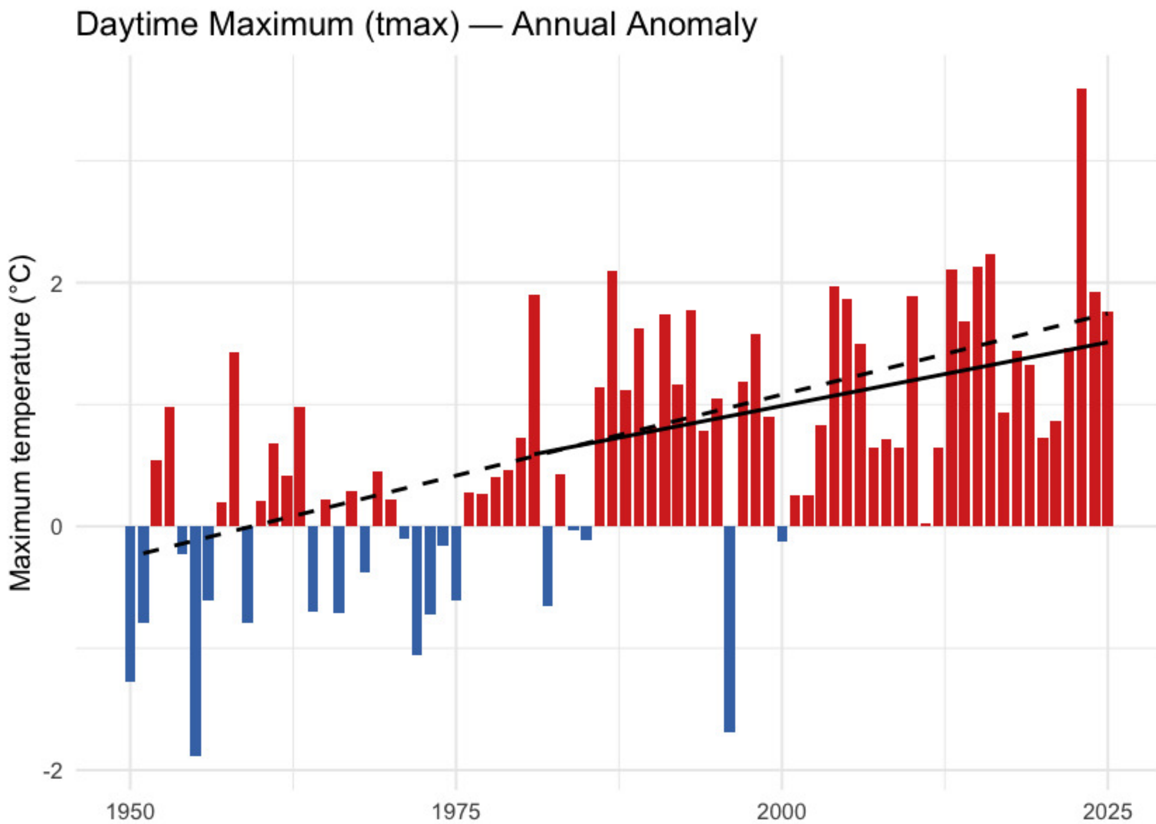


Figure 5.4: Annual daytime maximum temperature (tmax) anomaly relative to the 1951-1980 baseline.

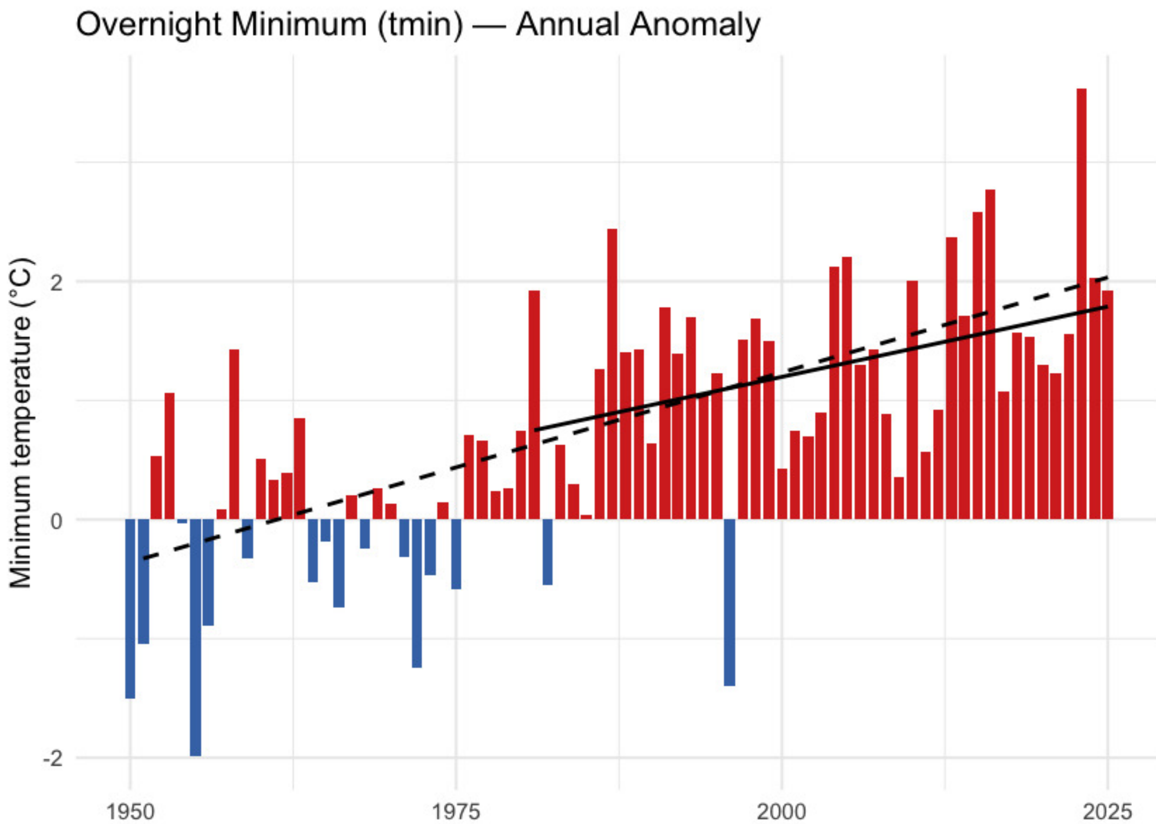


Figure 5.5: Annual overnight minimum temperature (tmin) anomaly relative to the 1951-1980 baseline.

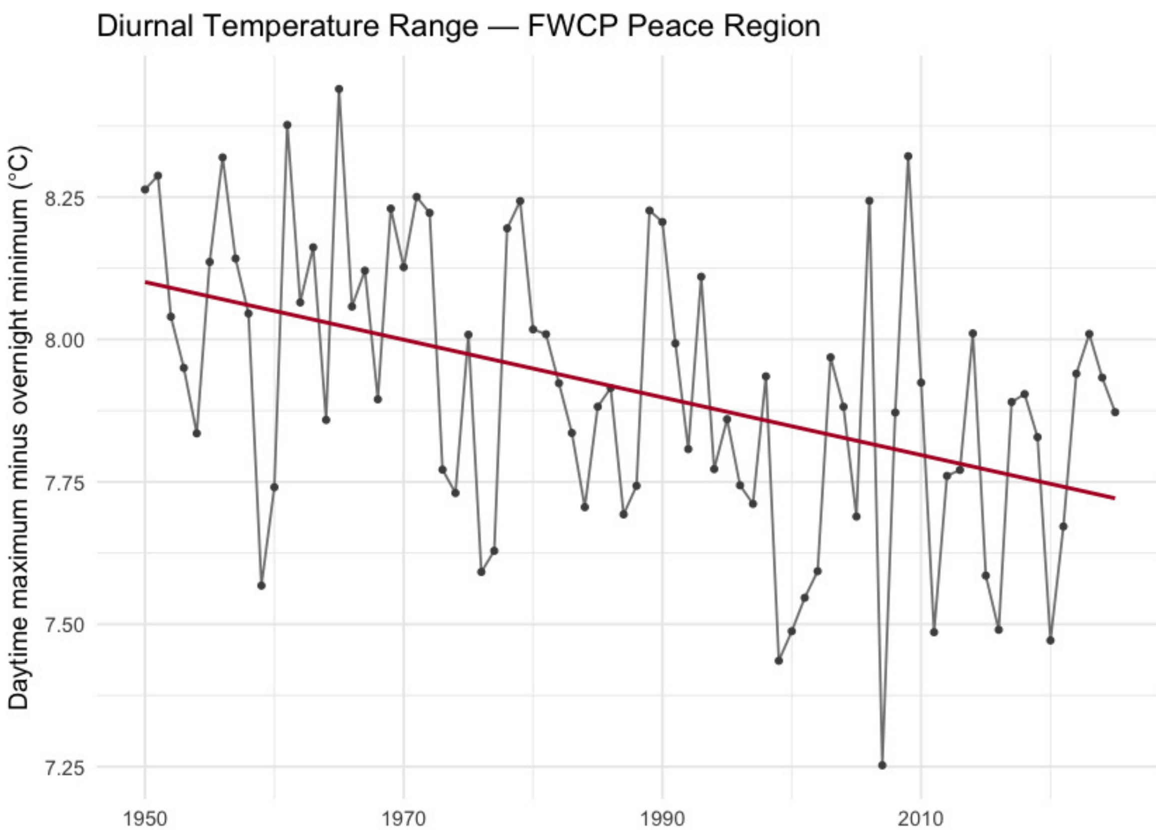


Figure 5.6: Diurnal temperature range (daytime maximum minus overnight minimum) annual mean for the FWCP Peace Region. The downward slope is the day-night asymmetry — overnight lows are warming faster than daytime highs.

Snowpack

Snowpack drives the seasonal water supply across British Columbia: winter precipitation falls as snow, accumulates on the ground, and releases as meltwater across spring and summer. That seasonal storage is the difference between a late-summer creek that still flows and one that does not. It also sets the timing of the spring freshet — the annual flood pulse that salmonids time their up-river migrations to.

For the FWCP Peace Region the snowpack signal is unambiguous and concentrated in the warm shoulders of the year. Annual snow water equivalent declined about 10 % (135 → 122 mm), but the seasonal breakdown is sharper: **summer snow water equivalent collapsed by 75 %** (21.5 → 5.3 mm) and **spring snowmelt rose 37 %** (212 → 290 mm) as the freshet shifted earlier in the calendar. Total annual snowfall barely changed (-6 %) — the snow water equivalent decline is mostly about warming removing snow earlier, not less snow falling.

Seasonal snowpack curve

Aggregated to the four standard meteorological seasons — winter (December–February), spring (March–May), summer (June–August), and fall (September–November) — the table below compares the recent decade (2015–2025) directly against the pre-warming reference (1951–1980) for each of the four monthly-native snow variables. The headline numbers (summer snow water equivalent collapse, spring snowmelt rise) are in the **summer** and **spring** rows.

Table 5.5: Seasonal snowpack: recent decade versus pre-warming reference, FWCP Peace Region. Summer snow water equivalent collapse (-75 %) and spring snowmelt rise (+37 %) carry the headline signals.

Variable	Season	Pre-warming (1951–1980)	Recent (2015–2025)	Δ %
Snow water equivalent (mm)	winter	195.9	192.6	-1.7
Snow water equivalent (mm)	spring	292.8	259.6	-11.4
Snow water equivalent (mm)	summer	21.5	5.3	-75.5
Snow water equivalent (mm)	fall	30.2	29.1	-3.6
Snowfall (mm)	winter	176.3	170.0	-3.6
Snowfall (mm)	spring	109.0	94.1	-13.7
Snowfall (mm)	summer	12.0	4.9	-59.5
Snowfall (mm)	fall	135.0	136.6	1.2
Snowmelt (mm)	winter	1.1	1.4	37.2
Snowmelt (mm)	spring	212.2	289.9	36.6
Snowmelt (mm)	summer	165.0	63.8	-61.3
Snowmelt (mm)	fall	30.9	25.9	-16.2

Annual snowpack signals

Four derived annual scalars capture the snowpack signals the literature treats as headline metrics for snow hydrology: peak snowpack (Mote et al. 2018, 2005), the date of melt midpoint (Stewart et al. 2005; Cayan et al. 2001), freshet flashiness, and the fraction of precipitation falling as snow (Knowles et al. 2006).

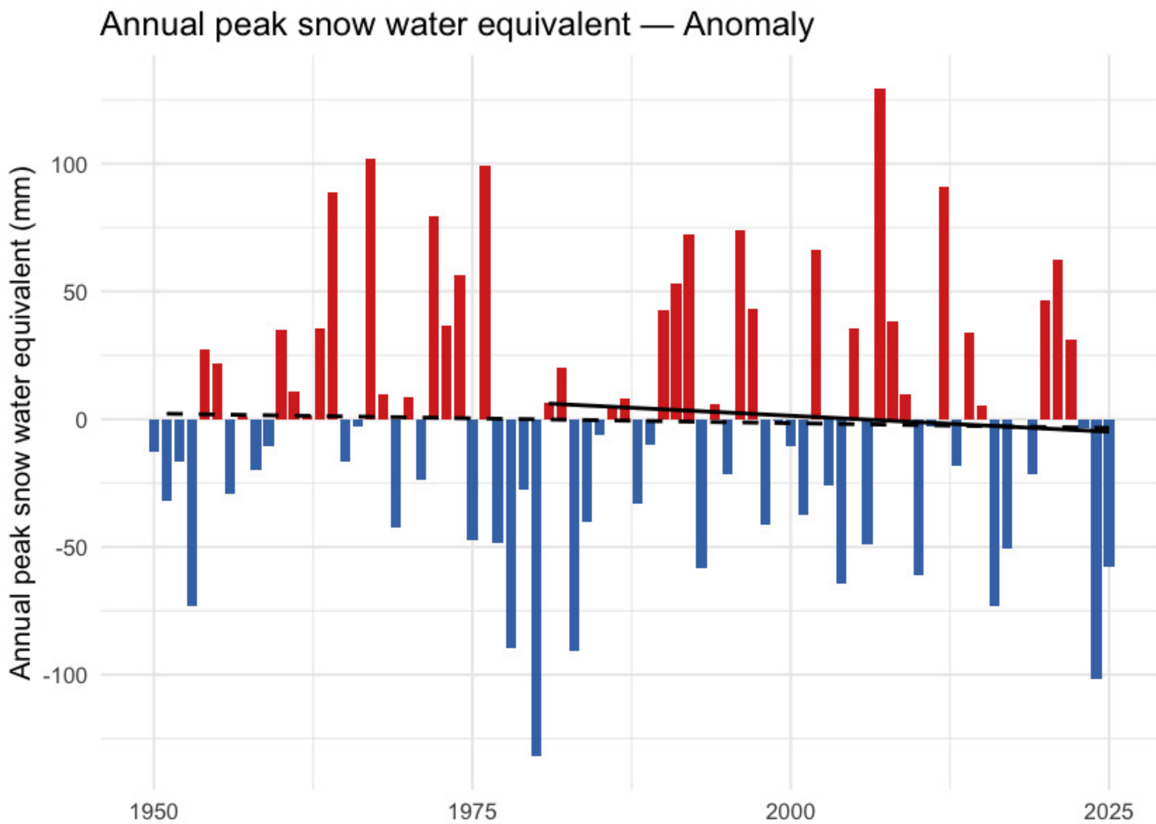


Figure 5.7: Annual peak snow water equivalent (swe_max) for the FWCP Peace Region. ERA5-Land mm snow water equivalent, regional spatial mean.

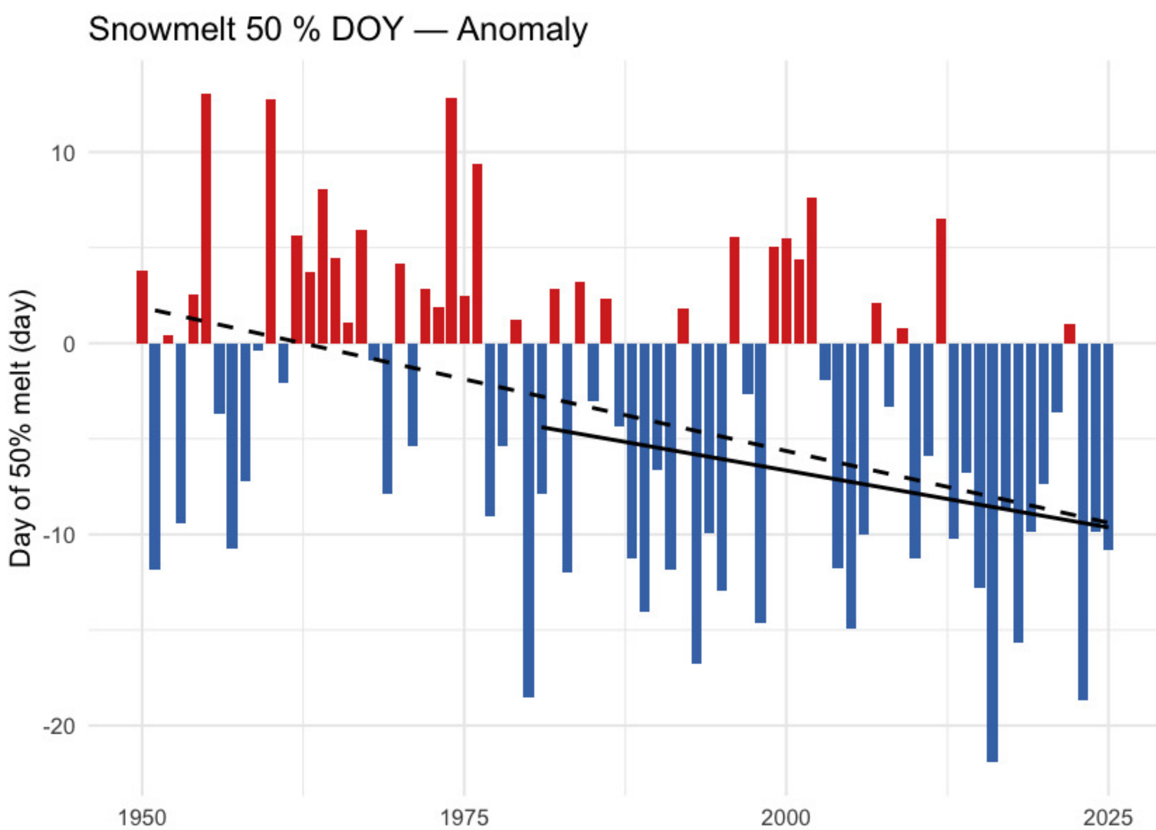


Figure 5.8: Day of year when half the annual snowmelt has accumulated (snowmelt_doy_50). Earlier dates indicate an earlier freshet centroid — directly relevant to crossing-design hydrology and to migration timing.

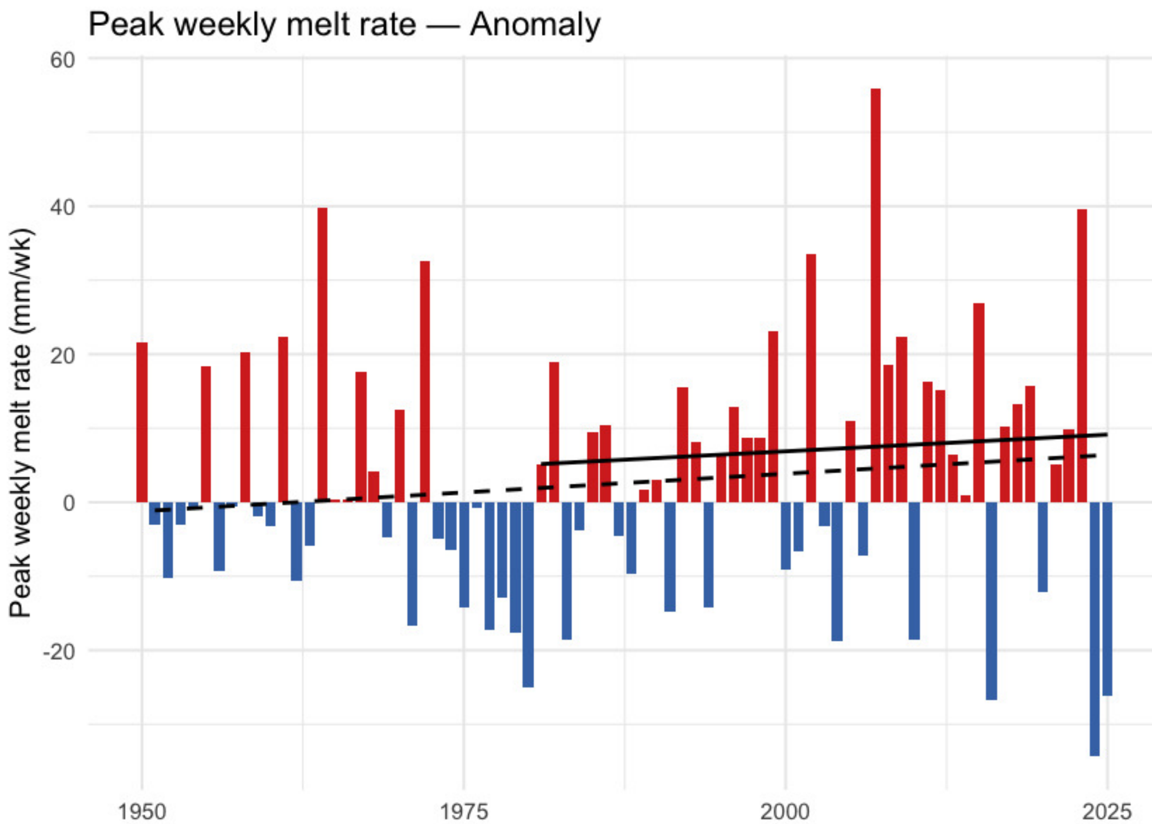


Figure 5.9: Annual maximum of 7-day rolling daily snowmelt (snowmelt_rate_peak). Higher values indicate more concentrated freshet pulses on the snowpack side, upstream of routing through soil and channel storage.

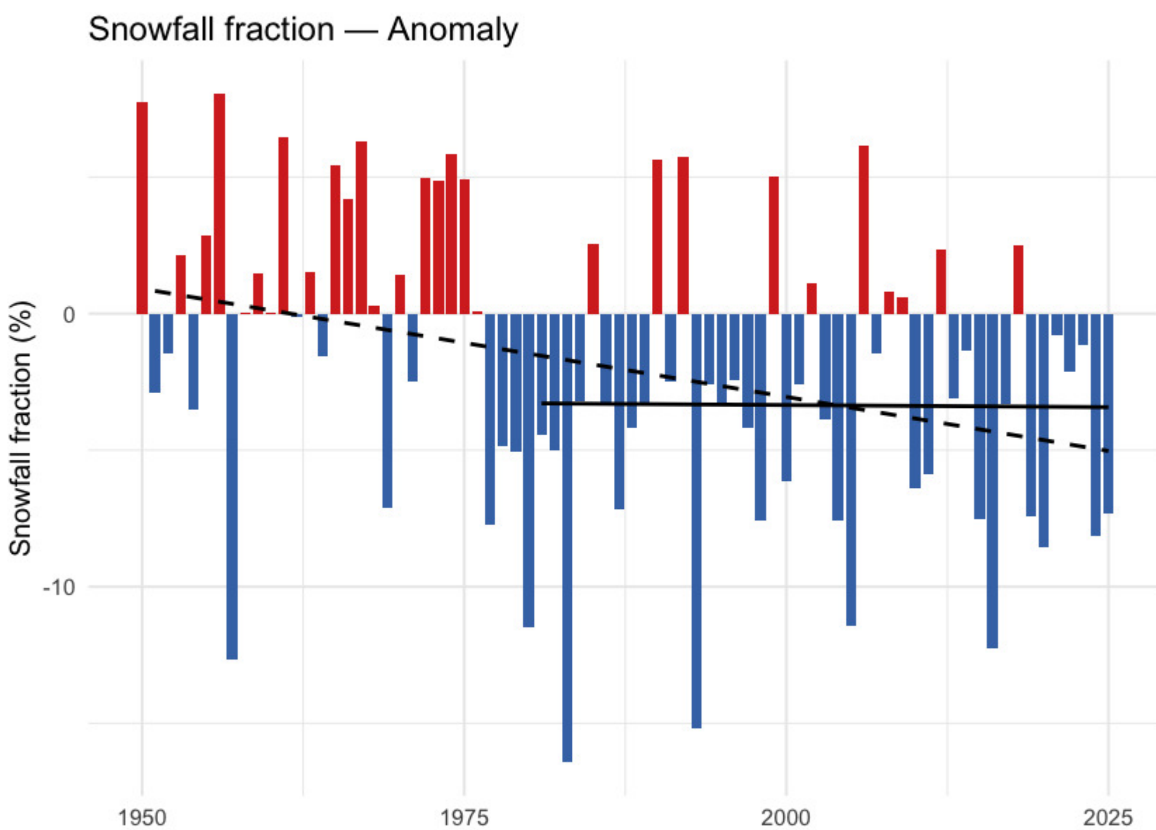


Figure 5.10: Annual snowfall fraction: the percent of annual precipitation that fell as snow. Anomaly is in percentage points relative to the 1951-1980 baseline.

What this means for the FWCP Peace Region

Snow is leaving the region earlier each year, not falling less. Total annual snowfall is roughly stable (-6 %); the snowpack decline is dominated by *earlier* melt, not by *less* snow. The seasonal data make this concrete: spring snowmelt is up 37 % while summer snowmelt is down 61 % — same total melt, redistributed earlier into the year. This matches the broader Pacific-Northwest signal documented since Mote et al. (2005).

The freshet is shifting into spring. Spring snowmelt up 37 % is direct evidence of an earlier freshet centroid — consistent in direction and rough magnitude with the 1–4 week earlier streamflow timing across western North America in Stewart et al. (2005), and with the 10-day Fraser freshet advance in Kang et al. (2016) for a basin whose southern boundary is just south of the Peace.

Summer is becoming snow-free. Summer snow water equivalent has collapsed by 75 % and summer snowmelt is down 61 %. The high-elevation snowpack that historically lingered into summer no longer does in the recent decade. For aquatic ecosystems downstream this is a loss of late-season cold-water input to streams during the warmest part of the year — the weeks where stream-temperature stress is highest for cold-water salmonids.

Spatial pattern of warming

The regional zonal mean reduces a region this size to a single number; the spatial pattern carries the rest of the story.

Warming is not spatially uniform. Total departures range from about +1.5 °C in the southeast to over +2.1 °C in the west and centre (Figure 5.11). The dominant gradient runs east-to-west — the western half of the region averages roughly 0.2 °C more warming than the eastern half — with a smaller south-to-north component. The west-of-Rockies pattern is consistent with windward-slope amplification along the Continental Divide (Pepin et al. 2015).

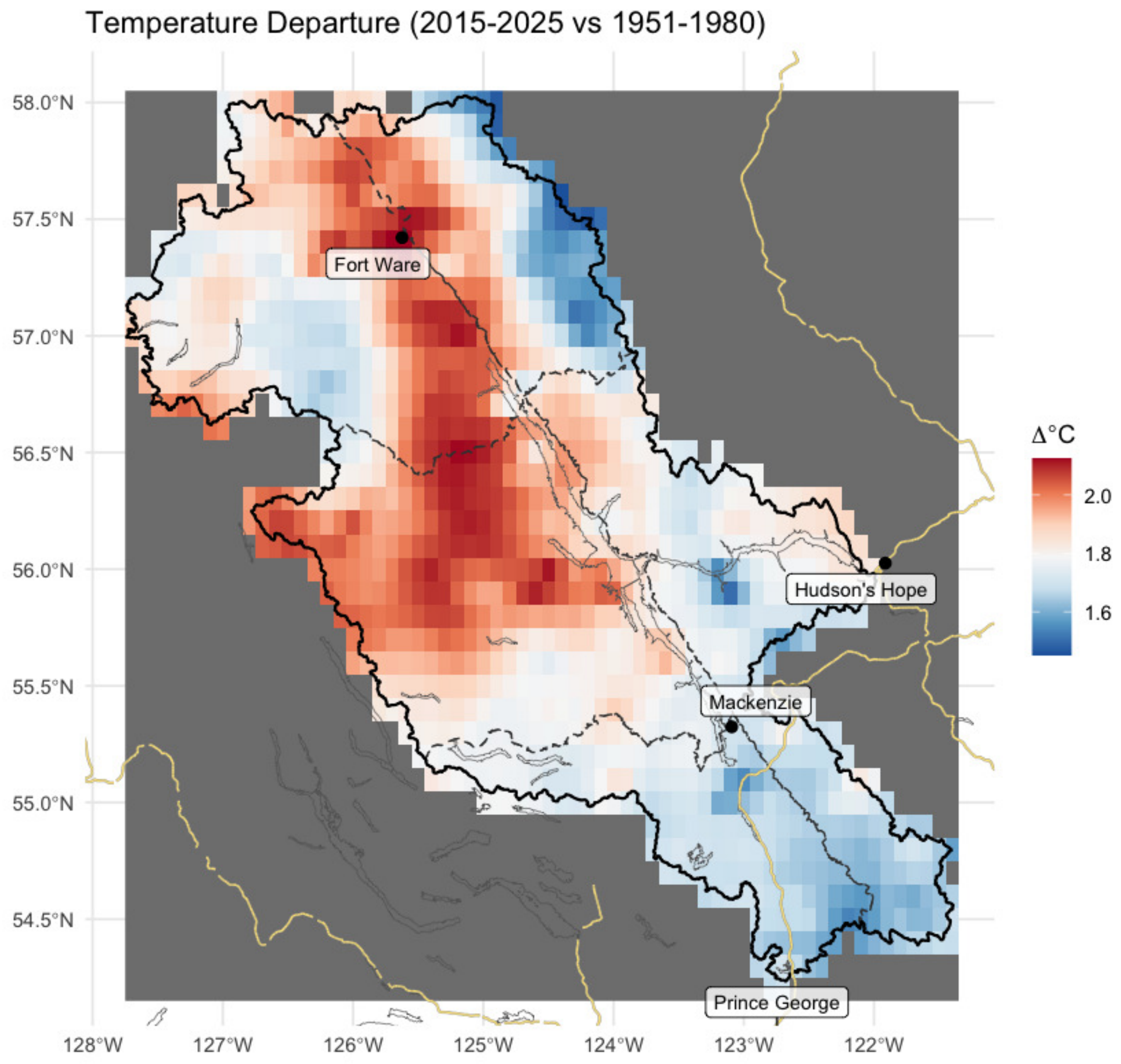


Figure 5.11: Spatial pattern of annual mean temperature departure across the FWCP Peace Region (2015-2025 mean minus 1951-1980 mean, degrees C). Warming is broad but uneven — the west-of-Rockies pattern is consistent with windward-slope amplification along the Continental Divide.

Per-ecoregion variation

The regional zonal mean averages over five ecoregions with different elevations and exposures. Running the same pipeline on each ecoregion polygon tests whether the regional story holds within each ecoregion — and where it does not.

All five ecoregions warmed at roughly the same rate. Precipitation is the variable that diverges: the two northernmost ecoregions (Boreal Mountains and Plateaus, Northern Canadian Rocky Mountains) show statistically significant precipitation increases, while the southern and central ecoregions do not. Vapour pressure deficit is up significantly in all five.

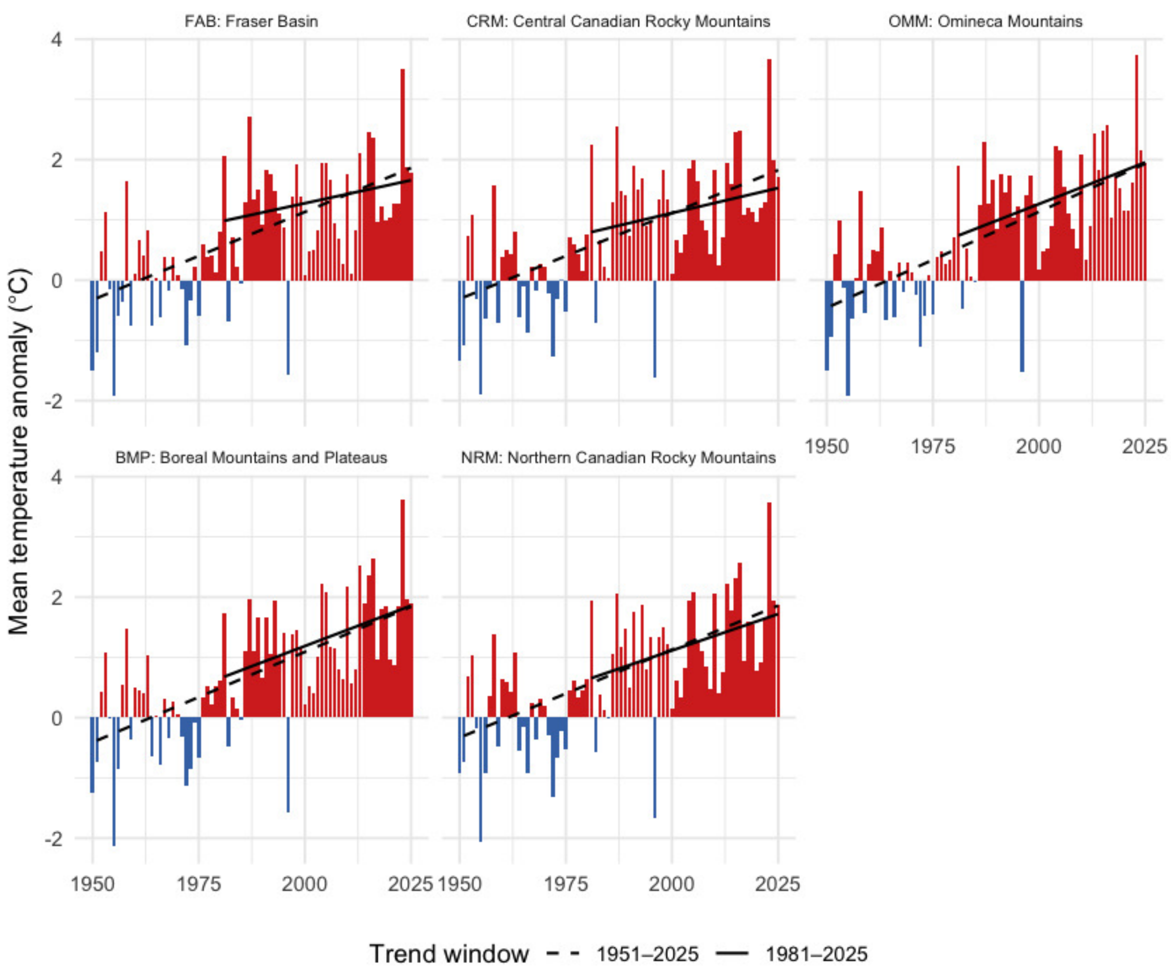


Figure 5.12: Annual mean temperature anomaly relative to the 1951-1980 baseline, by ecoregion. Dashed line is the 75-year Theil-Sen trend (1951-present); solid line is the 45-year trend (1981-present). A solid line steeper than the dashed line indicates accelerating warming.

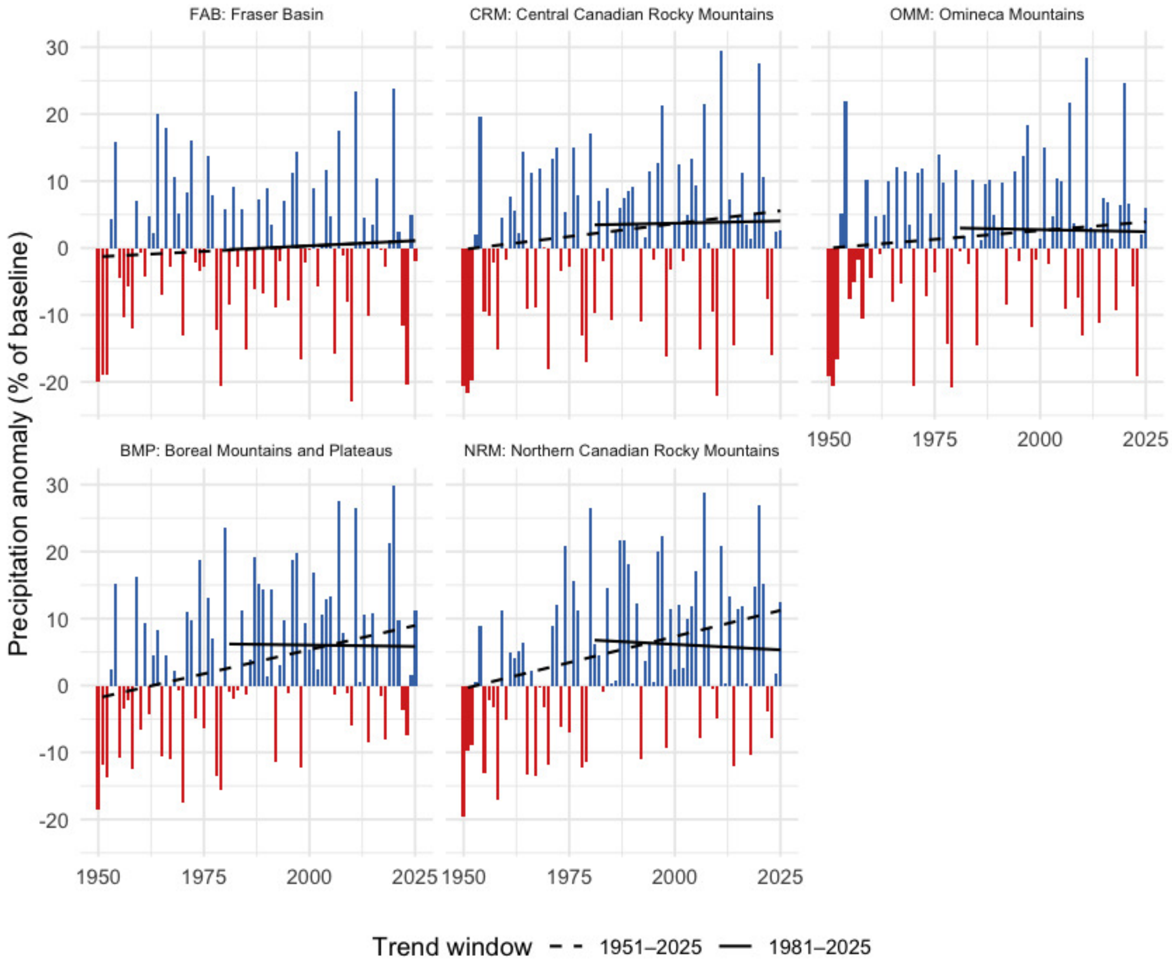


Figure 5.13: Annual precipitation anomaly (% of the 1951-1980 baseline) by ecoregion. The two northernmost ecoregions (BMP, NRM) show statistically significant precipitation increases over the full record; the others do not.

Snow by ecoregion

Two snow metrics are worth viewing per-ecoregion: peak snow water equivalent (annual snowpack magnitude) and DOY-50 (the day-of-year by which half the year's snowmelt has already happened — earlier values mean an earlier freshet). Pair them with the Watershed Groups Across Ecoregions section below to read each watershed group's dominant ecoregion signal.

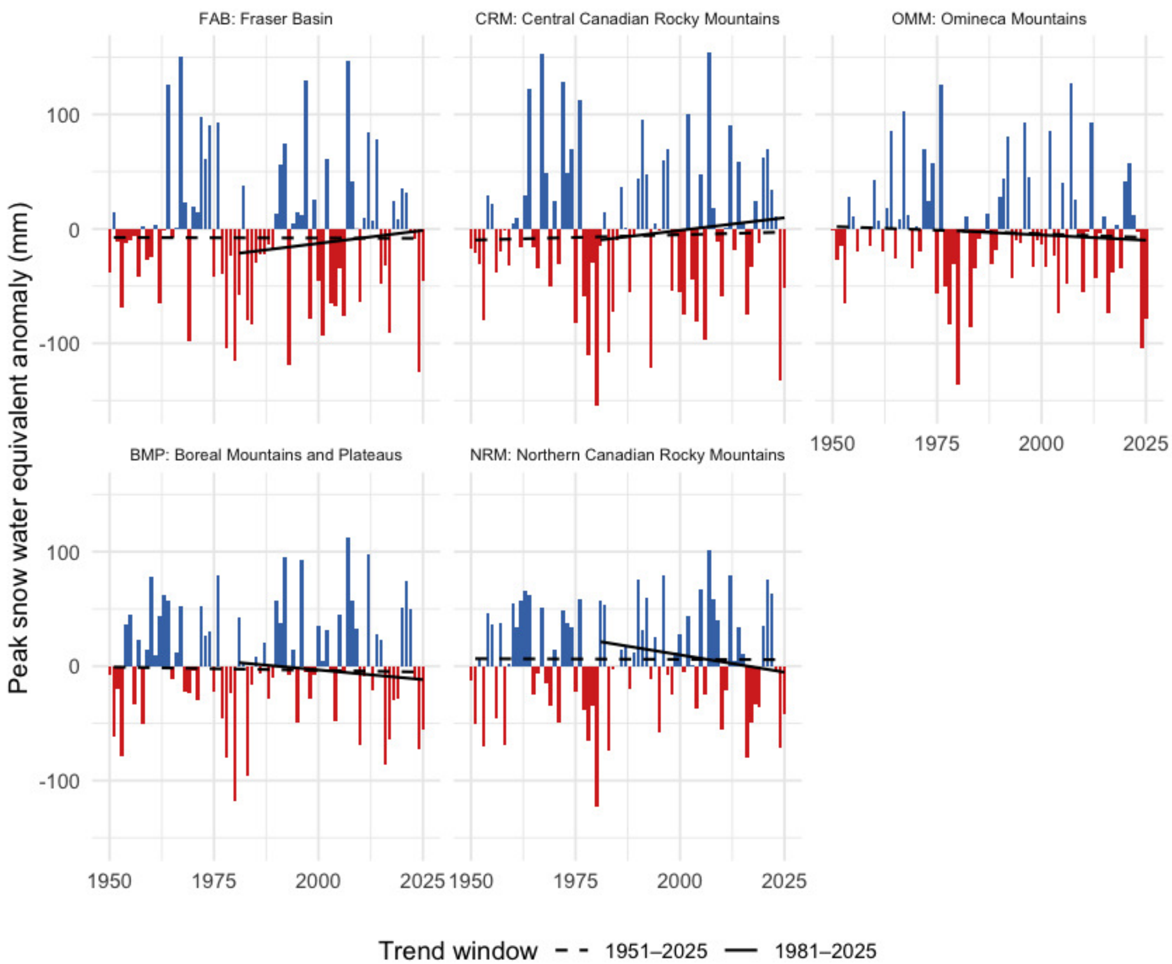


Figure 5.14: Annual peak snow water equivalent anomaly by ecoregion, relative to the 1951-1980 baseline. Bars are mm of water-equivalent snowpack departure from baseline.

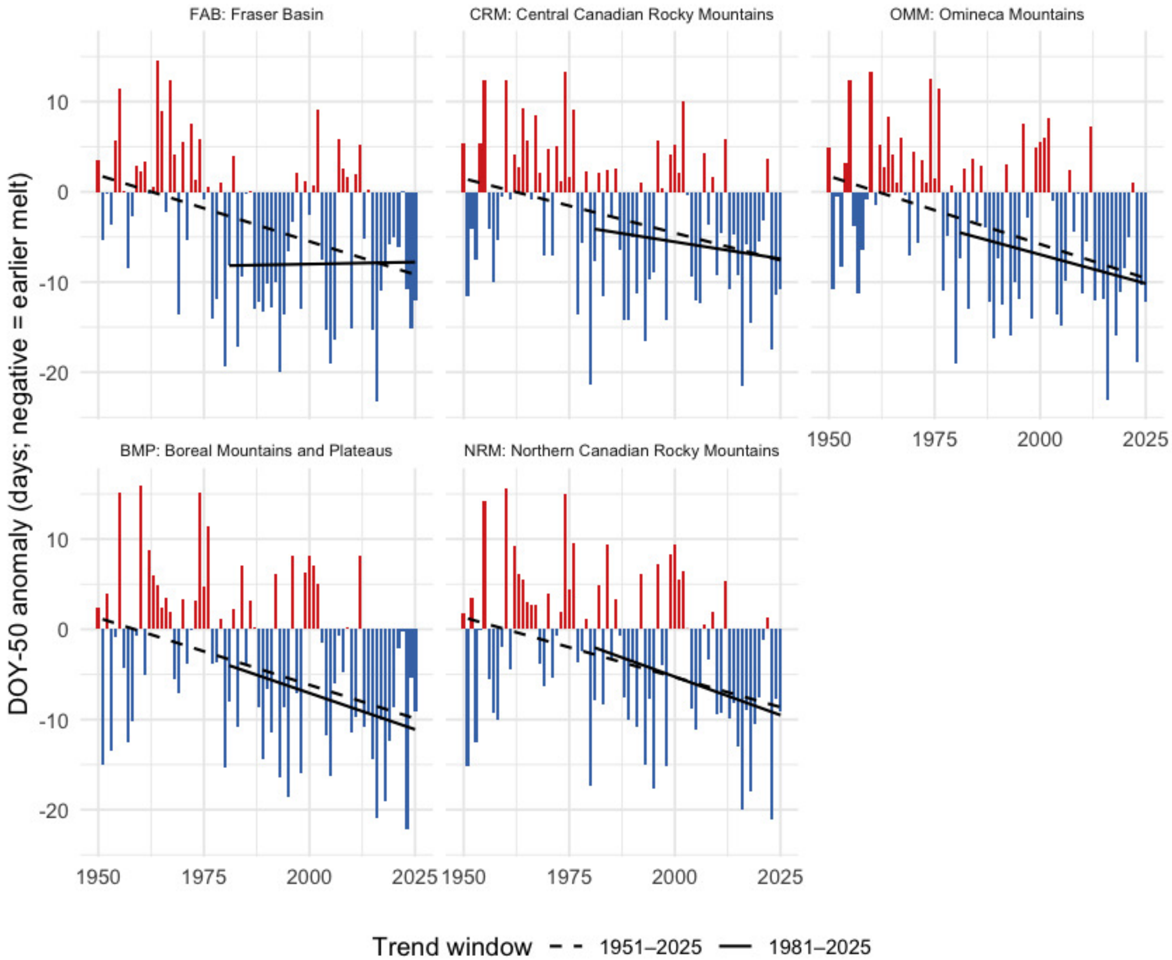


Figure 5.15: Snowmelt 50 % day-of-year (DOY-50) anomaly by ecoregion, relative to the 1951-1980 baseline. Negative values (red) mean the freshet midpoint shifted earlier in the year.

Table 5.6: Per-ecoregion roll-up over the 75-year window (1951-present): annual mean / daytime maximum / overnight minimum temperature trends (°C per decade); annual precipitation trend (mm per year) and Mann-Kendall p-value; annual vapour pressure deficit trend (hPa per decade) and p-value; and recent (2015-2025) versus pre-warming (1951-1980) percent change for precipitation and soil moisture. All temperature p-values are below 0.001. A tmin slope greater than the tmax slope indicates the day-night asymmetry.

Ecoregion	tmean °C/dec	tmax °C/dec	tmin °C/dec	prcp mm/yr	prcp p	vpd hPa/dec	vpd p	prcp % change	soil moisture % chg
FAB	0.29	0.26	0.30	0.032	0.634	0.061	0.002	0.8	-1.3
CRM	0.29	0.25	0.30	0.077	0.253	0.045	0.001	4.1	-0.5
OMM	0.32	0.29	0.34	0.052	0.421	0.054	0.000	2.5	-0.7
BMP	0.30	0.27	0.33	0.144	0.023	0.033	0.003	6.3	0.3
NRM	0.29	0.26	0.31	0.156	0.015	0.030	0.004	6.6	0.6

Ecoregion codes: FAB — Fraser Basin; CRM — Central Canadian Rocky Mountains; OMM — Omineca Mountains; BMP — Boreal Mountains and Plateaus; NRM — Northern Canadian Rocky Mountains.

Watershed groups across ecoregions

The FWCP Peace Region is reported and managed at the watershed- group scale — the British Columbia Freshwater Atlas hydrological reporting unit, and the unit the FWCP funds work in. Sixteen watershed groups make up the canonical FWCP Peace list. They span the five ecoregions in different ways: some sit entirely within one, others split across two or three (Figure 5.16). The table that follows gives, for each watershed group, the share of its area falling in each ecoregion — the cross-reference that maps the regional and ecoregion climate signals onto the per-watershed prioritisation unit.

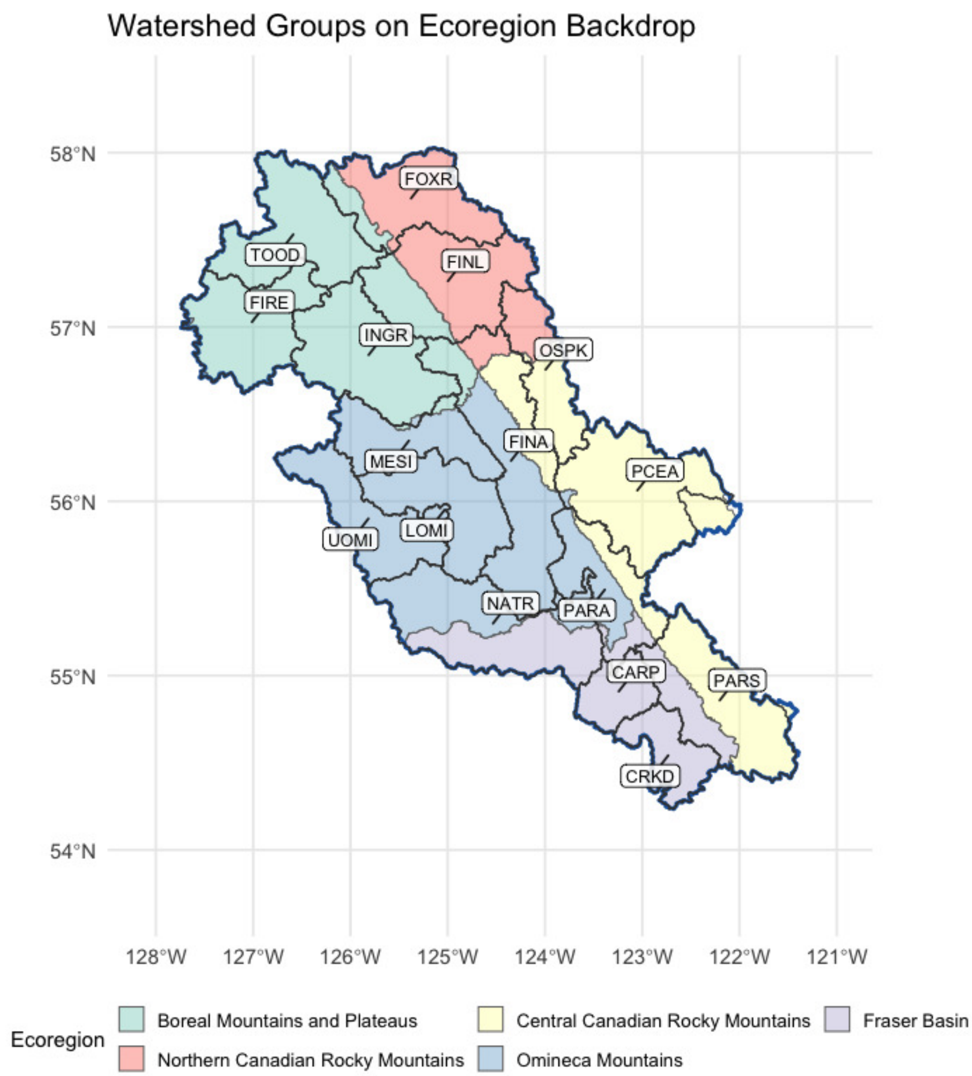


Figure 5.16: Watershed groups intersecting the FWCP Peace Region (full extent, including parts spilling outside the FWCP boundary), labelled with their codes, on top of ecoregion fills.

Table 5.7: Watershed groups in the FWCP Peace Region with the percent of each group's area falling in each of the five ecoregions. Rows where one ecoregion is at or near 100 % indicate watershed groups contained within a single climate-physiography zone; rows split across two or more ecoregions indicate watershed groups whose climate departure can pull from more than one regional pattern.

Code	Name	km² in FWCP	FAB %	CRM %	OMM %	BMP %	NRM %
CARP	Carp Lake	1794	100.0	0.0	0.0	0.0	0.0
CRKD	Crooked River	2172	100.0	0.0	0.0	0.0	0.0
FINA	Finlay Arm	7383	0.3	20.2	61.8	10.3	7.4
FINL	Finlay River	5479	0.0	0.0	0.0	30.4	69.6
FIRE	Firesteel River	4376	0.0	0.0	1.0	99.0	0.0
FOXR	Fox River	4279	0.0	0.0	0.0	16.8	83.2
INGR	Ingenika River	5328	0.0	0.0	0.7	99.3	0.0
LOMI	Lower Omineca River	4013	0.0	0.0	100.0	0.0	0.0
MESI	Mesilinka River	3294	0.0	0.0	94.8	5.2	0.0
NATR	Nation River	6889	49.3	0.0	50.7	0.0	0.0

Interpretation for fish passage

Four findings tie the regional climate-departure analysis to per-watershed barrier prioritisation.

Warming is broad and uniform; the gradient is small. All five ecoregions warmed +1.7 to +1.9 °C since 1951 — temperature is not the variable that *separates* watershed groups within the Peace. What it *does* set is the regional context: every reach in the project area is operating in air ~1.8 °C warmer than the conditions the channel, floodplain, and fish community evolved under. The within-region thermal differentiation comes from elevation, aspect, riparian cover, and groundwater contribution — not from climate-departure magnitude.

Precipitation is increasing significantly only in the two northernmost ecoregions. Watershed groups dominated by Boreal Mountains and Plateaus (Toodoggone, Firesteel, Ingenika) or Northern Canadian Rocky Mountains (Finlay, Fox) sit on statistically defensible positive precipitation trends. The other eleven watershed groups sit in Fraser Basin, Central Canadian Rockies, or Omineca Mountains, where precipitation has not changed in a way distinguishable from natural variability. For barrier prioritisation in those eleven, the climate-driven hydrology story is dominated by *atmospheric drying* (rising VPD, flat soil moisture, snowpack timing) rather than precipitation magnitude.

The atmosphere is drying despite, in places, more precipitation. Vapour pressure deficit rose significantly in every ecoregion. Soil moisture is flat across the region even where precipitation rose

— the warmer atmosphere is pulling the extra water back through evaporation and transpiration. The ecological implication is that streamflow during the summer low-flow period — the thermal-stress weeks where cold-water rearing habitat is at the margin — is not getting the relief that the precipitation increase, where it occurred, would otherwise suggest.

Snow is leaving the region earlier, not falling less. The snowpack-timing signal is regionally uniform — DOY-50 shifted earlier in every ecoregion. The freshet — the dominant high-flow event that mobilises spawning gravels, refills off-channel rearing habitat, and that crossing structures are sized against — is arriving weeks earlier across all 16 watershed groups. Summer snow water equivalent collapsed 75 %; the cold-water input that high-elevation snowpack provides to streams during the warmest weeks of summer is dropping in parallel. Both signals act on barrier prioritisation through hydrology and stream temperature rather than through which watershed they hit hardest.

For the cold-water resident salmonids the FWCP Peace supports — bull trout, Arctic grayling, mountain whitefish, rainbow trout, kokanee — the implication for barrier prioritisation is that two fundamentally different climate-departure stories play out across the 16 watershed groups, and that the priority weight on barrier removal should respond to both: warmer headwater reaches that gain growing-degree-days have *more* habitat to recover above the barrier, while reaches sitting closer to the upper thermal niche have less. The watershed-group ecoregion mapping in the table above is the bridge between the regional climate-departure summary in the main body of this report and the per-watershed weight applied to each barrier.

Appendix - Floodplain Delineation

This appendix details the floodplain delineation for the Parsnip River Watershed Group. The rationale for mapping floodplain extent alongside the fish passage barrier inventory is described in the report background — in brief, the off-channel habitats that floodplains sustain are often as important to fish productivity as the mainstem passage that crossing remediations restore, and infrastructure corridors can compromise floodplain function in ways that structure replacement alone does not address.

Floodplain footprints were delineated using the `flooded` R package, which implements the Valley Confinement Algorithm of Nagel et al. (2014). The algorithm combines a digital elevation model with the stream network and a modelled bankfull flood surface to distinguish valley bottoms from confined canyon reaches. Bankfull depth — the flow depth at which a stream begins to spill onto its floodplain — is estimated from upstream watershed area and mean annual precipitation using the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams.

We mapped the **functional floodplain**: the area a river actively inundates and reworks during regular high-flow events. This sits between two other scales — the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain matters most for habitat planning because it is the ground where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Inputs were the MRDEM-30 30 m DEM, the bull trout accessible stream network (stream order ≥ 3) from bcfishpass, and BC Freshwater Atlas lakes and wetlands on that network. Waterbodies were included so the valley confinement model fills cells that gradient and cost-distance masks would otherwise exclude.

Table 5.8: Parsnip River Watershed Group — modelled functional floodplain and the off-channel habitat units within it.

Metric	Value
Watershed group area (km²)	5,597
Floodplain area (ha)	48,116
Floodplain as % of watershed group	8.6
Bull trout accessible streams, order 3+ (km)	2,310
Streams within floodplain (km)	1,580
Lakes within floodplain (count)	198
Lakes within floodplain (ha)	1,603
Wetlands within floodplain (count)	464
Wetlands within floodplain (ha)	6,582

The Parsnip River Watershed Group covers roughly 5,597 km². Modelled functional floodplain (Table 5.8) covers about 48,116 ha, or 8.6 % of the watershed group, attached to 1,580 km of the 2,310 km bull trout accessible order-3+ network. Within that footprint sit 198 lakes and 464 mapped wetland polygons — the off-channel habitat units that lateral floodplain connectivity supports.

The watershed-wide footprint is shown in Figure 5.17; a detail view of the south-east headwaters around Arctic Lake is shown in Figure 5.18.

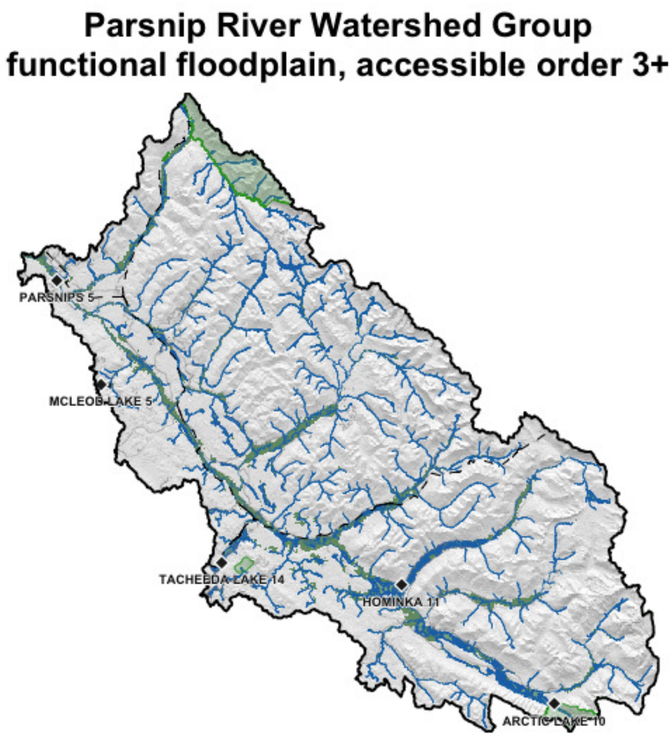


Figure 5.17: Parsnip River Watershed Group functional floodplain (green) over hillshaded terrain. Bull trout accessible order-3+ streams in blue; lakes and wetlands in light blue; parks light green; First Nations reserves light grey with diamond markers and labels; municipalities outlined in red; forest service / resource roads grey; railways black dashed; watershed boundary heavy black.

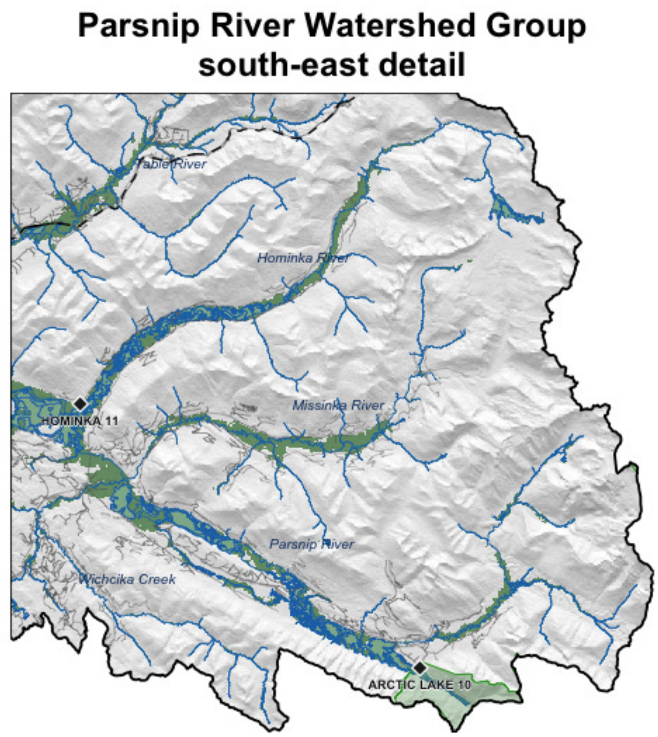


Figure 5.18: South-east corner of the Parsnip River Watershed Group at full resolution — the headwaters around Arctic Lake on the continental divide. Symbology as Figure 5.17; named streams labelled in italic blue; municipalities outlined and labelled in red where present.

The watershed-wide view shows the floodplain following the trunk Parsnip River north toward Williston Reservoir, with substantial off-channel area in the broader valley reaches and at tributary confluences. The detail view resolves the per-reach floodplain pattern in the Arctic Lake headwaters on the continental divide and gives a sense of the resolution available within any sub-area for downstream analyses — barrier prioritisation, land-cover change tracking, or restoration scoping. In coming years we plan to extend this mapping to additional FWCP Peace watersheds so that floodplain context can inform barrier prioritisation across the region.

Appendix - UAV Imagery

UAV (drone) imagery products generated for the Peace project area since 2020 are stored as Cloud Optimized GeoTIFFs on AWS. Per-image tile-service viewer and direct download links are provided online within the UAV Imagery appendix of this report found here.

Appendix - Collaborative GIS Layers

Background spatial layers and tables loaded to the collaborative GIS project (sern_peace_fwcp_2023) at the time of writing are listed online within the Collaborative GIS Layers appendix of this report found here. The project serves as the spatial integration layer for fieldwork — viewable in QGIS in the office and synced to mobile devices for navigation and standardized data collection in the field via Mergin Maps.

Appendix - Parsnip River Habitat and Connectivity Modelling

For any stream, we want to know three things: can a fish get there, is it good habitat, and what — if anything — blocks the way. We model this for every segment of the stream network in a watershed group, and for each species we work out:

- **Access** — can the fish reach the segment, or is the route downstream likely too steep (gradient), or cut off by a barrier such as a dam, waterfall, or perched culvert?
- **Habitat** — if it is reachable, is the segment likely large enough (channel width) and its gradient gentle enough for the modelled species to use it as **spawning** habitat, **rearing** habitat, or simply passable water with neither?
- **Barriers** — the most significant obstacle downstream: a **dam**, a **known** barrier that has been assessed in the field (a road culvert, weir, etc.), a **modelled** crossing (predicted from road–stream intersections but not yet field-checked), or one that has since been **remediated** (fixed).

bcfishpass (Hillcrest Geographics) produces this per-segment classification province-wide. We reproduce it, and because we re-express the methodology in our own tools we can re-parameterise the rules to our own values and extend the method to species `bcfishpass` does not yet model.

We demonstrate both here for the Parsnip River Watershed Group: reproducing the bull-trout (`BT`) classification, and a net-new classification for Arctic grayling (`GR`).

The Parsnip River Watershed Group sits between Prince George and Mackenzie, BC. The Parsnip flows north into the southern arm of Williston Reservoir, joining the Peace River system; from there the drainage runs Peace → Slave → Mackenzie, ultimately discharging to the Arctic Ocean via the Mackenzie Delta. Both bull trout and Arctic grayling are cold-water species whose distributions we resolve through gradient, channel width, and access. Bull trout — provincially blue-listed and a COSEWIC species of special concern in the Western Arctic population — spawn as adfluvial migrants in cold, low-gradient tributaries such as the Misinchinka and Anzac. Arctic grayling, at the southern edge of their range in the Williston watershed, hold to cooler, larger (fourth-order and up) clear-water reaches and spawn over fine gravels.

We build on the canonical `fwapg` / `bcfishpass` / `bcfishobs` stack from Hillcrest Geographics — porting the processing methodology into our own `fresh` and `link` packages while still drawing on the canonical `bcfishpass` inputs (crossing overrides and the like). That lets us experiment with the rules for species already modelled and extend to species not yet modelled — here, Arctic grayling — while staying checkable, segment for segment, against the canonical outputs.

Modelling parameters

The main parameters are the maximum gradients for access, spawning, and rearing and a minimum channel width for habitat (we apply further rules too — waterbody and stream-edge types, clustering of rearing with spawning habitat, and others). The values we applied are below.

Table 5.9: Access, spawning, and rearing gradient ceilings and minimum channel widths applied for the two species. Grayling's tighter access ceiling and gradient windows give it the smaller modelled network.

species	access grad max	spawn grad max	rear grad max	spawn CW min (m)	rear CW min (m)
BT	0.25	0.0549	0.1049	2	1.5
GR	0.15	0.0249	0.0349	4	1.5

Reproducing bcfishpass (parity)

We compare our per-segment classification against the `bcfishpass` snapshot, segment by segment. Of the two species, only bull trout occurs in the watershed group, so this is a single-species comparison.

Table 5.10: Per-segment classification parity for bull trout, our values vs the bcfishpass snapshot.

species	segments	match %	n diffs
BT	43120	99.04	416

We reproduce **99.04%** of bcfishpass's per-segment bull-trout classification across 43,120 segments, with 416 disagreements, consistent with the ~99.7% study-area median we see across the Peace.

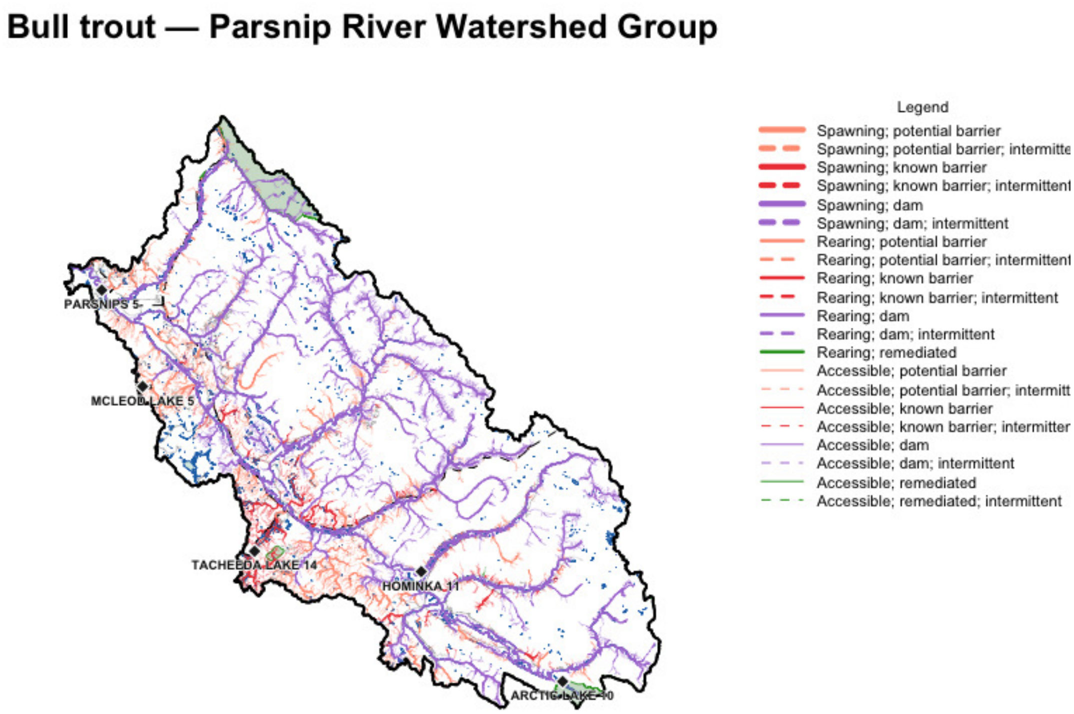


Figure 5.19: Bull-trout habitat and connectivity classification across the Parsnip River Watershed Group. Each stream segment is coloured by the most significant barrier downstream and weighted by habitat value; the legend gives the full key.

Arctic grayling — extending the method

`bcfishpass` does not yet model Arctic grayling, so this is net-new output — there is nothing to compare against. We apply the same method, re-parameterised for grayling, and map the result with the same symbology so the two species are directly comparable.

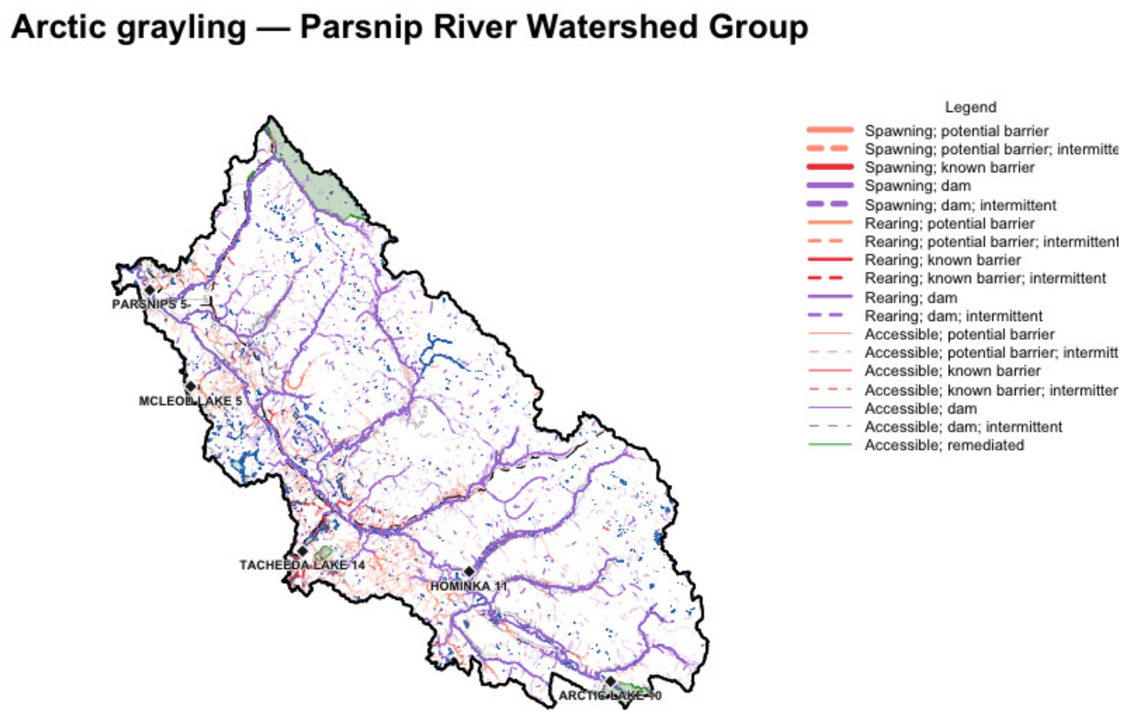


Figure 5.20: Arctic grayling habitat and connectivity classification across the Parsnip River Watershed Group — a species bcfishpass does not yet model. Same symbology as the bull-trout map, so the two are directly comparable; grayling's modelled network is the smaller of the two.

Maps — detail comparison

The full-watershed views compress a lot of network. Cropping to a sub-reach puts bull trout and grayling side by side at full resolution — the reaches where grayling carries a classification that bull trout does not are where the extension does genuinely new work.

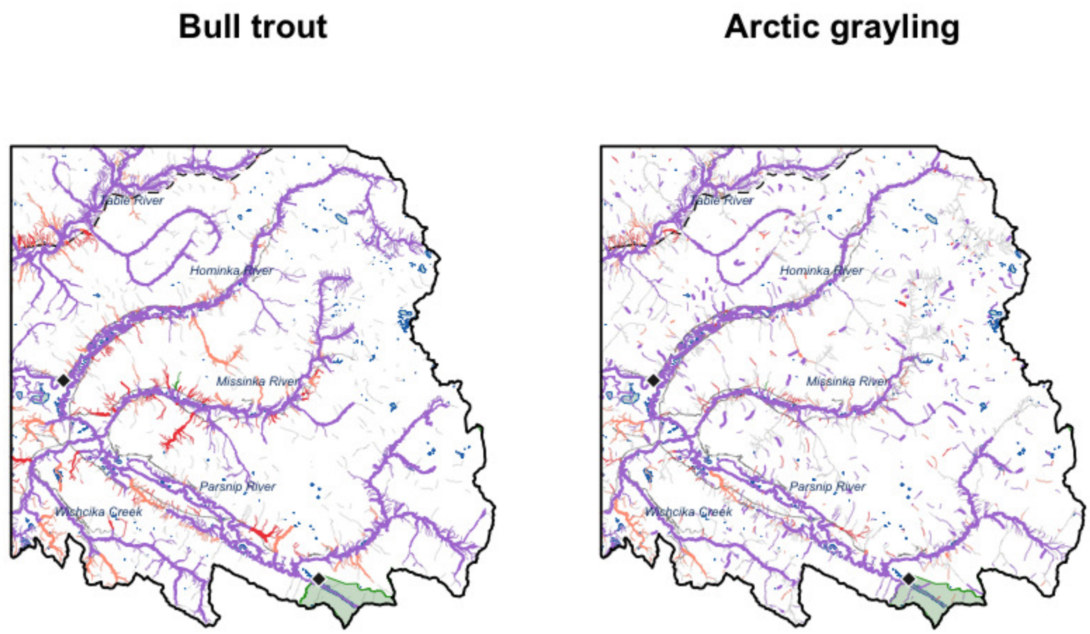


Figure 5.21: South-east corner of the Parsnip River Watershed Group at full resolution: bull trout (left) and Arctic grayling (right), same extent and symbology. Grey background streams are the full modelled network, so the coloured overlay shows where each species' classification reaches.

Appendix - Environmental DNA Results

This appendix contains site-level detail for the 2025 environmental DNA (eDNA) sampling program in the Peace project area. Headline per-species detection counts are presented in the main Results chapter; the tables below show which sites detected which species. Field blanks are tracked separately as quality-assurance samples — they test for protocol contamination, not site eDNA. Office blanks (samples filtered at the field accommodation rather than at site) are excluded from these tables because their UTM coordinates are inherited from related sites and do not represent real sampling locations.

The interactive map of these results is linked from the main Results chapter and at the end of this appendix.

Per-site detection results

Format: `CODE(max_droplets)`, with $*$ flagging samples that UNBC reran. Detected = ≥4 positive droplets per the standard ddPCR call threshold; Sub-threshold = 1-3 droplets, not interpretable as species presence per ddPCR convention; Not detected = 0 droplets across all runs.

Table 5.11: Site-level eDNA detection results across real environmental samples in the Peace project area. UNBC ID is the lab sample identifier; Pos. control = TRUE indicates the sample was collected at a site where the target species was also confirmed by electrofishing.

Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
125000_ds_ed1	Tributary To Parsnip River	M41347		RAIN(980)	—	BULT(0), GRAY(0), SOCK(0)
125000_us_ed1	Tributary To Parsnip River	M41348	–	RAIN(143)	—	BULT(0), GRAY(0), SOCK(0)
125179_ds_ed1a	Tributary To Missinka River	M41349	–	RAIN(309)	BULT(1)*	GRAY(0), SOCK(0)
125179_us_ed1a	Tributary To Missinka River	M41351	yes	RAIN(383)*	—	BULT(0), GRAY(0), SOCK(0)
125180_ds_ed1	Tributary To Missinka River	M41353		BULT(4), RAIN(330)	—	GRAY(0), SOCK(0)
125180_us_ed1	Tributary To Missinka River	M41354		RAIN(293)	BULT(3)	GRAY(0), SOCK(0)
125194_ds_ed1a	Tributary To Missinka River	M21242		—	BULT(1), RAIN(3)*	—
125231_ds_ed1c	Tributary To Table River	M41356	–	RAIN(205)	—	BULT(0), GRAY(0), SOCK(0)
125231_us_ed1	Tributary To Table River	M41357	–	RAIN(5)	—	BULT(0), GRAY(0), SOCK(0)

Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
125261_ds_ed1	Fern Creek	M41358	–	RAIN(258)	—	BULT(0), GRAY(0), SOCK(0)
125261_us_ed1a	Fern Creek	M41359	–	RAIN(533)	—	BULT(0), GRAY(0), SOCK(0)
125261_us_ed1b	Fern Creek	M41360	–	RAIN(567)	—	BULT(0), GRAY(0), SOCK(0)
198692_ds_ed1b	Tributary To Kerry Lake	M41362	–	RAIN(119)	BULT(1)	GRAY(0), SOCK(0)
198692_ds_ed2	Tributary To Kerry Lake	M41363	–	RAIN(630)	—	BULT(0), GRAY(0), SOCK(0)
198692_us_ed1	Tributary To Kerry Lake	M41364	–	RAIN(235)	—	BULT(0), GRAY(0), SOCK(0)
199663_ds_ed1	Tributary To Parsnip River	M10139	–	BURB(195), RAIN(112)	—	BULT(0), GRAY(0)
199663_us_ed1	Tributary To Parsnip River	M10140	–	RAIN(65)	—	BULT(0), BURB(0), GRAY(0)
203597_ds_ed1b	Tributary To Nation River	M41366		RAIN(173)*	BULT(3)*	GRAY(0), *SOCK(0)*
203597_us_ed1	Tributary To Nation River	M41367		RAIN(213)*	—	BULT(0), *GRAY(0)*, SOCK(0)*
203605_ds_ed1	Tributary To Williston Reservoir	M41368	–	RAIN(455)	BULT(3)*	GRAY(0), SOCK(0)
203605_us_ed1	Tributary To Williston Reservoir	M41369	–	RAIN(429)	BULT(1)	GRAY(0), SOCK(0)
24718557_ed1	Missinka River	M41370	–	RAIN(4)	BULT(2), *GRAY(1)*	SOCK(0)
57690_ds_ed1	Tributary To Wichcika Creek	M41371	–	RAIN(10)	—	BULT(0), GRAY(0), SOCK(0)
57690_us_ed1	Tributary To Wichcika Creek	M41372	–	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
57695_ds_ed1	Tributary To Wichcika Creek	M41373	–	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
57695_us_ed1	Tributary To Wichcika Creek	M21244	–	—	RAIN(1)	BULT(0)
57696_ds_ed1	Tributary To Wichcika Creek	M21245		—	—	BULT(0), RAIN(0)
57696_us_ed1	Tributary To Wichcika Creek	M21246		—	—	BULT(0), RAIN(0)

Field blanks

Field blanks filter distilled water at the field site to test for protocol contamination during collection and processing. **Detections in blanks indicate potential cross-contamination, not species presence at the location** — these rows are presented for QA transparency only, not as eDNA detections.

Table 5.12: Field blanks collected during the 2025 eDNA program in the Peace project area. Detections in blanks reflect protocol-side contamination, not site eDNA.

Site ID	Stream	UNBC ID	Detected	Sub-threshold	Clean (0 droplets)
125231_ds_ed1a	Tributary To Table River	M41355	—	—	BULT(0), GRAY(0), RAIN(0), SOCK(0)
198692_ds_ed1a	Tributary To Kerry Lake	M41361	RAIN(5)*	—	BULT(0), *GRAY(0)*, SOCK(0)*
203597_ds_ed1a	Tributary To Nation River	M41365	RAIN(4)*	—	BULT(0), *GRAY(0)*, SOCK(0)*

Retests

UNBC reran a subset of samples to confirm borderline results. The table below lists every (site × species) combination that was rerun, including reruns from both real environmental samples and field blanks.

Table 5.13: UNBC retests for the 2025 Peace eDNA samples. Runs is the total number of times the sample x species combination was assayed; Max droplets is the highest positive-droplet count observed across all runs.

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
125179_ds_ed1a	Tributary To Missinka River	Bull Trout	BULT	environmental	4	1	Sub-threshold (1-3 droplets, NOT a detection)
125179_us_ed1a	Tributary To Missinka River	Rainbow Trout	RAIN	environmental	4	383	Detected (≥4 droplets)
125194_ds_ed1a	Tributary To Missinka River	Rainbow Trout	RAIN	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
198692_ds_ed1a	Tributary To Kerry Lake	Bull Trout	BULT	field blank	4	0	No detection (0 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Arctic Grayling	GRAY	field blank	4	0	No detection (0 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Rainbow Trout	RAIN	field blank	6	5	Detected (≥4 droplets)
198692_ds_ed1a	Tributary To Kerry Lake	Kokanee	SOCK	field blank	6	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Bull Trout	BULT	field blank	4	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Arctic Grayling	GRAY	field blank	4	0	No detection (0 droplets)
203597_ds_ed1a	Tributary To Nation River	Rainbow Trout	RAIN	field blank	4	4	Detected (≥4 droplets)

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
203597_ds_ed1a	Tributary To Nation River	Kokanee	SOCK	field blank	4	0	No detection (0 droplets)
203597_ds_ed1b	Tributary To Nation River	Bull Trout	BULT	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
203597_ds_ed1b	Tributary To Nation River	Arctic Grayling	GRAY	environmental	4	0	No detection (0 droplets)
203597_ds_ed1b	Tributary To Nation River	Rainbow Trout	RAIN	environmental	4	173	Detected (≥4 droplets)
203597_ds_ed1b	Tributary To Nation River	Kokanee	SOCK	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Bull Trout	BULT	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Arctic Grayling	GRAY	environmental	4	0	No detection (0 droplets)
203597_us_ed1	Tributary To Nation River	Rainbow Trout	RAIN	environmental	4	213	Detected (≥4 droplets)
203597_us_ed1	Tributary To Nation River	Kokanee	SOCK	environmental	4	0	No detection (0 droplets)
203605_ds_ed1	Tributary To Williston Reservoir	Bull Trout	BULT	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
24718557_ed1	Missinka River	Bull Trout	BULT	environmental	4	2	Sub-threshold (1-3 droplets, NOT a detection)
24718557_ed1	Missinka River	Arctic Grayling	GRAY	environmental	4	1	Sub-threshold (1-3 droplets, NOT a detection)

Interactive map

The interactive Peace eDNA map shows site-level detection results with toggleable per-species layers, an opt-in sub-threshold-signals layer (off by default), and an opt-in Controls layer for field blanks. See the [interactive Peace eDNA map].

Tributary To Parsnip River - 199663 - Appendix

Site Location

PSCIS crossing 199663 is located on Tributary To Parsnip River, approximately 20km northeast of community of McLeod Lake, BC, in the Parsnip River watershed group (Figure 5.22). The crossing is located 780m upstream of the stream's confluence with the Parsnip River on Chco 12000 road which is the responsibility of BC Timber Sales.

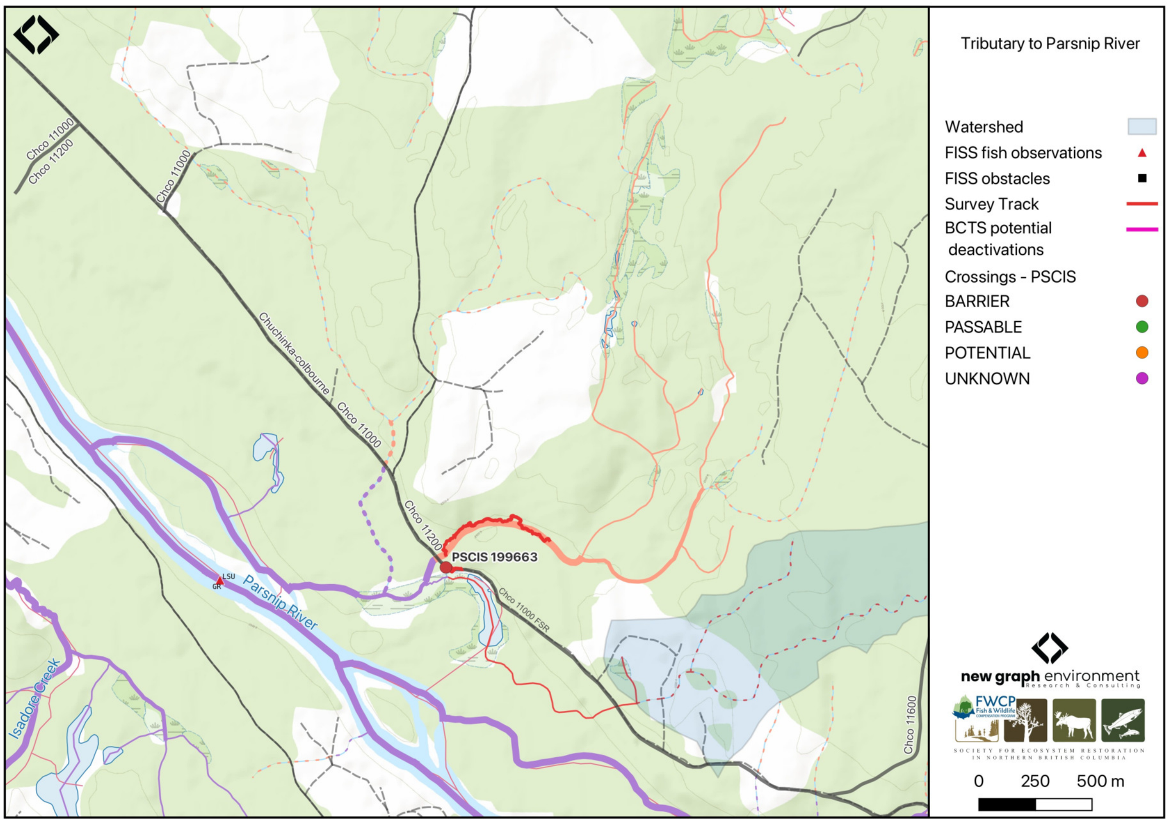


Figure 5.22: Map of Tributary To Parsnip River

Background

At PSCIS crossing 199663, Tributary To Parsnip River is a third order stream and drains a watershed of approximately 1km^2. The watershed ranges in elevation from a maximum of 849m to 713m near the crossing (Table 5.14).

PSCIS crossing 199663 was first assessed in 2024 as part of this project and was ranked as a barrier to fish passage with medium value habitat documented (Irvine and Schick 2025). The crossing was reassessed in 2025, and a habitat confirmation was completed due to the presence of medium value habitat and a significant outlet drop at the culvert. Just upstream of the confluence of this tributary and the Parsnip River, grayling have been documented (Norris [2018] 2024; MoE 2024a).

A summary of habitat modelling outputs for the crossing are presented in Table 5.15.

Table 5.14: Summary of derived upstream watershed statistics for PSCIS crossing 199663.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
199663	1	713	707	849	784	774	SW

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.15: Summary of fish habitat modelling for PSCIS crossing 199663.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	1.6	1.6	100
BT Spawning (km)	0.7	0.7	100
BT Network (km)	9.9	9.3	94
BT Stream (km)	8.0	7.4	92
BT Lake Reservoir (ha)	0.5	0.5	100
BT Wetland (ha)	12.8	12.8	100
BT Slopeclass03 (km)	3.6	3.0	83
BT Slopeclass05 (km)	3.6	3.6	100
BT Slopeclass08 (km)	0.9	0.9	100
BT Slopeclass15 (km)	0.2	0.2	100

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 199663

At the time of the 2025 assessment, PSCIS crossing 199663 on Chco 12000 was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table 5.16). The culvert had a 0.6m outlet drop and a 0.4m deep outlet pool.

The water temperature was 7°C, pH was 8.2 and conductivity was 310 uS/cm.

Table 5.16: Summary of fish passage assessment for PSCIS crossing 199663.

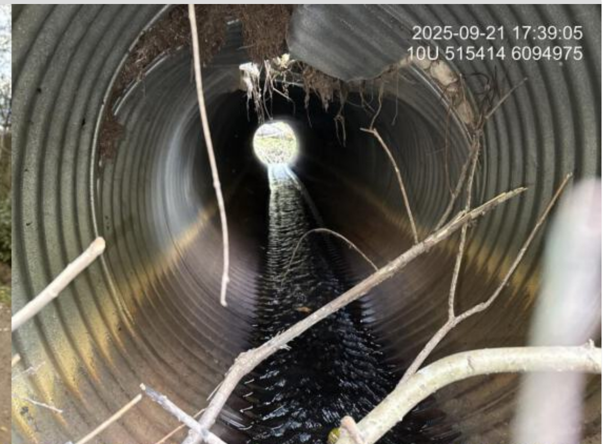
Location and Stream Data	•	**Crossing Characteristics**	**–**
Date	2025-09-21	Crossing Sub Type	Round Culvert
PSCIS ID	199663	Diameter (m)	1.2
External ID	–	Length (m)	13
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	515416	Resemble Channel	–
Northing	6094963	Backwatered	No
Stream	Tributary To Parsnip River	Percent Backwatered	–
Road	Chco 12000	Fill Depth (m)	0.5
Road Tenure	BCTS	Outlet Drop (m)	0.6
Channel Width (m)	0	Outlet Pool Depth (m)	0.4
Stream Slope (%)	1	Inlet Drop	No
Beaver Activity	No	Slope (%)	3
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	30	Barrier Result	Barrier
Fix type	Replace Structure with Streambed Simulation CBS	Fix Span / Diameter	3

Comments: Only photos and downstream width measurements were collected in 2025 due to limited time; all other information was transcribed from the 2024 survey. Habitat confirmations and eDNA samples were collected both upstream and downstream. A large wetland was located approximately 70–80m downstream of the culvert outlet. Abundant gravels were present within the first 20–30m below the outlet pool, where flow was strong, and a substantial outlet drop was observed.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Stream Characteristics at Crossing 19…

Location and Stream Data •	**Crossing Characteristics** –



Stream Characteristics Downstream of Crossing 199663

The stream was surveyed downstream from crossing 199663 for 80m (Figure 5.23). The habitat was rated as medium value for salmonid spawning and rearing. The stream had good flow and abundant gravels for 30–40m downstream of the culvert. A wetland was located ~80m downstream of the FSR. The average channel width was 2.6m and the average wetted width was 2.8m. The dominant substrate was cobbles with gravels sub-dominant.Total cover amount was rated as abundant with deep pools dominant. Cover was also present as small woody debris, large woody debris, undercut banks, and overhanging vegetation.

Stream Characteristics Upstream of Crossing 199663

The stream was surveyed upstream from crossing 199663 for 600m (Figure 5.23). The habitat was rated as medium value. Total cover amount was rated as abundant with overhanging vegetation dominant. Cover was also present as small woody debris, large woody debris, and undercut banks.The average channel width was 3m, the average wetted width was 2.1m, and the average gradient was 2.8%.The dominant substrate was fines with gravels sub-dominant. The stream was low gradient with significant riparian tree blowdown throughout. Wide floodplain within a well defined canyon (approximately 15m deep with slopes of ~45°). Healthy riparian consisting of herb and shrub cover within sparse spruce forest that appears to have been impacted by some sort of beetle or other affliction. Most of the spruce trees were dead with many already fallen down.

Environmental DNA

eDNA samples were collected both upstream and downstream of crossing 199663 on Chco 12000, with sampling techniques summarized in Table 5.17. Samples were submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, Kokanee, and Burbot assays (Burbot was tested only at this crossing among the Peace 2025 sites). Per-position results are summarized in Table 5.18. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran.

Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results. as well as within the interactive Peace eDNA map.

Table 5.17: Summary of eDNA samples collected at site 199663.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
199663_ds_ed1	Tributary To Parsnip						

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
River	riffle	instream	mce	1	2.0	Sample collected 15m downstream of culvert right bank. Site was chosen close to the culvert because of the wetland that was just downstream of this area - and wanted some flow.	
199663_us_ed1	Tributary To Parsnip River	riffle	instream	mce	1	2.9	Sample collected 25m upstream of culvert inlet on the right bank

Table 5.18: eDNA detection results at crossing 199663.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
199663_ds_ed1	M10139	BURB(195), RAIN(112)	—	BULT(0), GRAY(0)
199663_us_ed1	M10140	RAIN(65)	—	BULT(0), BURB(0), GRAY(0)

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Chco 12000 crossing (199663) with a streambed simulation (3m span) is recommended. At the time of reporting in 2026, the cost of the work is estimated at $300,000.

Conclusion

The habitat upstream of PSCIS crossing 199663 on Chco 12000 was documented as medium value for salmonid spawning and rearing, and the crossing is rated as a moderate priority for replacement due to the large outlet drop. Environmental DNA sampling at the site confirmed rainbow trout both upstream and downstream of the crossing; no bull trout were detected. Burbot were detected downstream but not upstream.

Since the assessment season, the FWCP 2026 proposal for replacement of this crossing was accepted and 50% project funding is confirmed. Replacement is targeted for 2026 in coordination with BC Timber Sales; an engineering firm is expected to collect the data required for the site plan.

Table 5.19: Summary of habitat details for PSCIS crossing 199663.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
199663	Downstream	80	2.6	2.8	–	–	abundant	–
199663	Upstream	600	3.0	2.1	0.2	2.8	abundant	medium

Figure 5.23: Left: Typical habitat downstream of PSCIS crossing 199663. Right: Typical habitat upstream of PSCIS crossing 199663.

Tributary To Nation River - 203597 - Appendix

Site Location

PSCIS crossing 203597 is located on Tributary To Nation River, approximately 65km west of McKenzie, BC, as the crow flies, and is in the Nation River watershed group (Figure 5.24). The crossing is located 1.9m upstream of the stream's confluence with the Nation River on Thut 15000 road and is situated under Sekani Forest Products Ltd. road tenure R07214 8.

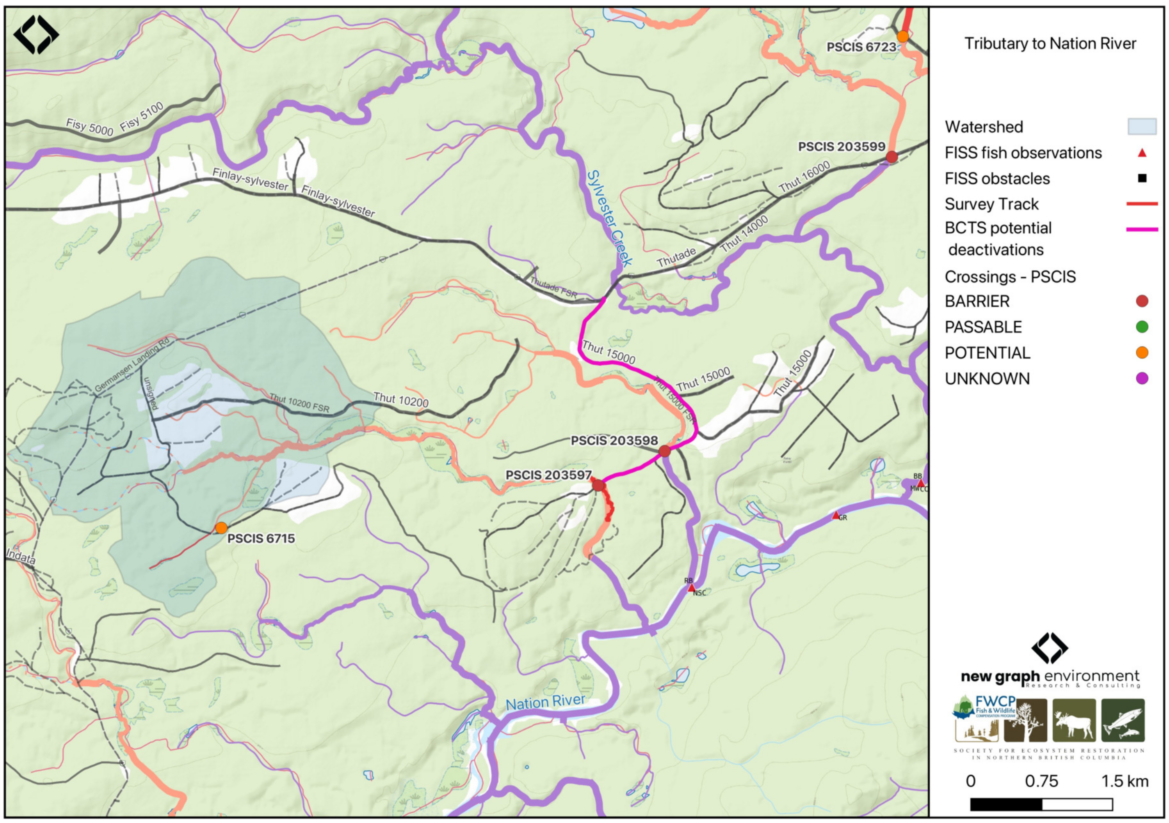


Figure 5.24: Map of Tributary To Nation River

Background

At PSCIS crossing 203597, Tributary To Nation River is a second order stream and drains a watershed of approximately 8.2km^2. The watershed ranges in elevation from a maximum of 1065m to 900m near the crossing (Table 5.20).

This was the first assessment of PSCIS crossing 203597, in part because the site is situated on a road identified by BC Timber Sales as a potential deactivation candidate. The stream exhibited high-quality habitat, so a habitat confirmation was completed. Bull trout are documented in the Nation River near the confluence with this stream (Norris [2018] 2024; MoE 2024a).

A summary of habitat modelling outputs for the crossing are presented in Table 5.21.

Table 5.20: Summary of derived upstream watershed statistics for PSCIS crossing 203597.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
203597	8.2	900	907	1065	980	971	S

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.21: Summary of fish habitat modelling for PSCIS crossing 203597.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	5.6	5.6	100
BT Spawning (km)	0.7	0.7	100
BT Network (km)	16.8	9.5	57
BT Stream (km)	12.2	5.2	43
BT Lake Reservoir (ha)	1.0	1.0	100
BT Wetland (ha)	47.1	46.2	98
BT Slopeclass03 (km)	7.9	2.5	32
BT Slopeclass05 (km)	4.0	2.5	62
BT Slopeclass08 (km)	0.5	0.5	100
BT Slopeclass15 (km)	0.0	0.0	–

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 203597

At the time of the 2025 assessment, PSCIS crossing 203597 on Thut 15000 was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table 5.22).

Stream Characteristics at Crossing 20…

The water temperature was 6.5°C, pH was 7.5 and conductivity was 188 uS/cm.

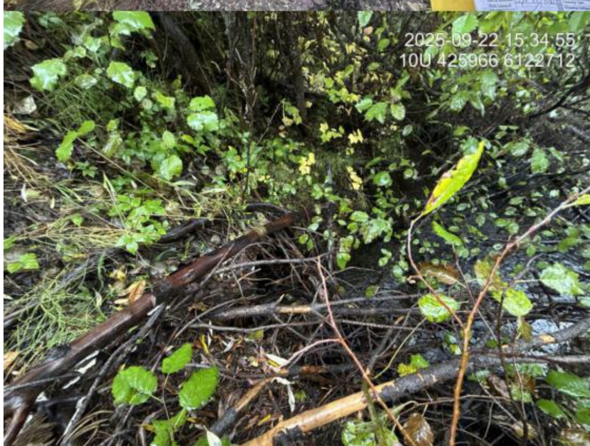
Table 5.22: Summary of fish passage assessment for PSCIS crossing 203597.

Location and Stream Data	•	Crossing Characteristics	–
Date	2025-09-22	Crossing Sub Type	Round Culvert
PSCIS ID	203597	Diameter (m)	0.8
External ID	15201563	Length (m)	19
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	425959	Resemble Channel	–
Northing	6122703	Backwatered	No
Stream	Tributary To Nation River	Percent Backwatered	–
Road	Thut 15000	Fill Depth (m)	1.5
Road Tenure	Sekani R07214 8	Outlet Drop (m)	0.25
Channel Width (m)	2.5	Outlet Pool Depth (m)	0.5
Stream Slope (%)	2.5	Inlet Drop	Yes
Beaver Activity	No	Slope (%)	1
Habitat Value	High	Valley Fill	Deep Fill
Final score	29	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	15

Comments: High-quality stream with abundant gravels and cover, directly connected to the Nation River. Bull trout noted near the confluence. eDNA samples collected upstream and downstream of the crossing. This crossing is located on a road identified by BCTS as a candidate for potential deactivation.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Stream Characteristics at Crossing 20…

Location and Stream Data	**Crossing Characteristics**
	
	
	

Stream Characteristics Downstream of Crossing 203597

The stream was surveyed downstream from crossing 203597 for 400m (Figure 5.25). The stream had good flow, likely elevated due to significant precipitation the previous night. Abundant gravels were present throughout, and cover was provided by multiple sources including woody debris and undercut banks. No barriers were observed. The average channel width was 2.8m, the average wetted width was 2.6m, and the average gradient was 3%.Total cover amount was rated as abundant with undercut banks dominant. Cover was also present as small woody debris, large woody debris, boulders, deep pools, and overhanging vegetation.The dominant substrate was cobbles with gravels sub-dominant.

Stream Characteristics Upstream of Crossing 203597

The stream was surveyed upstream from crossing 203597 for 350m up to the beginning of the wetlands (Figure 5.25). The habitat was rated as NA value. The average channel width was 2.5m, the average wetted width was 2.4m, and the average gradient was 3%.Total cover amount was rated as moderate with small woody debris dominant. Cover was also present as large woody debris, undercut banks, and overhanging vegetation.The dominant substrate was fines with cobbles sub-dominant. The stream had good flow, although recent rainfall may have influenced discharge. Pockets of gravels were present within the area surveyed. Wetlands adjacent to the channel appeared shallow, suggesting they may freeze to the bottom during winter.

Environmental DNA

eDNA samples were collected both upstream and downstream of crossing 203597 on Thut 15000, with sampling techniques summarized in Table 5.23. Samples were submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, and Kokanee assays. Per-position results from real environmental samples are summarized in Table 5.24. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with * flagging samples UNBC reran.

Field blank at this crossing. Sample `203597_ds_ed1a` (UNBC ID M41365) was a field blank — distilled water filtered at the field site to test for protocol contamination, not site eDNA. UNBC's analysis returned a low-level Rainbow Trout signal at this blank (4 positive droplets, ≈0.27 copies/µL). Per ddPCR convention this represents protocol-side contamination during collection or processing, not species presence at this location. The blank is presented in the Controls layer of the interactive map (off by default) and in the field-blanks table within Appendix - Environmental DNA Results.

Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results. as well as within the interactive Peace eDNA map.

Table 5.23: Summary of eDNA samples collected at site 203597.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
203597_ds_ed1b	Tributary To Nation River	riffle	grab	mce	1	1.7	Sample collected 350 - 400m downstream of culvert. Right bank. Accessed via historic road which is now wildlife trail.
203597_us_ed1	Tributary To Nation River	riffle	inatream	mce	1	2.0	Sample collected 5m upstream of the culvert on the left bank

Table 5.24: eDNA detection results at crossing 203597 (real environmental samples).

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
203597_ds_ed1b	M41366	RAIN(173)*	BULT(3)*	GRAY(0), *SOCK(0)*
203597_us_ed1	M41367	RAIN(213)*	—	BULT(0), *GRAY(0)*, SOCK(0)*

Structure Remediation and Cost Estimate

Since the time of field assessements this crossing was deactivated by BC Timber Sales.

Conclusion

The habitat upstream of PSCIS crossing 203597 on Thut 15000 was documented as medium value for salmonid spawning and rearing, and the crossing was rated as a high for deactivation.

Since the assessment season, the FWCP 2026 proposal for riparian enhancements at this site was accepted. Planned work consists of soil decompaction along the deactivated road prism, riparian planting with native species, and placement of coarse woody debris to advance recovery of the channel margin. Environmental DNA sampling confirmed rainbow trout both upstream and downstream of the crossing; an under-threshold bull trout detection is suspected to reflect real presence given the suitable size of the stream and its proximity to the Nation River, where bull trout detections are recorded in provincial records near the subject stream’s confluence with the Nation.

Table 5.25: Summary of habitat details for PSCIS crossing 203597.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
203597	Downstream	400	2.8	2.6	0.3	3	abundant	high
203597	Upstream	350	2.5	2.4	0.3	3	moderate	–

Figure 5.25: Left: Typical habitat downstream of PSCIS crossing 203597. Right: Wetland habitat located 350m upstream of PSCIS crossing 203597.

Tributary To Williston Reservoir - 203605 - Appendix

Site Location

PSCIS crossing 203605 is located on Tributary To Williston Reservoir, on the western side of the reservoir, approximately 10km southeast of MacKenzie, BC, in the Parsnip Arm watershed group (Figure 5.26). The crossing is located 1.3m upstream of the stream's confluence with the Williston Reservoir on Finlay FSR road which is the responsibility of the Ministry of Forests.

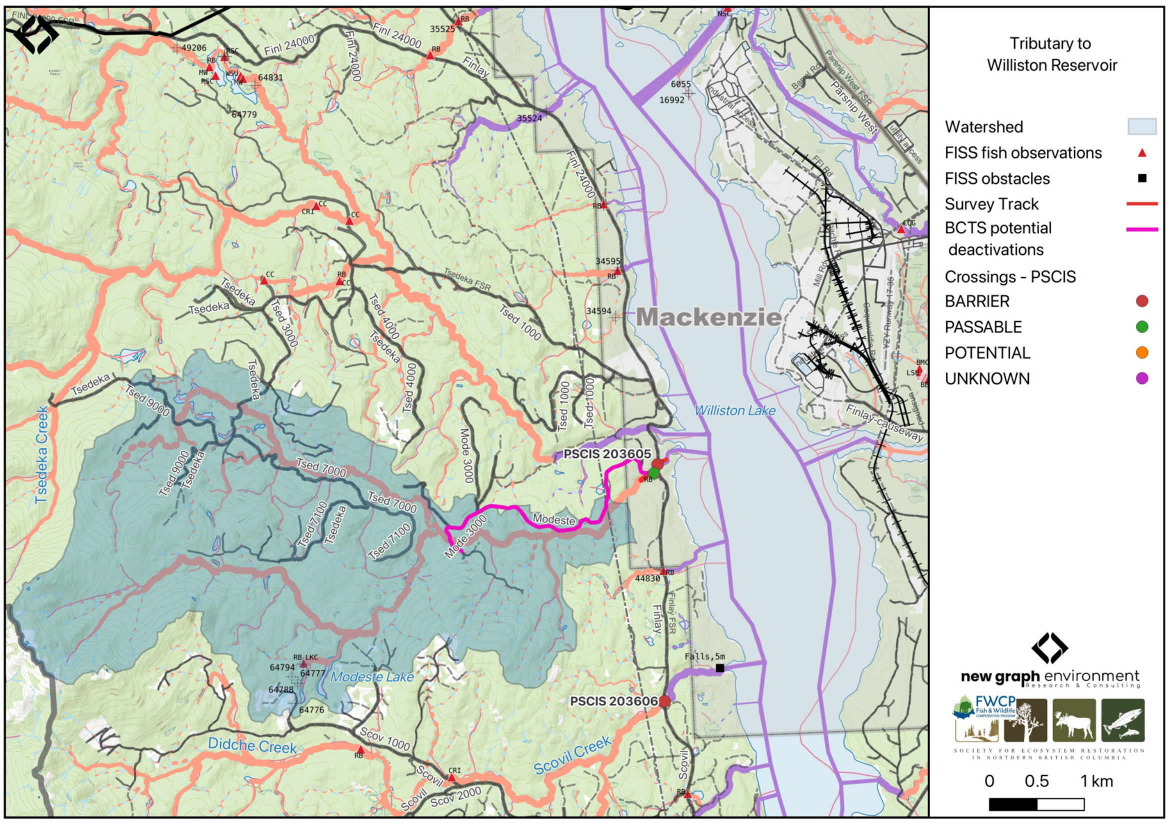


Figure 5.26: Map of Tributary To Williston Reservoir

Background

At PSCIS crossing 203605, Tributary To Williston Reservoir is a fourth order stream and drains a watershed of approximately 55.6km^2. The watershed ranges in elevation from a maximum of 1644m to 681m near the crossing (Table 5.26).

This was the first assessment of PSCIS crossing 203605, selected because it is located very close to Williston Reservoir, which supports documented bull trout. Approximately 35km of bull trout rearing habitat is modelled upstream with no additional modelled barriers, and rainbow trout have previously been recorded in the system (Norris [2018] 2024; MoE 2024a).The stream exhibited high-quality habitat, therefore a habitat confirmation was completed.

Approximately 215m upstream of Finlay FSR, the stream crosses Modeste FSR at a bridge, which was also assessed for the first time in 2025 (PSCIS crossing 203611). Modeste FSR provides access to the lower half of the watershed, and is identified by BC Timber Sales as a potential deactivation candidate.

A summary of habitat modelling outputs for the crossing are presented in Table 5.27.

Table 5.26: Summary of derived upstream watershed statistics for PSCIS crossing 203605.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
203605	55.6	681	671	1644	980	937	SE

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.27: Summary of fish habitat modelling for PSCIS crossing 203605.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	35.9	30.8	86
BT Spawning (km)	22.0	19.5	89
BT Network (km)	100.5	76.4	76
BT Stream (km)	86.7	64.1	74
BT Lake Reservoir (ha)	51.1	42.4	83
BT Wetland (ha)	51.9	50.3	97
BT Slopeclass03 (km)	16.3	14.4	88
BT Slopeclass05 (km)	21.8	10.4	48
BT Slopeclass08 (km)	17.6	12.2	69
BT Slopeclass15 (km)	28.2	24.4	87

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 203605

At the time of the 2025 assessment, PSCIS crossing 203605 on Finlay FSR was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table 5.28). The culvert had a 0.95m outlet drop and a 0.5m deep outlet pool.

The water temperature was 6.8°C, pH was 7.6 and conductivity was 251 uS/cm.

Table 5.28: Summary of fish passage assessment for PSCIS crossing 203605.

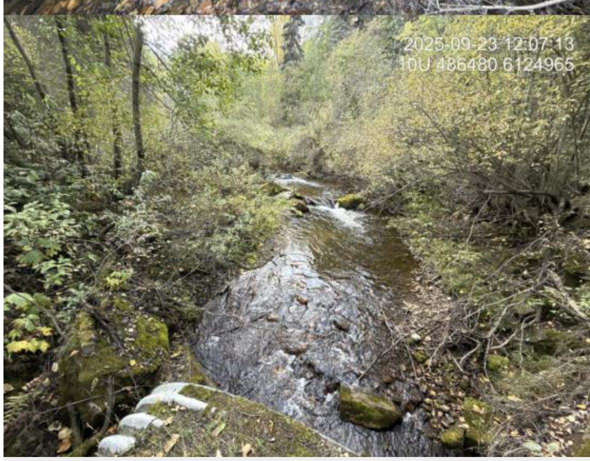
Location and Stream Data	•	Crossing Characteristics	–
Date	2025-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	203605	Diameter (m)	3.9
External ID	16401519	Length (m)	46
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	486504	Resemble Channel	–
Northing	6124990	Backwatered	No
Stream	Tributary To Williston Reservoir	Percent Backwatered	–
Road	Finlay FSR	Fill Depth (m)	2.5
Road Tenure	MoF	Outlet Drop (m)	0.95
Channel Width (m)	6	Outlet Pool Depth (m)	0.5
Stream Slope (%)	2	Inlet Drop	No
Beaver Activity	No	Slope (%)	2
Habitat Value	High	Valley Fill	Deep Fill
Final score	37	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	15

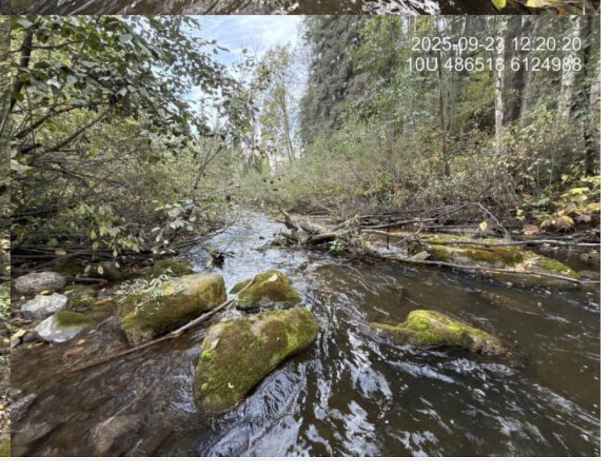
Comments: High value habitat on very large system with pipe that has large outlet drop near 1m. Habitat confirmation and eDNA sampling conducted upstream and downstream of the crossing.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Stream Characteristics at Crossing 20…

Location and Stream Data	•	Crossing Characteristics	–





Stream Characteristics Downstream of Crossing 203605

The stream was surveyed downstream from crossing 203605 for 250m to the mapped reservoir location, where the channel retained stream characteristics at the time of assessment (Figure 5.27). Abundant gravels and cover were present throughout, with numerous run sections providing suitable rearing habitat for juvenile westslope cutthroat trout and rainbow trout. No barriers to fish passage were observed in the area surveyed and the habitat was rated as high value for salmonid spawning and rearing. The average channel width was 8.5m, the average wetted width was 5.6m, and the average gradient was 1.8%.Total cover amount was rated as moderate with dominant. Cover was also present as large woody debris and overhanging vegetation.The dominant substrate was gravels with fines sub-dominant.

Stream Characteristics Upstream of Crossing 203605

The stream was surveyed upstream from crossing 203605 for 650m (Figure 5.27). The habitat was rated as high value. The dominant substrate was gravels with cobbles sub-dominant.Total cover amount was rated as moderate with undercut banks dominant. Cover was also present as small woody debris, large woody debris, boulders, deep pools, and overhanging vegetation.The average channel width was 6.7m, the average wetted width was 4.6m, and the average gradient was 2.9%. The stream was large with complex habitat and diverse cover including deep pools, boulders, large and small woody debris, and undercut banks. A canyon section ~100m long occurred ~100m upstream of the Modeste FSR bridge, containing multiple cascade steps up to 1m high with deep downstream pools. At the upstream end of the site, the channel deflected against steep gully walls. Abundant gravels were present, suitable for adfluvial and resident rainbow trout as well as bull trout.

Environmental DNA

eDNA samples were collected both upstream and downstream of crossing 203605 on Finlay FSR, with sampling techniques summarized in Table 5.29. Samples were submitted to UNBC for ddPCR analysis against Bull Trout, Arctic Grayling, Rainbow Trout, and Kokanee assays. Per-position results are summarized in Table 5.30. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran. Sub-threshold signals (1-3 droplets) are tracked separately as they are not interpretable as species presence per ddPCR convention.

Table 5.29: Summary of eDNA samples collected at site 203605.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
203605_ds_ed1	Tributary To Williston Reservoir	riffle	grab	mce	1	–	Sample collected 15m downstream of the culvert outlet on the right bank

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
203605_us_ed1	Tributary To Williston Reservoir	riffle	instream	mce	1	2	Approximately 45 m upstream of culvert inlet on the left bank

Table 5.30: eDNA detection results at crossing 203605.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
203605_ds_ed1	M41368	RAIN(455)	BULT(3)*	GRAY(0), SOCK(0)
203605_us_ed1	M41369	RAIN(429)	BULT(1)	GRAY(0), SOCK(0)

Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results. The interactive Peace eDNA map shows toggleable per-species layers.

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Finlay FSR crossing (203605) with a bridge (15m span) is recommended. At the time of reporting in 2026, the cost of the work is estimated at $450,000.

Conclusion

The habitat upstream of PSCIS crossing 203605 on Finlay FSR was documented as high value for salmonid spawning and rearing, and the crossing is rated as a high priority for replacement due to the large outlet drop. There is a significant amount of bull trout rearing habitat modelled upstream, and since the road that provides access to lower half of the may be deactivated, this crossing could be a good candidate for replacement.

Table 5.31: Summary of habitat details for PSCIS crossing 203605.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
203605	Downstream	250	8.5	5.6	0.2	1.8	moderate	high
203605	Upstream	650	6.7	4.6	3.7	2.9	moderate	high

Figure 5.27: Left: Typical habitat downstream of PSCIS crossing 203605. Right: Typical habitat upstream of PSCIS crossing 203605.

Tributary to Missinka River - 125179 - Monitoring Appendix

Effectiveness monitoring was conducted in 2025 at PSCIS crossing 125179 on the Chuchinka-Missinka FSR. Site location, background, prior-year assessment and replacement history, and earlier effectiveness monitoring results are documented in Irvine (2020), Irvine and Winterscheidt (2023), and Irvine and Schick (2025); the 2024 site appendix is available at https://www.newgraphenvironment.com/fish_passage_peace_2024_reporting/trib-to-missinka.html.

Monitoring Form

2025 effectiveness monitoring form results are summarized in Table 5.32.

Table 5.32: Summary of effectiveness monitoring form results for site 125179 in 2025.

Variable	Value
Pscis Crossing Id	125179
Stream Name	Tributary to Missinka River
Road Name	Chuchinka-Missinka FSR
Crossing Subtype	BRIDGE
Span	16
Width	4.8
Assessment Comment	Routine effectiveness evaluation. Structure was replaced several years ago and monitoring is ongoing. Electrofishing with fish over 65mm in length PIT tagged, as well as EDNA sampling was conducted in 2025.
Dewatering Notes	Stream is flowing nicely through the construction footprint with no evidence of aggregation, or any sign of dewatering
Velocity Notes	Velocity within project footprint is slightly faster than naturalized areas upstream and downstream because of the construction from riprap however, overall they do not appear to prevent migration of juveniles larger than approximately 130mm. During higher flow times of the year velocity may be quite higher, however.
Constriction Notes	Channel is constricted within the 4 - 5m upstream and downstream of the bridge with channel widths ranging from 2.5 - 3.4m.
Substrate Notes	Substrate is filling in nicely with gravels and cobble/boulder stone lines with a composition comparable to upstream and downstream within natural areas of the stream.
Riparian Notes	Overall, the riparian footprint disturbed is quite small and shrub growth from the edges of the footprint is beginning to fill in the upstream and downstream edges.
Flow Depth Notes	Flow depth within the construction footprint are comparable to those upstream and downstream in natural areas with some deeper pockets scoured adjacent to stone lines and placed boulders within the channel.
Stability Notes	Structure appears stable with no indication of any issues.

Variable	Value
Revegetation Notes	Adjacent to the bridge there's only a small amount of herbaceous regrowth in between heavily rip wrapped areas. It is expected that as tsime passes fine substrates will infiltrate the riprap more during high flow events, and that those areas will colonize more with natural herb and shrub species over time.
Cover Notes	Placed riprap boulders and developed stone lines have created areas of bubble curtain and shallow pool cover throughout the construction footprint. Overhead cover from vegetation is extremely minimal still however some willow and other shrub regrowth indicates it will improve overtime.
Maintenance Notes	No maintenance required

Fish Sampling

Electrofishing was conducted downstream and upstream of PSCIS crossing 125179 in 2025. Per-site capture results are in Table 5.33; densities are in Table 5.34 and Figure 5.28.

Table 5.33: Fish sampling site summary for 125179.

site	passes	ef_length_m	ef_width_m	area_m2	enclosure
125179_us_ef1	1	45	2.2	99.0	partial enclosure
125179_ds_ef1	1	35	2.7	94.5	partial enclosure

Table 5.34: Fish sampling density results summary for 125179.

local_name	species_code	life_stage	catch	density_100m2	nfc_pass
125179_ds_ef1	Rainbow Trout	adult	2	2.1	FALSE
125179_ds_ef1	Rainbow Trout	fry	6	6.3	FALSE
125179_ds_ef1	Rainbow Trout	juvenile	2	2.1	FALSE
125179_ds_ef1	Rainbow Trout	parr	3	3.2	FALSE
125179_us_ef1	Rainbow Trout	adult	2	2.0	FALSE
125179_us_ef1	Rainbow Trout	fry	3	3.0	FALSE
125179_us_ef1	Rainbow Trout	juvenile	2	2.0	FALSE
125179_us_ef1	Rainbow Trout	parr	4	4.0	FALSE

* nfc_pass FALSE means fish were captured in final pass indicating more fish of this species/lifestage may have remained in site. Mark-recaptured required to reduce uncertainties.

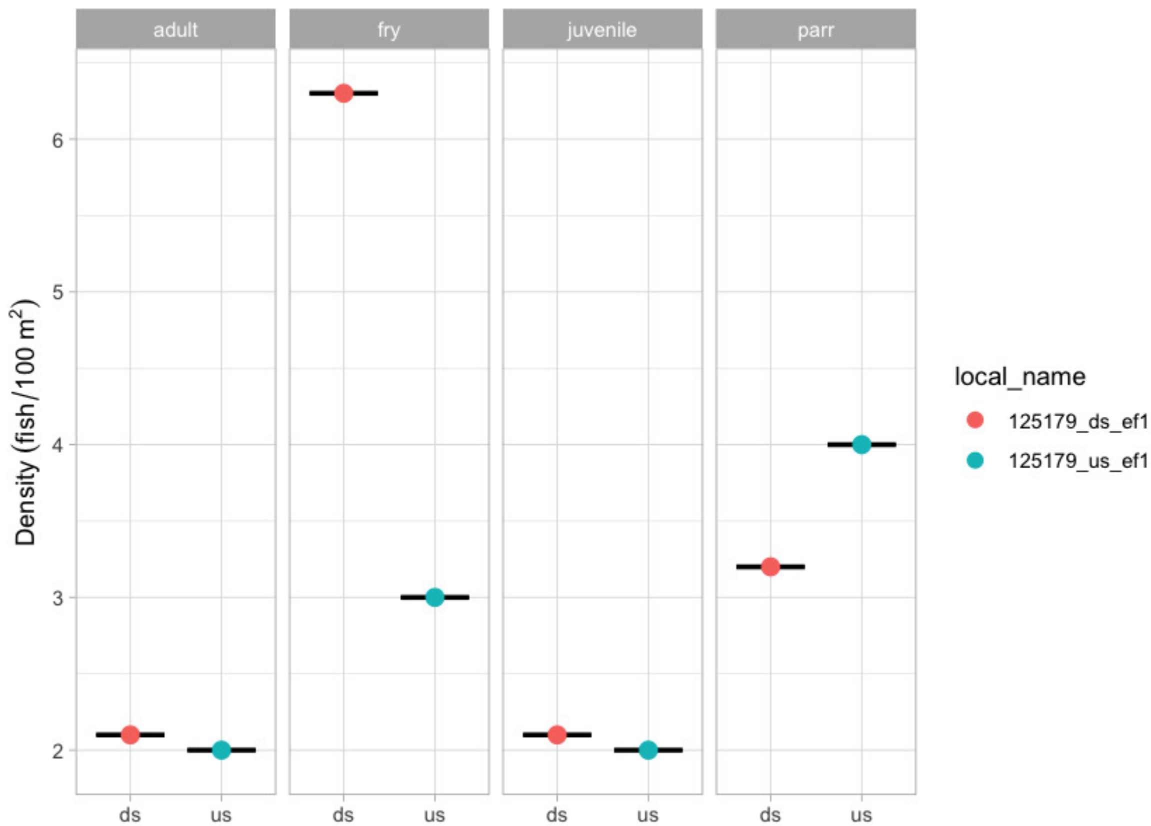


Figure 5.28: Densities of fish (fish/100m2) captured upstream and downstream of PSCIS crossing 125179 in 2025.

Environmental DNA

eDNA samples were collected upstream and downstream of PSCIS crossing 125179 in 2025 as part of the program-wide eDNA sampling effort. The reference site at PSCIS crossing 125180, located on the same road as documented in Irvine and Schick (2025), was also sampled for eDNA in 2025. Per-position detection results for both sites are presented in Table 5.35. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with ∗ flagging samples UNBC reran. Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results, and the interactive Peace eDNA map provides a toggleable per-species view.

Table 5.35: eDNA detection results at crossing 125179.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
125179_ds_ed1a	M41349	RAIN(309)	BULT(1)*	GRAY(0), SOCK(0)

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
125179_us_ed1a	M41351	RAIN(383)*	—	BULT(0), GRAY(0), SOCK(0)
125180_ds_ed1	M41353	BULT(4), RAIN(330)	—	GRAY(0), SOCK(0)
125180_us_ed1	M41354	RAIN(293)	BULT(3)	GRAY(0), SOCK(0)

Conclusion

At the time of reporting, the structure at PSCIS crossing 125179 appears stable and the site is showing positive signals of recovery. Continued effectiveness monitoring at this site is recommended in future years to extend the multi-year time series of structure performance and fish use.

Tributary to the Table River - 125231 - Monitoring Appendix

Effectiveness monitoring was conducted in 2025 at PSCIS crossing 125231 on the Chuchinka-Table FSR. Site location, background, and the 2023 replacement of the crossing with a clear-span bridge by Canfor (with engineering oversight by DWB Consulting Services Ltd. and 50% funding from FWCP via SERNbc coordination) are documented in Irvine and Winterscheidt (2024); the 2023 site appendix is available at https://www.newgraphenvironment.com/fish_passage_peace_2023_reporting/tributary-to-the-table-river—125231—appendix.html.

Monitoring Form

2025 effectiveness monitoring form results are summarized in Table 5.36.

Table 5.36: Summary of effectiveness monitoring form results for site 125231 in 2025.

Variable	Value
Pscis Crossing Id	125231
Stream Name	Tributary to Table River
Road Name	Table FSR
Crossing Subtype	BRIDGE
Span	–
Width	–
Assessment Comment	Overall, the site looks very similar to last year, in good condition with no evidence of any structural issues and the instream portion doing quite well. As there is a large amount of rock rip wrap adjacent to the stream that was not planted the riparian vegetation within the footprint is basically nonexistent. Electrofishing was conducted at sites downstream and upstream of the bridge with fish over 65mm PIT tagged. eDNA samples were collected as well.
Dewatering Notes	No degrading of the channel noted. Good flow throughout construction footprint.
Velocity Notes	Velocities in the construction footprint are similar to upstream and downstream with maturing stone lines and placed boulders, stepping out the gradient to natural conditions
Constriction Notes	Width of channel appears similar to upstream and downstream with room for channel migration.
Substrate Notes	Substrate is very similar to upstream and downstream with well placed riprap and boulders providing scour and creating pool habitat.
Riparian Notes	Basically no riparian regrowth within the construction footprint due to large amount of rip wrap placed.
Flow Depth Notes	Flow depths are acceptable with deep sections adjacent to stone lines and placed boulders.

Variable	Value
Stability Notes	Structure appears stable with no evidence of any movement
Revegetation Notes	Revegetation of the site is generally poor due to the large amount of rock placed in the construction footprint and a lack of planting and seeding.
Cover Notes	Cover is available adjacent to stone lines and placed boulders were scour and some shallow pools have been created. Fish were captured in these areas within the construction footprint, indicating that they are providing suitable habitat for local fish species.
Maintenance Notes	No maintenance required

Fish Sampling

Electrofishing was conducted downstream and upstream of PSCIS crossing 125231 in 2025. Per-site capture results are in Table 5.37; densities are in Table 5.38 and Figure 5.29.

Table 5.37: Fish sampling site summary for 125231.

site	passes	ef_length_m	ef_width_m	area_m2	enclosure
125231_ds_ef1	1	50	2.8	140	partial enclosure
125231_us_ef1	1	50	1.6	80	partial enclosure

Table 5.38: Fish sampling density results summary for 125231.

local_name	species_code	life_stage	catch	density_100m2	nfc_pass
125231_ds_ef1	Rainbow Trout	adult	1	0.7	FALSE
125231_ds_ef1	Rainbow Trout	fry	7	5.0	FALSE
125231_ds_ef1	Rainbow Trout	juvenile	1	0.7	FALSE
125231_ds_ef1	Rainbow Trout	parr	5	3.6	FALSE
125231_us_ef1	Rainbow Trout	adult	1	1.2	FALSE
125231_us_ef1	Rainbow Trout	fry	5	6.2	FALSE
125231_us_ef1	Rainbow Trout	juvenile	3	3.8	FALSE
125231_us_ef1	Rainbow Trout	parr	2	2.5	FALSE
125231_us_ef1	Sucker (General)	adult	1	1.2	FALSE

* nfc_pass FALSE means fish were captured in final pass indicating more fish of this species/lifestage may have remained in site. Mark-recaptured required to reduce uncertainties.

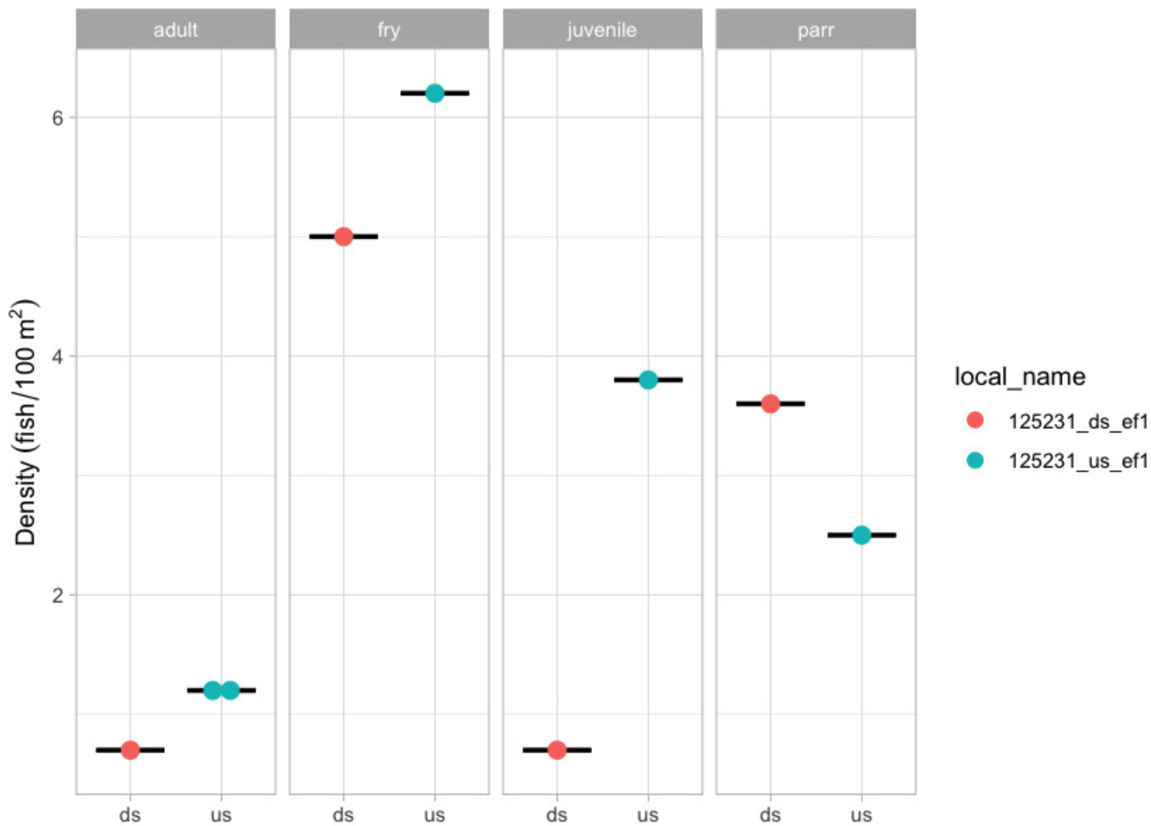


Figure 5.29: Densities of fish (fish/100m2) captured upstream and downstream of PSCIS crossing 125231 in 2025.

Environmental DNA

eDNA samples were collected upstream and downstream of PSCIS crossing 125231 in 2025 as part of the program-wide eDNA sampling effort. Per-position detection results are presented in Table 5.39. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with ∗ flagging samples UNBC reran. Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results, and the interactive Peace eDNA map provides a toggleable per-species view.

Table 5.39: eDNA detection results at crossing 125231.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
125231_ds_ed1c	M41356	RAIN(205)	—	BULT(0), GRAY(0), SOCK(0)
125231_us_ed1	M41357	RAIN(5)	—	BULT(0), GRAY(0), SOCK(0)

Conclusion

At the time of reporting, the structure at PSCIS crossing 125231 appears stable and the site is showing positive signals of recovery. Continued effectiveness monitoring at this site is recommended in future years to extend the multi-year time series of structure performance and fish use.

Tributary to Kerry Lake - 198692 - Monitoring Appendix

Baseline monitoring was conducted in 2025 at PSCIS crossing 198692 ahead of planned remediation, to inform future effectiveness monitoring once the structure is upgraded. Site location, background, and the 2024 habitat confirmation assessment are documented in Irvine and Schick (2025); the 2024 site appendix is available at https://www.newgraphenvironment.com/fish_passage_peace_2024_reporting/trib-to-kerry.html.

Fish Sampling

Electrofishing was conducted downstream and upstream of PSCIS crossing 198692 in 2025 to establish a pre-remediation baseline. Per-site capture results are in Table 5.40; densities are in Table 5.41.

Table 5.40: Fish sampling site summary for 198692.

site	passes	ef_length_m	ef_width_m	area_m2	enclosure
198692_us_ef1	1	16	–	–	partial enclosure
198692_us_ef2	1	12	–	–	open
198692_ds_ef1	1	34	–	–	partial enclosure
198692_ds_ef2	1	4	–	–	open

Table 5.41: Fish sampling density results summary for 198692.

local_name	species_code	life_stage	catch	density_100m2	nfc_pass
198692_ds_ef1	Rainbow Trout	fry	6	–	FALSE
198692_ds_ef1	Rainbow Trout	parr	4	–	FALSE
198692_ds_ef2	Rainbow Trout	fry	1	–	FALSE
198692_ds_ef2	Rainbow Trout	juvenile	4	–	FALSE
198692_ds_ef2	Rainbow Trout	parr	9	–	FALSE
198692_us_ef1	Rainbow Trout	fry	1	–	FALSE
198692_us_ef2	Rainbow Trout	fry	3	–	FALSE
198692_us_ef2	Rainbow Trout	parr	7	–	FALSE

* nfc_pass FALSE means fish were captured in final pass indicating more fish of this species/lifestage may have remained in site. Mark-recaptured required to reduce uncertainties.

Environmental DNA

eDNA samples were collected upstream and downstream of PSCIS crossing 198692 in 2025 as part of the program-wide eDNA sampling effort. Per-position detection results are presented in Table 5.42. Detections are based on the standard ddPCR call threshold of ≥4 positive droplets; format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran. Site-level detail across all sampled crossings is in Appendix - Environmental DNA Results, and the interactive Peace eDNA map provides a toggleable per-species view.

Table 5.42: eDNA detection results at crossing 198692.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
198692_ds_ed1b	M41362	RAIN(119)	BULT(1)	GRAY(0), SOCK(0)
198692_ds_ed2	M41363	RAIN(630)	—	BULT(0), GRAY(0), SOCK(0)
198692_us_ed1	M41364	RAIN(235)	—	BULT(0), GRAY(0), SOCK(0)

Conclusion

Baseline fish and eDNA data were collected at PSCIS crossing 198692 in 2025 to inform future effectiveness monitoring once remediation completes. Resampling at the site once the structure is upgraded will allow direct comparison of fish abundance and habitat conditions before and after the intervention.

References

Amazon Web Services. 2025. *Aws/Aws-Cli: Universal Command Line Interface for Amazon Web Services*. V. 2.18.15. Released. https://github.com/aws/aws-cli.

BC Species & Ecosystem Explorer. 2020. “Salvelinus Confluentus (Bull Trout).”https://a100.gov.bc.ca/pub/eswp/reports.do?elcode=AFCHA05020.

Beaudry, P. 2014. *Assessment and Assignment of Sensitivity Ratings to Sub-Basins of the Table Watershed in Parsnip Drainage – Ominieca Region. Contract Number: GS15823011*. https://a100.gov.bc.ca/pub/acat/documents/r51634/3_Beaudry_2014_Table_FSW_51634_1480368942206_0367832493.pdf.

Beaudry, Pierre G. 2013a. *Assessment and Assignment of Sensitivity Ratings to Sub-Basins of the Anzac Watershed in Parsnip Drainage – Ominieca Region. Contract Number: GS14823018*. Report. http://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=51630.

Beaudry, Pierre G. 2013b. *Assessment and Assignment of Sensitivity Ratings to Sub-Basins of the Missinka Watershed in Parsnp Drainage – Ominieca Region. Contract Number: GS14FWH-006*. Report. P. Beaudry and Associates Ltd.http://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=51631.

Beaudry, Pierre G. 2014. *Assessment and Assignment of Sensitivity Ratings to Sub-Basins of the Hominka Watershed in Parsnip Drainage – Ominieca Region. Contract Number: GS15823011*. Report. http://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=51633.

Bell, M. C. 1991. *Fisheries Handbook of Engineering Requirements and Biological Criteria*. https://www.fs.fed.us/biology/nsaec/fishxing/fplibrary/Bell_1991_Fisheries_handbook_of_engineering_requirements_and.pdf.

Bourne, Christina, Dan Kehler, Yolanda Wiersma, and David Cote. 2011. “Barriers to Fish Passage and Barriers to Fish Passage Assessments: The Impact of Assessment Methods and Assumptions on Barrier Identification and Quantification of Watershed Connectivity.”*Aquatic Ecology* 45: 389–403. https://doi.org/10.1007/s10452-011-9362-z.

Bramblett, Robert G., Mason D. Bryant, Brenda E. Wright, and Robert G. White. 2002. “Seasonal Use of Small Tributary and Main-Stem Habitats by Juvenile Steelhead, Coho Salmon, and Dolly Varden in a Southeastern Alaska Drainage Basin.”*Transactions of the American Fisheries Society* 131 (3): 498–506. https://doi.org/10.1577/1548-8659(2002)131<0498:SUOSTA>2.0.CO;2.

Busch, D.Shallin, Mindi Sheer, Kelly Burnett, Paul McElhany, and Tom Cooney. 2011. “Landscape-Level Model to Predict Spawning Habitat For Lower Columbia River Fall Chinook Salmon (*Oncorhynchus Tshawytscha*): Intrinsic Potential Model for Spawning Fall Chinook Salmon.”*River Research and Applications* 29 (3): 297–312. https://doi.org/10.1002/rra.1597.

Cayan, Daniel R., Michael D. Dettinger, Susan A. Kammerdiener, Joseph M. Caprio, and David H. Peterson. 2001. “Changes in the Onset of Spring in the Western United States.”*Bulletin of the American Meteorological Society* 82 (3): 399–415. https://doi.org/10.1175/1520-0477(2001)082<0399:citoos>2.3.co;2.

Clarkin, K, A Connor, M Furniss, et al. 2005. *National Inventory and Assessment Procedure For Identifying Barriers to Aquatic Organism Passage at Road-Stream Crossings*. United States Department of Agriculture, Forest Service, National Technology; Development Program. https://www.fs.fed.us/biology/nsaec/fishxing/publications/PDFs/NIAP.pdf.

cogeotiff. (2018) 2025. *Cogeotiff/Rio-Cogeo*. March 9; Cogeotiff, released March 19. https://github.com/cogeotiff/rio-cogeo.

Cooney, Thomas, and Damon Holzer. 2006. *Appendix C: Interior Columbia Basin Stream Type Chinook Salmon and Steelhead Populations: Habitat Intrinsic Potential Analysis*. 21.

COSEWIC. 2012. *COSEWIC Assessment and Status Report on the Bull Trout Salvelinus Confluentus in Canada.* Report. Committee on the Status of Endangered Wildlife in Canada. https://www.sararegistry.gc.ca/virtual_sara/files/cosewic/sr_omble_tete_plat_bull_trout_1113_e.pdf.

Cote, David, P Frampton, M Langdon, and R Collier. 2005. *Fish Passage and Stream Habitat Restoration in Terra Nova National Park Highway Culverts*.

Development Seed. (2019) 2025. *Developmentseed/Titiler*. V. 0.21.1. June 28; Development Seed, released. https://github.com/developmentseed/titiler.

Diebel, M. W., M. Fedora, S. Cogswell, and J. R. O'Hanley. 2015. "Effects of Road Crossings on Habitat Connectivity for Stream-Resident Fish: STREAM-RESIDENT FISH HABITAT CONNECTIVITY." *River Research and Applications* 31 (10): 1251–61. https://doi.org/10.1002/rra.2822.

Fernandes, I. M., Y. F. Bastos, D. S. Barreto, L. S. Lourenço, and J. M. Penha. 2017. "The Efficacy of Clove Oil as an Anaesthetic and in Euthanasia Procedure for Small-Sized Tropical Fishes." *Brazilian Journal of Biology = Revista Brasleira De Biologia* 77 (3): 444–50. https://doi.org/10.1590/1519-6984.15015.

Finnegan, Noah J., Gerard Roe, David R. Montgomery, and Bernard Hallet. 2005. "Controls on the Channel Width of Rivers: Implications for Modeling Fluvial Incision of Bedrock." *Geology* 33 (3): 229–32. https://doi.org/10.1130/G21171.1.

Fish and Wildlife Compensation Program. 2020. *Peace Region Rivers, Lakes, & Reservoirs Action Plan*. https://fwcp.ca/app/uploads/2020/08/Action-Plan-Peace-Region-Rivers-Lakes-Reservoirs-Aug-11-2020.pdf.

Fish Passage Technical Working Group. 2011. *A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation Fo a Stream Crossing*. https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/checklist-for-fish-habitat-confirmation-201112.pdf.

FLNRORD. 2020a. "Digital Road Atlas (DRA) - Master Partially-Attributed Roads - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development. https://catalogue.data.gov.bc.ca/dataset/digital-road-atlas-dra-master-partially-attributed-roads.

FLNRORD. 2020b. "Forest Tenure Road Section Lines - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development (FLNRORD). https://catalogue.data.gov.bc.ca/dataset/forest-tenure-road-section-lines.

Forest and Range Practices Act. 2018. *Order - Fisheries Sensitive Watershed Prince George Forest District*. http://www.env.gov.bc.ca/wld/frpa/fsw/approved.html.

Forest Investment Account. 2003. *Guidelines for in-Stream and Off-Channel Routine Effectiveness Evaluation*. British Columbia Ministry of Forests. http://env.gov.bc.ca/wld/documents/fia_docs/ree_guidelines.pdf.

Hagen, John, Ray Pillipow, and Nikolaus Gantner. 2018. *Trend in Abundance of Arctic Grayling (Thymallus Arcticus) in Index Sites of the Parsnip River Watershed, 1995-2018.* 37.

Hagen, John, Ian Spendlow, and Ray Pillipow. 2020. *Critical Spawning Habitats and Abundance of Bull Trout in the Williston Reservoir Watershed, 2019*. 57.

Hagen, John, and Mike Stamford. 2021. *Arctic Grayling (Thymallus Arcticus) Abundance and Trend in the Parsnip River Watershed, 1995-2020.* https://a100.gov.bc.ca/pub/acat/documents/r59157/PEA_F21_F_3203_1622500728936_967641590A.pdf.

Hagen, John, and Mike Stamford. 2023. *Parsnip Arctic Grayling Abundance and Critical Habitats 2018-2022 Final Report*.

Hagen, John, and Susanne Weber. 2019. *Limiting Factors, Enhancement Potential, Critical Habitats, and Conservation Status for Bull Trout of the Williston Reservoir Watershed: Information*

Synthesis and Recommended Monitoring Framework. Report. http://fwcp.ca/app/uploads/2019/08/Bull-Trout-Info-Synthesis-Monitoring-Framework-_FINAL29Aug2019.pdf.

Hagen, John, Susanne Williamson, Mike Stamford, and Ray Pillipow. 2015. *Critical Habitats for Bull Trout and Arctic Grayling Within the Parsnip River and Pack River Watersheds*.

Hall, Jason E., Diane M. Holzer, and Timothy J. Beechie. 2007. "Predicting River Floodplain and Lateral Channel Migration for Salmon Habitat Conservation."*Journal of the American Water Resources Association* 43 (3): 786–97. https://doi.org/10.1111/j.1752-1688.2007.00063.x.

Hansen, James, Makiko Sato, and Reto Ruedy. 2012. "Perception of Climate Change."*Proceedings of the National Academy of Sciences* 109 (37). https://doi.org/10.1073/pnas.1205276109.

Hill, N. H., J. L. Thorley, and A. Irvine. 2024. *Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b*. Poisson Consulting Analysis Appendix. https://www.poissonconsulting.ca/f/1295467017.

Hill, N. H., J. L. Thorley, and A. Irvine. 2025. *Spatial Stream Network Analysis of Skeena Watershed Stream Temperatures*. Poisson Consulting Analysis Appendix. https://www.poissonconsulting.ca/f/1130667589.

Hirst, S. M. 1991. *Impacts of the Operations of Existing Hydroelectric Developments on Fishery Resources in British Columbia*. https://waves-vagues.dfo-mpo.gc.ca/Library/124832main.pdf.

"History Who We Are." 2023. McLeod Lake Indian Band. https://www.mlib.ca/about/History/.

Hobbs, Jared, Caren C. Helbing, and Caren Goldberg. 2025. *Environmental DNA Protocol for Freshwater Aquatic Ecosystems: Version 3.0*. BC Ministry of Environment, Ecosystems Branch.

Irvine. (2021) 2022. *Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021*. https://github.com/NewGraphEnvironment/fish_passage_skeena_2021_reporting.

Irvine. 2022. *PEA-F22-F-3577-DCA Restoring Fish Passage in the Peace Region*. https://newgraphenvironment.github.io/fish_passage_parsnip_2021_reporting/.

Irvine. (2025) 2025. *NewGraphEnvironment/Stac_uav_bc*. January 29, released. https://github.com/NewGraphEnvironment/stac_uav_bc.

Irvine, Allan. 2020. *Parsnip River Watershed – Fish Habitat Confirmations (PEA-F20-F-2967)*. https://newgraphenvironment.github.io/Parsnip_Fish_Passage/.

Irvine, and L Schick. 2025. *Restoring Fish Passage in the Peace Region - 2024*. https://newgraphenvironment.github.io/fish_passage_peace_2024_reporting/.

Irvine, and M Winterscheidt. 2023. *Restoring Fish Passage in the Peace Region - 2022 - PEA-F23-F-3761-DCA*. https://newgraphenvironment.github.io/fish_passage_peace_2022_reporting/.

Irvine, and M Winterscheidt. 2024. *Restoring Fish Passage in the Peace Region - 2023*. https://newgraphenvironment.github.io/fish_passage_peace_2023_reporting/.

Kang, Do Hyuk, Huilin Gao, Xiaogang Shi, Siraj ul Islam, and Stephen J. Déry. 2016. "Impacts of a Rapidly Declining Mountain Snowpack on Streamflow Timing in Canada's Fraser River Basin."*Scientific Reports* 6 (1). https://doi.org/10.1038/srep19299.

Karl, Thomas R., Richard W. Knight, Kevin P. Gallo, et al. 1993. "A New Perspective on Recent Global Warming: Asymmetric Trends of Daily Maximum and Minimum Temperature."*Bulletin of the American Meteorological Society* 74 (6): 1007–23. https://doi.org/10.1175/1520-0477(1993)074<1007:ANPORG>2.0.CO;2.

Kemp, P. S., and J. R. O'Hanley. 2010. "Procedures for Evaluating and Prioritising the Removal of Fish Passage Barriers: A Synthesis: EVALUATION OF FISH PASSAGE BARRIERS."*Fisheries Management and Ecology*, no–. https://doi.org/10.1111/j.1365-2400.2010.00751.x.

Kirsch, J M, Joseph D Buckwalter, and Daniel J Reed. 2014. *Fish Inventory and Anadromous Cataloging in the Susitna River, Matanuska River, and Knik River Basins, 2003 and 2011.* 149.

Knowles, Noah, Michael D. Dettinger, and Daniel R. Cayan. 2006. "Trends in Snowfall Versus Rainfall in the Western United States." *Journal of Climate* 19 (18): 4545–59. https://doi.org/10.1175/jcli3850.1.

Martins, E, B O'Connor, J Bottoms, et al. 2022. *Spatial Ecology of Arctic Grayling in the Parsnip Core Area*. https://a100.gov.bc.ca/pub/acat/documents/r59889/PEA_F22_F_3388_1664380292568_8FCE11DCCD.pdf.

MoE. 2011. *Field Assessment for Determining Fish Passage Status of Closed Bottom Structures*. BC Ministry of Environment (MoE). https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/field-assessment-for-determining-fish-passage-status-of-cbs.pdf.

MoE. 2024a. "Known BC Fish Observations and BC Fish Distributions." https://catalogue.data.gov.bc.ca/dataset/known-bc-fish-observations-and-bc-fish-distributions.

MoE. 2024b. "Provincial Obstacles to Fish Passage - Data Catalogue." https://catalogue.data.gov.bc.ca/dataset/provincial-obstacles-to-fish-passage.

Mote, Philip W., Alan F. Hamlet, Martyn P. Clark, and Dennis P. Lettenmaier. 2005. "DECLINING MOUNTAIN SNOWPACK IN WESTERN NORTH AMERICA*." *Bulletin of the American Meteorological Society* 86 (1): 39–50. https://doi.org/10.1175/bams-86-1-39.

Mote, Philip W., Sihan Li, Dennis P. Lettenmaier, Mu Xiao, and Ruth Engel. 2018. "Dramatic Declines in Snowpack in the Western US." *Npj Climate and Atmospheric Science* 1 (1). https://doi.org/10.1038/s41612-018-0012-1.

Muñoz Sabater, Joaquín. 2019. *ERA5-Land Hourly Data from 1950 to Present*. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). https://doi.org/10.24381/cds.e2161bac.

Murray, Brent W., and Barry Booth. 2023. *Report on the 2023 eDNA Survey of Study Creeks in Nechako Watershed from Prince George to the Upper Fraser*. University of Northern British Columbia, Department of Ecosystem Science and Management. https://www.newssociety.org/uploads/images/News%20report%20Images/e-DNA%20summary-%20UNBC-NEWSS%202023.pdf.

Nagel, David E., John M. Buffington, Sharon L. Parkes, Seth Wenger, and John R. Goode. 2014. *A Landscape Scale Valley Confinement Algorithm: Delineating Unconfined Valley Bottoms for Geomorphic, Aquatic, and Riparian Applications*. Gen. Tech. Rep. RMRS-GTR-321. U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station. https://www.fs.usda.gov/rmrs/publications/landscape-scale-valley-confinement-algorithm-delineating-unconfined-valley-bottoms.

Norris, Simon. (2018) 2024. *Smnorris/Bcfishobs*. May 11, released. https://github.com/smnorris/bcfishobs.

Norris, Simon. (2020) 2024. *Smnorris/Bcfishpass*. November 7, released January 12. https://github.com/smnorris/bcfishpass.

OpenDroneMap Authors. (2014) 2025. *OpenDroneMap/ODM*. V. 3.5.4. September 25; OpenDroneMap, released. https://github.com/OpenDroneMap/ODM.

Pearce, T, S Lieske, J Morgan, and A Salonas. 2019. *First Nations Information Gathering on Kokanee, Bull Trout and Arctic Grayling: : Report for the Tse'khene Nation (McLeod Lake Indian Band)*. https://a100.gov.bc.ca/pub/acat/documents/r57248/PEA-F19-F-2866-DCA-5-Final_Report-Arctic_North-Ko_1600902816644_0895426641.pdf.

Pepin, N., R. S. Bradley, H. F. Diaz, et al. 2015. "Elevation-Dependent Warming in Mountain Regions of the World." *Nature Climate Change* 5 (5): 424–30. https://doi.org/10.1038/nclimate2563.

Porter, Marc, Darcy Pickard, Katherine Wieckowski, and Katy Bryan. 2008. *Developing Fish Habitat Models for Broad-Scale Forest Planning in the Southern Interior of B.C.* ESSA Technologies Ltd. and

B.C. Ministry of the Environment (MOE) for B.C. Forest Science Program. https://www.for.gov.bc.ca/hfd/library/FIA/2008/FSP_Y081231.pdf.
PySTAC Authors. (2017) 2025. *Stac-Utils/Pystac*. V. 1.12.2. November 8; Stac-utils, released. https://github.com/stac-utils/pystac.
Radiantearth. 2024. "STAC: SpatioTemporal Asset Catalogs." https://stacspec.org/en/.
Resource Inventory Standards Committee. 2001. *Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Standards and Procedures*. https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/nr-laws-policy/risc/recce2c.pdf.
Roberge, M, J M B Hume, C K Minns, and T Slaney. 2002. *Life History Characteristics of Freshwater Fishes Occurring in British Columbia and the Yukon, with Major Emphasis on Stream Habitat Characteristics*. 262.
Rosenfeld, Jordan, Marc Porter, and Eric Parkinson. 2000. *Habitat Factors Affecting the Abundance and Distribution of Juvenile Cutthroat Trout (Oncorhynchus Clarki) and Coho Salmon (Oncorhynchus Kisutch)*. 57: 9.
Shrimpton, J. M., S. L. Roberts, and A. D. Clarke. 2012. *Genetic Analysis of Arctic Grayling Population Structure in the Williston Watershed*. http://a100.gov.bc.ca/appsdata/acat/documents/r38246/Report_311_GR_genetics_2012_1379090176074_c3e946726980918f9466d47b59978ad28ded2b969041dd7fe53b83cfe3011493.pdf.
Simoes, Rolf, Felipe Souza, Matheus Zaglia, Gilberto Ribeiro de Queiroz, Rafael dos Santos, and Karine Ferreira. 2021. "Rstac: An R Package to Access Spatiotemporal Asset Catalog Satellite Imagery." *2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 7674–77. https://doi.org/10.1109/IGARSS47720.2021.9553518.
Slaney, P A, and D Zaldokas. 1997. *Fish Habitat Rehabilitation Procedures*. https://www.for.gov.bc.ca/hfd/library/FFIP/Slaney_PA1997_A.pdf.
Sloat, Matthew R., Gordon H. Reeves, and Kelly R. Christiansen. 2017. "Stream Network Geomorphology Mediates Predicted Vulnerability of Anadromous Fish Habitat to Hydrologic Change in Southeast Alaska." *Global Change Biology* 23 (2): 604–20. https://doi.org/10.1111/gcb.13466.
Species Registry Canada. 2020. "Bull Trout (Salvelinus Confluentus), Western Arctic Populations - Species Search - Species at Risk Registry." https://species-registry.canada.ca/index-en.html#/species/1202-869.
stac-utils. 2025. *Stac-Utils/Stac-Fastapi-Pgstac: PostgreSQL Backend for Stac-Fastapi Using Pgstac (https://github.com/stac-utils/pgstac)*. V. 4.0.2. Released. https://github.com/stac-utils/stac-fastapi-pgstac.
Stamford, Mike, John Hagen, and Susanne Williamson. 2017. *FWCP Arctic Grayling Synthesis Report*. 148. http://fwcp.ca/app/uploads/2017/07/FWCP_Grayling_Synthesis_Final.pdf.
Stewart, D B, N J Mochnacz, J D Reist, T J Carmichael, and C D Sawatzky. 2007. "Fish Life History and Habitat Use in the Northwest Territories: Arctic Grayling (Thymallus Arcticus)." *Canadian Manuscript Report of Fisheries and Aquatic Sciences 2797*, 64. https://www.researchgate.net/publication/255580930_Fish_life_history_and_habitat_use_in_the_Northwest_Territories_Arctic_grayling_Thymallus_arcticus.
Stewart, Iris T., Daniel R. Cayan, and Michael D. Dettinger. 2005. "Changes Toward Earlier Streamflow Timing Across Western North America." *Journal of Climate* 18 (8): 1136–55. https://doi.org/10.1175/jcli3321.1.
Swales, S., and C. D. Levings. 1989. "Role of Off-Channel Ponds in the Life Cycle of Coho Salmon (*Oncorhynchus Kisutch*) and Other Juvenile Salmonids in the Coldwater River, British Columbia." *Canadian Journal of Fisheries and Aquatic Sciences* 46 (2): 232–42. https://doi.org/10.1139/f89-032.

Thompson, Richard. 2013. *Assessing Fish Passage at Culverts – the Method, Its Metrics and Preliminary Findings from over 4,000 Assessments.*https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fish-passage/assessing_fish_passage_at_culverts.pdf.

Thorley, J. L., S. Norris, and A. Irvine. 2021. *Channel Width 2021b*. https://www.poissonconsulting.ca/f/859859031.

Thorley, J, and A Irvine. 2021. "Channel Width 2021."https://www.poissonconsulting.ca/f/1792764180.

Toffolon, Marco, and Sebastiano Piccolroaz. 2015. "A Hybrid Model for River Water Temperature as a Function of Air Temperature and Discharge."*Environmental Research Letters* 10 (11): 114011. https://doi.org/10.1088/1748-9326/10/11/114011.

Washington Department of Fish & Wildlife. 2009. *Fish Passage Barrier and Surface Water Diversion Screening Assessment and Prioritization Manual*. Washington Department of Fish; Wildlife. Olympia, Washington. https://wdfw.wa.gov/sites/default/files/publications/00061/wdfw00061.pdf.

Williamson, S. A, and J. T. Zimmerman. 2005. *Region 7a, Omineca Arctic Grayling (Thymallus Arcticus): Data Consolidation Review and Gap Analysis.*B.C. Ministry of Water, Land, and Air Protection.

Woll, Christine, David Albert, and Diane Whited. 2017. *Salmon Ecological Systems*. The Nature Conservancy. https://www.conservationgateway.org/ConservationByGeography/NorthAmerica/UnitedStates/alaska/sw/cpa/Documents/TNC_A_Salmon_Ecological_Systems_Model_Nushagak_Kvichak.pdf.

Note that this reference was not actually cited in the body of the report but rather in the `index.Rmd` file `yaml` headerj under the `no_cite` key.

Session Info

Information about the computing environment is important for reproducibility. A summary of the computing environment is saved to `session_info.csv`, which can be viewed and downloaded from here.

```
## ─ Session info ───────────────────────────────────────────────────────────────
##  setting  value
##  version  R version 4.5.2 (2025-10-31)
##  os       macOS Tahoe 26.2
##  system   aarch64, darwin20
##  ui       X11
##  language (EN)
##  collate  en_US.UTF-8
##  ctype    en_US.UTF-8
##  tz       America/Vancouver
##  date     2026-06-26
##  pandoc   3.9.0.2 @ /opt/homebrew/bin/ (via rmarkdown)
##  quarto   1.4.553 @ /usr/local/bin/quarto
## 
## ─ Packages ───────────────────────────────────────────────────────────────────
##  package          * version    date (UTC) lib source
##  arrow              22.0.0.1   2025-12-23 [2] CRAN (R 4.5.2)
##  askpass            1.2.1      2024-10-04 [1] CRAN (R 4.5.0)
##  assertthat         0.2.1      2019-03-21 [2] CRAN (R 4.5.0)
##  base64enc          0.1-6      2026-02-02 [1] CRAN (R 4.5.2)
##  bit                4.6.0      2025-03-06 [1] CRAN (R 4.5.0)
##  bit64              4.8.0      2026-04-21 [1] CRAN (R 4.5.2)
##  blob               1.3.0      2026-01-14 [1] CRAN (R 4.5.2)
##  bookdown         * 0.46       2025-12-05 [2] CRAN (R 4.5.2)
##  boot               1.3-32     2025-08-29 [2] CRAN (R 4.5.2)
##  bslib              0.10.0     2026-01-26 [1] CRAN (R 4.5.2)
##  cachem             1.1.0      2024-05-16 [1] CRAN (R 4.5.0)
##  cd               * 0.3.0      2026-05-12 [1] Github
(NewGraphEnvironment/cd@a3e4733)
##  chk                0.10.0     2025-01-24 [1] CRAN (R 4.5.0)
##  class              7.3-23     2025-01-01 [2] CRAN (R 4.5.2)
##  classInt           0.4-11     2025-01-08 [1] CRAN (R 4.5.0)
##  cli                3.6.6      2026-04-09 [1] CRAN (R 4.5.2)
##  codetools          0.2-20     2024-03-31 [2] CRAN (R 4.5.2)
##  crayon             1.5.3      2024-06-20 [1] CRAN (R 4.5.0)
##  crosstalk          1.2.2      2025-08-26 [1] CRAN (R 4.5.0)
##  curl               7.1.0      2026-04-22 [1] CRAN (R 4.5.2)
##  data.table         1.18.2.1   2026-01-27 [1] CRAN (R 4.5.2)
```

```
## DBI              1.3.0      2026-02-25 [1] CRAN (R 4.5.2)
## dbplyr           2.5.2      2026-02-13 [1] CRAN (R 4.5.2)
## desc             1.4.3      2023-12-10 [2] CRAN (R 4.5.0)
## devtools         2.4.6      2025-10-03 [2] CRAN (R 4.5.0)
## digest           0.6.39     2025-11-19 [1] CRAN (R 4.5.2)
## doParallel       1.0.17     2022-02-07 [2] CRAN (R 4.5.0)
## dplyr          * 1.2.1      2026-04-03 [1] CRAN (R 4.5.2)
## e1071            1.7-17     2025-12-18 [1] CRAN (R 4.5.2)
## elevatr          0.99.1     2025-09-10 [2] CRAN (R 4.5.0)
## ellipsis         0.3.2      2021-04-29 [2] CRAN (R 4.5.0)
## english          1.2-6      2021-08-21 [2] CRAN (R 4.5.0)
## evaluate         1.0.5      2025-08-27 [1] CRAN (R 4.5.0)
## farver           2.1.2      2024-05-13 [1] CRAN (R 4.5.0)
## fasstr           0.5.3      2024-09-27 [2] CRAN (R 4.5.0)
## fastmap          1.2.0      2024-05-15 [1] CRAN (R 4.5.0)
## fishbc         * 0.2.1.9000 2026-01-15 [2] Github (lucy-
schick/fishbc@0607ffd)
## forcats        * 1.0.1      2025-09-25 [2] CRAN (R 4.5.0)
## foreach          1.5.2      2022-02-02 [2] CRAN (R 4.5.0)
## fpr            * 1.2.0      2026-02-03 [2] Github
(NewGraphEnvironment/fpr@2817533)
## fs               2.1.0      2026-04-18 [1] CRAN (R 4.5.2)
## generics         0.1.4      2025-05-09 [1] CRAN (R 4.5.0)
## ggplot2        * 4.0.3      2026-04-22 [1] CRAN (R 4.5.2)
## ggrepel          0.9.7      2026-02-25 [2] CRAN (R 4.5.2)
## gh               1.5.0      2025-05-26 [2] CRAN (R 4.5.0)
## glue             1.8.1      2026-04-17 [1] CRAN (R 4.5.2)
## gq             * 0.1.0      2026-06-25 [1] local
## gtable           0.3.6      2024-10-25 [1] CRAN (R 4.5.0)
## here             1.0.2      2025-09-15 [2] CRAN (R 4.5.0)
## hms              1.1.4      2025-10-17 [1] CRAN (R 4.5.0)
## htmltools        0.5.9      2025-12-04 [1] CRAN (R 4.5.2)
## htmlwidgets      1.6.4      2023-12-06 [1] CRAN (R 4.5.0)
## httr             1.4.8      2026-02-13 [1] CRAN (R 4.5.2)
## iterators        1.0.14     2022-02-05 [2] CRAN (R 4.5.0)
## janitor          2.2.1      2024-12-22 [1] CRAN (R 4.5.0)
## jquerylib        0.1.4      2021-04-26 [1] CRAN (R 4.5.0)
## jsonlite         2.0.0      2025-03-27 [1] CRAN (R 4.5.0)
## kableExtra     * 1.4.0      2024-01-24 [2] CRAN (R 4.5.0)
## Kendall          2.2.2      2025-12-22 [2] CRAN (R 4.5.2)
## KernSmooth       2.23-26    2025-01-01 [2] CRAN (R 4.5.2)
## knitr          * 1.51       2025-12-20 [1] CRAN (R 4.5.2)
## labeling         0.4.3      2023-08-29 [1] CRAN (R 4.5.0)
## lattice          0.22-9     2026-02-09 [2] CRAN (R 4.5.2)
## leafem         * 0.2.5      2025-08-28 [2] CRAN (R 4.5.0)
## leaflet        * 2.2.3      2025-09-04 [1] CRAN (R 4.5.0)
## lifecycle        1.0.5      2026-01-08 [1] CRAN (R 4.5.2)
```

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##  lubridate       * 1.9.5      2026-02-04 [1] CRAN (R 4.5.2)
##  magick            2.9.0      2025-09-08 [2] CRAN (R 4.5.0)
##  magrittr          2.0.5      2026-04-04 [1] CRAN (R 4.5.2)
##  Matrix            1.7-4      2025-08-28 [2] CRAN (R 4.5.2)
##  memoise           2.0.1      2021-11-26 [1] CRAN (R 4.5.0)
##  mgcv              1.9-3      2025-04-04 [2] CRAN (R 4.5.2)
##  ngr             * 0.0.1      2026-01-22 [2] Github
(newgraphenvironment/ngr@22c9714)
##  nlme              3.1-168    2025-03-31 [2] CRAN (R 4.5.2)
##  otel              0.2.0      2025-08-29 [1] CRAN (R 4.5.0)
##  pagedown        * 0.23       2025-08-20 [2] CRAN (R 4.5.0)
##  pak               0.10.0     2026-06-07 [1] CRAN (R 4.5.2)
##  pdftools        * 3.7.0      2026-01-30 [2] CRAN (R 4.5.2)
##  pillar            1.11.1     2025-09-17 [1] CRAN (R 4.5.0)
##  pkgbuild          1.4.8      2025-05-26 [2] CRAN (R 4.5.0)
##  pkgconfig         2.0.3      2019-09-22 [1] CRAN (R 4.5.0)
##  pkgload           1.5.0      2026-02-03 [2] CRAN (R 4.5.2)
##  plyr              1.8.9      2023-10-02 [1] CRAN (R 4.5.0)
##  png               0.1-9      2026-03-15 [1] CRAN (R 4.5.2)
##  prettyunits       1.2.0      2023-09-24 [1] CRAN (R 4.5.0)
##  processx          3.9.0      2026-04-22 [1] CRAN (R 4.5.2)
##  progress          1.2.3      2023-12-06 [2] CRAN (R 4.5.0)
##  progressr         0.18.0     2025-11-06 [2] CRAN (R 4.5.0)
##  proxy             0.4-29     2025-12-29 [1] CRAN (R 4.5.2)
##  purrr           * 1.2.2      2026-04-10 [1] CRAN (R 4.5.2)
##  qpdf              1.4.1      2025-07-02 [2] CRAN (R 4.5.0)
##  R6                2.6.1      2025-02-15 [1] CRAN (R 4.5.0)
##  rappdirs          0.3.4      2026-01-17 [1] CRAN (R 4.5.2)
##  raster            3.6-32     2025-03-28 [1] CRAN (R 4.5.0)
##  rayshader         0.37.3     2024-02-21 [2] CRAN (R 4.5.0)
##  rbbt            * 0.0.0.9000 2026-01-15 [2] Github
(paleolimbot/rbbt@212680c)
##  RColorBrewer      1.1-3      2022-04-03 [1] CRAN (R 4.5.0)
##  Rcpp              1.1.1-1.1  2026-04-24 [1] CRAN (R 4.5.2)
##  RcppRoll          0.3.1      2024-07-07 [2] CRAN (R 4.5.0)
##  readr           * 2.2.0      2026-02-19 [1] CRAN (R 4.5.2)
##  readwritesqlite * 0.2.0.9021 2026-01-15 [2] Github
(poissonconsulting/readwritesqlite@ccdb297)
##  remotes           2.5.0      2024-03-17 [2] CRAN (R 4.5.0)
##  rgl               1.3.31     2025-11-14 [2] CRAN (R 4.5.2)
##  rlang             1.2.0      2026-04-06 [1] CRAN (R 4.5.2)
##  rmarkdown       * 2.31       2026-03-26 [1] CRAN (R 4.5.2)
##  RPostgres       * 1.4.10     2026-02-16 [1] CRAN (R 4.5.2)
##  rprojroot         2.1.1      2025-08-26 [2] CRAN (R 4.5.0)
##  RSQLite           3.52.0     2026-05-10 [1] CRAN (R 4.5.2)
##  rstudioapi        0.18.0     2026-01-16 [2] CRAN (R 4.5.2)
##  rvest             1.0.5      2025-08-29 [2] CRAN (R 4.5.0)
```

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##  s2                  1.1.9      2025-05-23 [1] CRAN (R 4.5.0)
##  S7                  0.2.2      2026-04-22 [1] CRAN (R 4.5.2)
##  sass                0.4.10     2025-04-11 [1] CRAN (R 4.5.0)
##  scales              1.4.0      2025-04-24 [1] CRAN (R 4.5.0)
##  sessioninfo         1.2.3      2025-02-05 [2] CRAN (R 4.5.0)
##  sf                * 1.1-0      2026-02-24 [1] CRAN (R 4.5.2)
##  snakecase           0.11.1     2023-08-27 [1] CRAN (R 4.5.0)
##  sp                  2.2-1      2026-02-13 [1] CRAN (R 4.5.2)
##  staticimports     * 0.0.0.9001 2026-01-15 [2] Github
(newgraphenvironment/staticimports@19cd73c)
##  stringi             1.8.7      2025-03-27 [1] CRAN (R 4.5.0)
##  stringr           * 1.6.0      2025-11-04 [1] CRAN (R 4.5.0)
##  svglite             2.2.2      2025-10-21 [2] CRAN (R 4.5.0)
##  systemfonts         1.3.1      2025-10-01 [2] CRAN (R 4.5.0)
##  terra             * 1.9-11     2026-03-26 [1] CRAN (R 4.5.2)
##  textshaping         1.0.4      2025-10-10 [2] CRAN (R 4.5.0)
##  tibble            * 3.3.1      2026-01-11 [1] CRAN (R 4.5.2)
##  tidyhydat           0.7.2      2025-10-23 [2] CRAN (R 4.5.0)
##  tidyr             * 1.3.2      2025-12-19 [1] CRAN (R 4.5.2)
##  tidyselect          1.2.1      2024-03-11 [1] CRAN (R 4.5.0)
##  tidyterra         * 1.1.0      2026-03-11 [2] CRAN (R 4.5.2)
##  tidyverse         * 2.0.0      2023-02-22 [2] CRAN (R 4.5.0)
##  tidyxl              1.0.10     2026-01-15 [2] Github
(nacnudus/tidyxl@7e2fbe7)
##  timechange          0.4.0      2026-01-29 [1] CRAN (R 4.5.2)
##  tzdb                0.5.0      2025-03-15 [1] CRAN (R 4.5.0)
##  units               1.0-1      2026-03-11 [1] CRAN (R 4.5.2)
##  usethis             3.2.1      2025-09-06 [2] CRAN (R 4.5.0)
##  vctrs               0.7.3      2026-04-11 [1] CRAN (R 4.5.2)
##  viridisLite         0.4.3      2026-02-04 [1] CRAN (R 4.5.2)
##  vroom               1.7.1      2026-03-31 [1] CRAN (R 4.5.2)
##  withr               3.0.2      2024-10-28 [1] CRAN (R 4.5.0)
##  wk                  0.9.5      2025-12-18 [1] CRAN (R 4.5.2)
##  xfun                0.57       2026-03-20 [1] CRAN (R 4.5.2)
##  xml2                1.5.2      2026-01-17 [1] CRAN (R 4.5.2)
##  yaml                2.3.12     2025-12-10 [1] CRAN (R 4.5.2)
##  zyp                 0.11-1     2023-03-22 [2] CRAN (R 4.5.0)
##
##  [1] /Users/airvine/Library/R/arm64/4.5/library
##  [2] /Library/Frameworks/R.framework/Versions/4.5-
arm64/Resources/library
##  * ── Packages attached to the search path.
##
##
```

Attachment - Phase 1 Data and Photos

Data and photos for all Phase 1 fish passage assessments are provided online, with a PDF version available here.

Attachment - Habitat Assessment and Fish Sampling Data

All field data collected is available here.

Habitat assessment data (including fish sampling and PIT tagging information) is available for download here https://github.com/NewGraphEnvironment/fish_passage_peace_2025_reporting/blob/main/data/habitat_confirmations.xls.

Raw fish data is available for download here https://github.com/NewGraphEnvironment/fish_passage_peace_2025_reporting/blob/main/data/fish_data_tags_joined.csv.